T2K 2.315×10^{20}-POT 3-Flavour Muon-Antineutrino Disappearance Analysis

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Abstract

We report the results of the 3-flavour $\bar{\nu}_\mu$-disappearance analysis, performed by the *VALOR* group, on the combined Run 5+6 dataset, using only data taken in antineutrino beam mode. This analysis adapts the work presented in Refs. [1], [2], [3], [4], [5] and [6] to this dataset. The Run 5+6 dataset corresponds to an integrated J-PARC neutrino beam exposure of 2.315×10^{20} POT.

This analysis predicts 59.8 ± 9.2 (syst) single μ-like ring events in Super K in the absence of any oscillation, but only 17 were observed. The observed deficit has a strong energy dependence: the ratio of the number of observed events to the number expected under the no-oscillation hypothesis is ∼18% $<$ 0.5 GeV, ∼9% between 0.5 and 1 GeV and ∼69% $>$ 1 GeV.

In order to be able to investigate whether the oscillation parameters for $\bar{\nu}_\mu$ disappearance are different from those for ν_μ disappearance, two separate sets of oscillation parameters are used. One set is applied to the neutrino interaction modes, while the other set is applied to the antineutrino modes.

A $\bar{\nu}_\mu$-disappearance analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single μ-like ring events was fitted, and separate fits were made for the normal and the inverted mass hierarchies. In these fits, $\sin^2\overline{\theta}_{23}$ and either $|\overline{\Delta m^2_{32}}|$ (normal hierarchy) or $|\overline{\Delta m^2_{31}}|$ (inverted hierarchy) were allowed to float, while all other oscillation parameters in both sets were fixed to T2K run1-4 best-fit values. All 44 systematic parameters considered in this analysis were also allowed to float in the fits.

The 3-flavour ν_μ-disappearance fit of the combined Run 5+6 dataset for the normal mass hierarchy gives $\sin^2\overline{\theta}_{23} = 0.511$ and $|\overline{\Delta m^2_{32}}| = 2.318 \times 10^{-3}$ eV^2/c^4 as the best-fit values (p-value = 0.40). Assuming the inverted mass hierarchy, the fit yields $\sin^2\overline{\theta}_{23} = 0.514$ and $|\overline{\Delta m^2_{31}}| = 2.291 \times 10^{-3}$ eV^2/c^4 as the best-fit values (p-value = 0.42). The p-value for the normal hierarchy best-fit values given a null hypothesis of the normal hierarchy best-fit values of the equivalent parameters from the official Run 1-4 neutrino mode analysis is 0.74; thus the results of this analysis considered together with the Run 1-4 neutrino mode results are consistent with CPT being conserved. Confidence regions in the parameter space $\sin^2\overline{\theta}_{23}$ - $|\overline{\Delta m^2_{32}}|$ (normal hierarchy) and $\sin^2\overline{\theta}_{23}$ - $|\overline{\Delta m^2_{31}}|$ (inverted hierarchy) are shown using the constant-$\Delta\chi^2$ method.

1. Introduction

In this technical note, we report the results of a 3-flavour $\bar{\nu}_\mu$-disappearance analysis, performed by the *VALOR* group, on the combined Run 5+6 dataset, using only data taken in antineutrino beam mode (2.315×10^{20} POT). This dataset was collected in two run periods: May - June 2014 (Run 5, 0.506×10^{20} POT in antineutrino beam mode) and November 2014 - March 2015 (Run 6, 1.809×10^{20} POT in antineutrino beam mode). This analysis builds on our ν_μ-disappearance analyses presented in Refs. [1], [2], [3], [4], [5] and [6], and adapts these analyses to antineutrino beam mode data.

One of the objectives of this analysis is to determine whether the oscillation parameters for $\bar{\nu}_\mu$ disappearance are different from those for ν_μ disappearance. For this reason, we use two separate sets of oscillation parameters in this analysis. The first set is $\sin^2\overline{\theta}_{23}$, $\sin^2\overline{\theta}_{12}$, $\sin^2\overline{\theta}_{13}$, $|\overline{\Delta m^2_{32}}|$ (normal hierarchy) or $|\overline{\Delta m^2_{31}}|$ (inverted hierarchy) , $\overline{\Delta m^2_{21}}$ and δ_{CP}, and these are applied to the $\bar{\nu}_\mu$, $\bar{\nu}_e$ and oscillated $\bar{\nu}_e$ interaction modes. The second set of oscillation parameters is $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), Δm^2_{21} and δ_{CP}, which are applied to the ν_μ, ν_e and oscillated ν_e interaction modes. All these parameters are fixed in the fits of the Super K data except $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ (normal hierarchy) or $|\overline{\Delta m^2_{31}}|$ (inverted hierarchy).

These oscillation parameters are determined by fitting the predicted reconstructed energy spectrum of 1 μ-like ring events at SuperK to the observed spectrum. The method of calculation of this predicted spectrum is presented in Sec. 2. This analysis uses flux tuning based on recent NA61 data [13][35]. It also uses an indirect extrapolation from ND280 to SuperK, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of the muon momentum-angle distributions of $\bar{\nu}_\mu$ and ν_μ CC0π-like, CC1π-like and CC other events whose vertices are in FGD1[24]. Updated nominal predictions of the reconstructed energy spectrum of single μ-like ring events in SuperK are presented for Run 5+6 in Sec. 2

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is made using the likelihood-ratio method [39]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. We consider a total of 44 energy scale, neutrino flux, neutrino cross-section, efficiency, initial-state nuclear environment and final-state rescattering systematics (nuisance parameters). In this Run 5+6 analysis, we allow all these nuisance parameters to float. The nuisance parameters and their effects on the extrapolated single μ-like ring event reconstructed energy spectrum are described in Sec. 3. The oscillation fitting method is presented in detail in Sec. 4.1, while the chosen goodness-of-fit test, suitable for the current statistics, is described in Sec. 4.2. The performance of the fitter was tested in detail for the Run 5+6 fit using ensembles of toy MC experiments, and a summary of the results of these tests is presented in Sec. 5.

The 2D confidence regions in $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}|$ are constructed using the constant-$\Delta\chi^2$ method. Systematic errors are included in the oscillation contours using the profiling method.

The results of the measurements of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}|$ using the Run 5+6 dataset are given in Sec. 6.1 of this note, while the goodness-of-fit tests are presented in Sec. 6.2. The best-fit spectra are shown in Sec. 6.1 for each of the normal and inverted mass hierarchy assumptions for the Run 5+6 dataset. The confidence regions are shown in Sec. 6.3. In that section, we also show comparisons of our T2K Run 5+6 results with results from MINOS as well as with the expected T2K sensitivity for an integrated exposure of 2.315×10^{20} POT.

Finally, a summary is given in Sec. 7.

2. Predictions of single μ-like ring event reconstructed energy spectrum in SuperK

2.1. Methodology

In this analysis, the oscillation parameters are determined by fitting the reconstructed energy (E_r) spectrum of single μ-like ring events observed at SuperK with the predicted spectrum. The predicted number $N_{SK;r}$ of single μ-like ring events in the rth reconstructed energy bin is computed as follows:

$$N_{SK;r} = \sum_{m} \sum_{t} \sum_{r\prime} P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime;\vec{f}} \cdot N^{MC}_{SK;m;r\prime;t}. \quad (1)$$

In Eq. 1, $N^{MC}_{SK;m;r\prime;t}$ is the input SuperK Monte Carlo (MC) template containing the number of events in the single μ-like ring MC sample with true reaction mode m in the true energy bin t and the reconstructed energy bin $r\prime$. The construction of the nominal MC templates is discussed in Sec. 2.2. Both flux-tuning weights based on NA61 hadron-production data (see Sec. 2.3) and flux and cross-section tuning weights based on ND280 data (see Sec. 2.4) are applied to the nominal MC templates. $S_{m;t;r\prime;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$ and a vector of nuisance (systematic) parameters $\vec{f}$. A complete description of the nuisance parameters considered in this analysis is given in Sec. 3. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the SuperK reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the SuperK MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in Sec. 2.5.

2.2. Construction of nominal SuperK Monte Carlo templates

The nominal SuperK single μ-like ring MC templates $N^{MC}_{SK;m;r;t}$ are constructed by applying the single μ-like ring selection cuts to the official SuperK MC samples. Details of the MC samples are given in Sec. 2.2.1. The calculation of the normalization of each MC sample is described in Sec. 2.2.2. The official single μ-like ring selection cuts are listed in Sec. 2.2.3. The list of MC templates used in this analysis is shown in Sec. 2.2.4, while the true and reconstructed energy binning of the templates is discussed in Sec. 2.2.5.

2.2.1. Input Monte Carlo samples

This analysis uses the official 14a SuperK MC samples. These were generated using the 13a JNUBEAM flux, NEUT 5.3.2, skdetsim v13p90 and the 14b SuperK software [27]. Six MC samples were generated for antineutrino beam mode running (horn current -250kA). Four of the samples ($\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$ and ν_e) were generated using the nominal JNUBEAM flux without oscillations. The fifth sample is $\bar{\nu}_e$ interactions generated with the $\bar{\nu}_\mu$ flux, and will be denoted the "oscillated $\bar{\nu}_e$" sample (it corresponds to the "signal-$\bar{\nu}_e$" sample in the $\bar{\nu}_e$-appearance analysis). The sixth sample is ν_e interactions generated with the ν_μ flux, and will be denoted the "oscillated ν_e" sample.

2.2.2. Normalization of Monte Carlo samples

The normalization (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dS dI dE \cdot \frac{d^3\Phi_{SK}}{dS dI dE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \quad (2)$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, σ_{H_2O} is the total interaction cross section in water for the given neutrino species, I is the beam intensity in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume.

The SuperK MC samples are normalised to the Run 5+6 POT (2.315×10^{20}) using the numbers in Tab. 1. Assuming that all the MC files are used, each sample is first normalised to 10^{21} POT by weighting it with the ratio of the number of events per 22.5 kt per 10^{21} POT (second column) to the number of generated events in the true fiducial volume in all the available MC (third column). Then each sample is normalised to 2.315×10^{20} POT by weighting the result by 0.25 (the Run 5+6 POT in units of 10^{21} POT).

Sample	Number of events per 22.5 kt fiducial per 10^{21} POT	Number of generated events in true fiducial volume in all available MC
$\bar{\nu}_\mu$	353.789	633988
ν_μ	208.369	634659
$\bar{\nu}_e$	6.811	127013
ν_e	8.941	127262
Oscillated $\bar{\nu}_e$	376.420	126812
Oscillated ν_e	215.099	127278

Table 1: Normalization of input SuperK MC samples [27]

2.2.3. Single μ-like ring event selection cuts

The SuperK MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard SuperK selection cuts for single μ-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is μ-like (`ip[0] = 3`),
- the reconstructed muon momentum is above 200 MeV (`amomm[0] > 200`), and
- there is at most 1 decay e^- in the event (`nmue <= 1`).

2.2.4. List of Monte Carlo templates

For each SuperK MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template granularity depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 12 MC templates were used for our 2010 Run 1+2 2-flavour disappearance analysis [2], while 15 MC templates were used for our 2010 Run 1+2 3-flavour disappearance analysis [3], 28 MC templates were used for our 2012 Run 1+2+3 3-flavour disappearance analysis [4] and 32 MC templates were used for our 2014 Run 1+2+3+4 3-flavour disappearance analysis [6].

In our previous ν_μ-disappearance analyses, NC events were divided into NC1$\pi^\pm$ and NC other. For this $\bar{\nu}_\mu$-disappearance analysis, the NC interaction modes have been further subdivided into NC1$\pi^\pm$, NC1π^0, NC coherent and NC other. We have also added 5 CC MEC templates for $\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$, ν_e and oscillated ν_e and 5 templates for the oscillated $\bar{\nu}_e$ MC sample. This gives a total of 46 templates:

- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC MEC,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC coherent,

- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- ν_μ CCQE,
- ν_μ CC MEC,
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1π^0,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC coherent,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC MEC,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- ν_e CCQE,
- ν_e CC MEC,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1$\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1π^0,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC coherent,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC MEC,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,

- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CCQE,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC MEC,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC1π,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC coherent,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC other.

2.2.5. Binning of Monte Carlo templates

Each of the 46 MC templates has 6132 2-dimensional bins (= 84 true energy bins × 73 reconstructed energy bins). The 84 true energy bins are arranged as follows:

- 6 50-MeV bins from 0-0.3 GeV,
- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV.

The 73 reconstructed energy bins are the following:

- 60 50-MeV bins from 0-3 GeV,
- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and
- 1 bin from 10-30 GeV.

The same reconstructed neutrino energy binning is used for the fit of the single μ-like ring reconstructed energy spectrum.

A study was performed to decide the optimum binning scheme balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4].

The binning described here is unchanged with respect to the Run 1+2+3+4 ν_μ-disappearance analysis,

2.3. Applying flux tuning based on NA61 data

In this analysis, the nominal MC templates, which were constructed using a MC event sample generated with the 13a flux, are reweighted to the 13av1.1 flux which was tuned on NA61 hadron production measurements [29][34]. The reweighted MC templates are denoted the *NA61-tuned* MC templates in this document. The applied weights were obtained from [29] and are shown as a function of the true neutrino energy in Figs. 1-4. The effect of this flux tune on the predicted reconstructed energy spectrum of single μ-like ring events is discussed in Sec. 2.6 (see Figs. 1).

	N_{SK} Nominal MC	N_{SK} Flux-tuned MC	N_{SK} BANFF prefit MC (flux + NIWG tune)	N_{SK} BANFF postfit MC
Total	55.658	57.949	50.551	59.778
ν_μ CCQE	8.188	8.377	8.237	8.904
ν_μ CC 1π	2.766	2.897	2.897	2.116
ν_μ CC coherent	0.116	0.122	0.122	0.141
ν_μ CC MEC	1.518	1.564	0.422	1.746
ν_μ CC other	0.692	0.753	0.753	0.803
ν_μ/ν_τ NC $1\pi^{+/-}$	0.286	0.302	0.302	0.196
ν_μ/ν_τ NC $1\pi^0$	0.047	0.050	0.050	0.029
ν_μ/ν_τ NC coherent	0.001	0.001	0.001	0.001
ν_μ/ν_τ NC other	0.250	0.271	0.271	0.406
$\bar{\nu}_\mu$ CCQE	30.698	32.206	29.332	33.434
$\bar{\nu}_\mu$ CC 1π	4.332	4.413	4.413	3.888
$\bar{\nu}_\mu$ CC coherent	1.391	1.451	1.451	1.777
$\bar{\nu}_\mu$ CC MEC	4.261	4.439	1.199	5.211
$\bar{\nu}_\mu$ CC other	0.528	0.514	0.514	0.580
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	0.330	0.335	0.335	0.220
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	0.047	0.047	0.047	0.028
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	0.003	0.004	0.004	0.004
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	0.153	0.150	0.150	0.237
ν_e CCQE	0.002	0.002	0.002	0.002
ν_e CC 1π	0.002	0.002	0.002	0.002
ν_e CC coherent	0.000	0.000	0.000	0.000
ν_e CC MEC	0.000	0.000	0.000	0.000
ν_e CC other	0.001	0.001	0.001	0.001
ν_e NC $1\pi^{+/-}$	0.012	0.012	0.012	0.008
ν_e NC $1\pi^0$	0.002	0.002	0.002	0.001
ν_e NC coherent	0.000	0.000	0.000	0.000
ν_e NC other	0.012	0.013	0.013	0.020
Osc. ν_e CCQE	0.000	0.000	0.000	0.000
Osc. ν_e CC 1π	0.000	0.000	0.000	0.000
Osc. ν_e CC coherent	0.000	0.000	0.000	0.000
Osc. ν_e CC MEC	0.000	0.000	0.000	0.000
Osc. ν_e CC other	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ CCQE	0.003	0.003	0.003	0.003
$\bar{\nu}_e$ CC 1π	0.001	0.001	0.001	0.001
$\bar{\nu}_e$ CC coherent	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ CC MEC	0.000	0.000	0.000	0.001
$\bar{\nu}_e$ CC other	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ NC $1\pi^{+/-}$	0.009	0.009	0.009	0.006
$\bar{\nu}_e$ NC $1\pi^0$	0.001	0.001	0.001	0.001
$\bar{\nu}_e$ NC coherent	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ NC other	0.007	0.006	0.006	0.010
Osc. $\bar{\nu}_e$ CCQE	0.000	0.000	0.000	0.000
Osc. $\bar{\nu}_e$ CC 1π	0.000	0.000	0.000	0.000
Osc. $\bar{\nu}_e$ CC coherent	0.000	0.000	0.000	0.000
Osc. $\bar{\nu}_e$ CC MEC	0.000	0.000	0.000	0.000
Osc. $\bar{\nu}_e$ CC other	0.000	0.000	0.000	0.000

Table 2: Calculated predicted numbers of 1-ring μ-like events without oscillations using the nominal MC templates, the NA61-tuned MC templates, the NIWG-tuned MC templates and the BANFF-tuned MC templates. The BANFF prefit column is the NA61 tuned data, with the CCQE samples reweighted according to the NIWG's external data fits [30]. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. These numbers were calculated for an exposure of 2.315×10^{20}POT.

	N_{SK} Nominal MC	N_{SK} Flux-tuned MC	N_{SK} BANFF prefit MC (flux + NIWG tune)	N_{SK} BANFF postfit MC
Total	19.139	19.744	17.712	19.943
ν_μ CCQE	3.618	3.767	3.722	3.967
ν_μ CC 1π	1.821	1.931	1.931	1.393
ν_μ CC coherent	0.079	0.084	0.084	0.096
ν_μ CC MEC	0.788	0.824	0.223	0.911
ν_μ CC other	0.605	0.662	0.662	0.704
ν_μ/ν_τ NC $1\pi^{+/-}$	0.286	0.302	0.302	0.196
ν_μ/ν_τ NC $1\pi^0$	0.047	0.050	0.050	0.029
ν_μ/ν_τ NC coherent	0.001	0.001	0.001	0.001
ν_μ/ν_τ NC other	0.250	0.271	0.271	0.406
$\bar{\nu}_\mu$ CCQE	7.054	7.255	6.720	7.655
$\bar{\nu}_\mu$ CC 1π	2.053	2.047	2.047	1.756
$\bar{\nu}_\mu$ CC coherent	0.326	0.331	0.331	0.401
$\bar{\nu}_\mu$ CC MEC	1.141	1.161	0.313	1.354
$\bar{\nu}_\mu$ CC other	0.465	0.451	0.451	0.508
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	0.330	0.335	0.335	0.220
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	0.047	0.047	0.047	0.028
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	0.003	0.004	0.004	0.004
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	0.153	0.150	0.150	0.237
ν_e CCQE	0.002	0.002	0.002	0.002
ν_e CC 1π	0.002	0.002	0.002	0.001
ν_e CC coherent	0.000	0.000	0.000	0.000
ν_e CC MEC	0.000	0.000	0.000	0.000
ν_e CC other	0.001	0.001	0.001	0.001
ν_e NC $1\pi^{+/-}$	0.012	0.012	0.012	0.008
ν_e NC $1\pi^0$	0.002	0.002	0.002	0.001
ν_e NC coherent	0.000	0.000	0.000	0.000
ν_e NC other	0.012	0.013	0.013	0.020
Osc. ν_e CCQE	0.004	0.004	0.004	0.004
Osc. ν_e CC 1π	0.002	0.002	0.002	0.002
Osc. ν_e CC coherent	0.000	0.000	0.000	0.000
Osc. ν_e CC MEC	0.001	0.001	0.000	0.001
Osc. ν_e CC other	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ CCQE	0.003	0.003	0.002	0.003
$\bar{\nu}_e$ CC 1π	0.001	0.001	0.001	0.001
$\bar{\nu}_e$ CC coherent	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ CC MEC	0.000	0.000	0.000	0.001
$\bar{\nu}_e$ CC other	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ NC $1\pi^{+/-}$	0.009	0.009	0.009	0.006
$\bar{\nu}_e$ NC $1\pi^0$	0.001	0.001	0.001	0.001
$\bar{\nu}_e$ NC coherent	0.000	0.000	0.000	0.000
$\bar{\nu}_e$ NC other	0.007	0.006	0.006	0.010
Osc. $\bar{\nu}_e$ CCQE	0.009	0.009	0.009	0.010
Osc. $\bar{\nu}_e$ CC 1π	0.002	0.002	0.002	0.002
Osc. $\bar{\nu}_e$ CC coherent	0.001	0.001	0.001	0.001
Osc. $\bar{\nu}_e$ CC MEC	0.001	0.001	0.000	0.002
Osc. $\bar{\nu}_e$ CC other	0.000	0.000	0.000	0.000

Table 3: Calculated predicted numbers of 1-ring μ-like events with oscillations using the nominal MC templates, the NA61-tuned MC templates, the NIWG-tuned MC templates and the BANFF-tuned MC templates. The BANFF prefit column is the NA61 tuned data, with the CCQE samples reweighted according to the NIWG's external data fits [30]. The total numbers of events and the numbers of events from each mode considered in this analysis are shown. The normal hierarchy was assumed, with oscillation parameters give the values listed in table 11. These numbers were calculated for an exposure of 2.315×10^{20}POT.

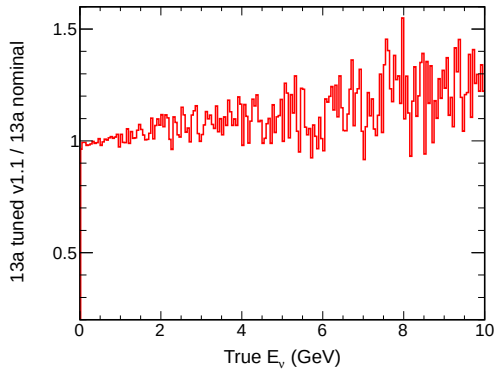


Figure 1: ν_μ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_μ and oscillated-ν_e MC templates [29].

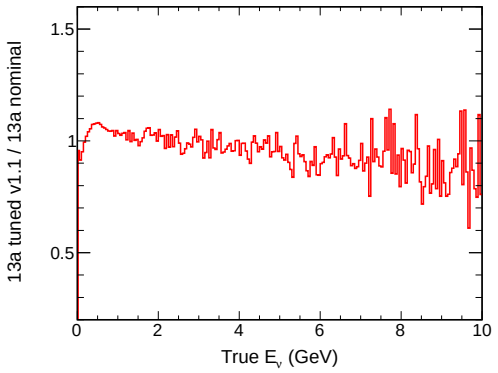


Figure 2: $\bar{\nu}_\mu$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [29].

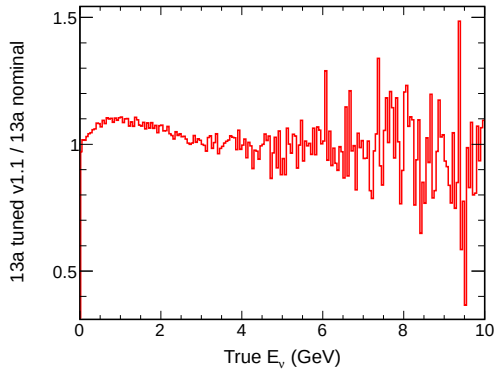


Figure 3: ν_e 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_e MC templates [29].

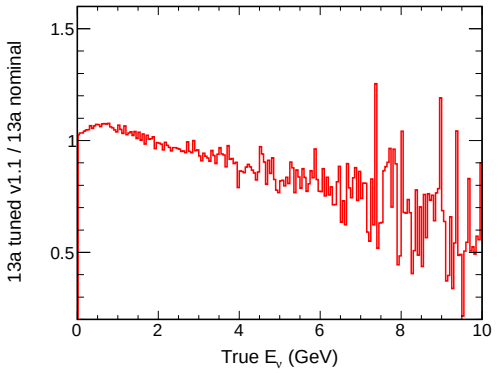


Figure 4: $\bar{\nu}_e$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [29].

2.4. Applying flux and cross-section (BANFF) tuning based on ND280 data

The BANFF group has made a fit of the $\bar{\nu}_\mu$ and ν_μ CC0π-like, CC1π-like and CC other ND280 samples to find the best-fit values of the flux normalisation and cross-section parameters[24]. This analysis uses these best-fit values to reweight the NA61-tuned MC templates, and the templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. A list of BANFF parameters, their best-fit values and their 1σ prefit and postfit errors is given in Tab. 4.

The effects of the BANFF tune on the predicted reconstructed energy spectrum of single μ-like ring events is discussed in Sec. 2.6 (see Figs. 10 and Tabs. 2 and 3.

2.5. Applying neutrino oscillations

The MC templates are constructed from the unoscillated MC samples and must be weighted with the appropriate survival or oscillation probability. Oscillations are applied as a function of true energy to the MC templates for all CC interactions.

In this $\bar{\nu}_\mu$-disappearance analysis, two separate sets of oscillation parameters are used when calculating these survival or oscillation probabilities. The MC templates calculated from the $\bar{\nu}_\mu$ MC sample are weighted with $\mathrm{P}(\bar{\nu}_\mu \to \bar{\nu}_\mu)$, with this survival probability being calculated using the first set of oscillation parameters, namely $\sin^2\overline{\theta}_{23}$, $\sin^2\overline{\theta}_{12}$, $\sin^2\overline{\theta}_{13}$, $|\overline{\Delta m^2_{32}}|$ (normal hierarchy) or $|\overline{\Delta m^2_{31}}|$ (inverted hierarchy), $|\overline{\Delta m^2_{21}}|$ and δ_{CP}. The $\bar{\nu}_e$ templates are weighted with $\mathrm{P}(\bar{\nu}_e \to \bar{\nu}_e)$, while the oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ templates are weighted with $\mathrm{P}(\bar{\nu}_\mu \to \bar{\nu}_e)$ (the sample was generated assuming 100% of $\bar{\nu}_\mu$ transform to $\bar{\nu}_e$). Again these survival and oscillation probabilities are calculated using the first set of oscillation paramters.

The ν_μ templates are weighted with $P(\nu_\mu \to \nu_\mu)$, and this survival probability is calculated using the second set of oscillation parameters, which are $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), $|\Delta m^2_{21}|$ and δ_{CP}. The ν_e templates are weighted with $P(\nu_e \to \nu_e)$ and the oscillated $\nu_\mu \to \nu_e$ templates with $P(\nu_\mu \to \nu_e)$, with these survival and oscillation probabilities being calculated using the second set of oscillation parameters.

All these survival and oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code were first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [53], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

In the standard 3-flavour oscillation framework, oscillations of ν_e and ν_μ can yield ν_τ, while oscillations of $\bar{\nu}_e$ and $\bar{\nu}_\mu$ can yield $\bar{\nu}_\tau$. In this analysis, we neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect is negligible[1]. Accordingly, this analysis uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ SuperK MC samples are available).

It should be emphasized here that the ν_μ NC MC templates for a mode m are proxies for the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m. The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

Also it should be noted that there are no explicit NC MC templates made from the oscillated $\bar{\nu}_e$ and ν_e samples. If they were used, the oscillated $\bar{\nu}_e$ (i.e. $\bar{\nu}_e$ coming from oscillations of $\bar{\nu}_\mu$) would be double counted since they are already included in the $\bar{\nu}_\mu$ NC MC templates. The same applies to oscillated ν_e.

2.6. Predictions of nominal and tuned-MC single μ-like ring spectra

In this section, we present the expected numbers of events and single μ-like ring reconstructed energy spectra in SuperK for various oscillation scenarios. We also present estimates of the effect of the NA61 and BANFF tunings and of assumptions made in the 3-flavour neutrino oscillation framework used in this analysis. Unless explicitly stated otherwise, the oscillation parameters are set to the values shown in Tab. 11 and the normal mass hierarchy is assumed. All plots were generated for an integrated exposure of 2.315×10^{20}POT.

Predicted single μ-like ring SuperK reconstructed energy spectra are shown in Fig. 6 for no oscillations, and in Fig. 8 for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 11.

A MC tune derived using NA61 data is applied to the nominal MC templates, and another tune derived using the ND280 data is applied to the NA61-tuned templates as described in Secs. 2.3 and 2.4. The fractional difference in the expected number of single μ-like ring events when switching from the nominal MC to the NA61-tuned MC templates is shown in Fig. 9 as a function of the oscillation parameters $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$. In Fig. 10 is shown the fractional difference in the expected number of single-ring μ-like events when switching from the NA61-tuned MC templates to the BANFF-tuned MC templates. The effects of the NA61 and BANFF tunes on the total expected number of single-ring μ-like events and on the numbers in each of the 46 component interaction modes are shown in Tab. 2 for no oscillations, and in Tab. 3 for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 11. Finally, the effects of the NA61 and BANFF tunes on the reconstructed energy spectrum of single μ-like ring events are shown in Figs. 10. In each case, spectra are shown for both no oscillations and for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 11. Also shown are the ratio of flux-tuned spectrum to nominal spectrum, and BANFF-tuned spectrum to flux-tuned spectrum in Figs. 10.

[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like SuperK sample. A SuperK MC study is required to determine whether the SuperK efficiency assumptions made there are valid.

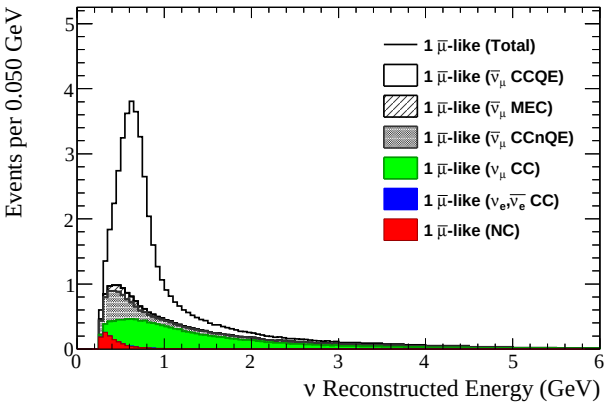


Figure 5: Predicted reconstructed-energy spectrum of 1 μ-like ring events, and contributions from various true neutrino reaction modes, for no oscillations and for an exposure of 2.315×10^{20}POT in antineutrino beam mode. The 32 components of the spectrum are calculated separately in the actual analysis, but, for this plot, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnonQE, $\bar{\nu}_\mu$ CC, $\nu_e/\bar{\nu}_e$ CC and NC. The spectrum was generated using the BANFF prefit MC templates.

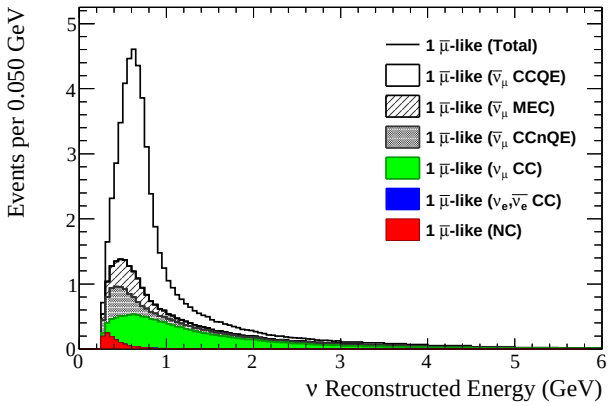


Figure 6: Predicted reconstructed-energy spectrum of 1 μ-like ring events, and contributions from various true neutrino reaction modes, for no oscillations and for an exposure of 2.315×10^{20}POT in antineutrino beam mode. The 32 components of the spectrum are calculated separately in the actual analysis, but, for this plot, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnonQE, $\bar{\nu}_\mu$ CC, $\nu_e/\bar{\nu}_e$ CC and NC. The spectrum was generated using the BANFF postfit-tuned MC templates.

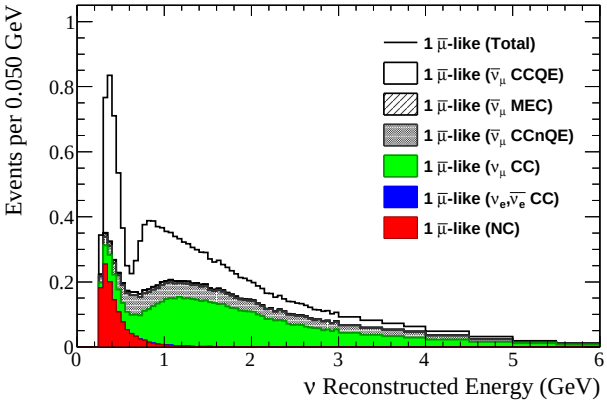


Figure 7: Predicted reconstructed-energy spectrum of 1 μ-like ring events, and contributions from various true neutrino reaction modes, with oscillations for an exposure of 2.315×10^{20}POT in antineutrino beam mode. The spectrum was generated using the BANFF prefit MC templates. The spectrum shown was generated assuming the normal hierarchy, and the oscillation parameters shown in table 11. Note that the vertical axis is zoomed in by a factor of 4 compared with Fig. 5.

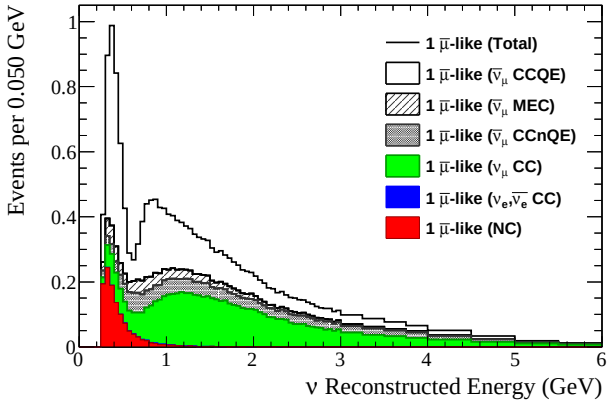


Figure 8: Predicted reconstructed-energy spectrum of 1 μ-like ring events, and contributions from various true neutrino reaction modes, with oscillations for an exposure of 2.315×10^{20}POT in antineutrino beam mode. The 32 components of the spectrum are calculated separately in the actual analysis, but, for this plot, are grouped into just 5 categories: ν_μ CCQE, ν_μ CCnonQE, $\bar{\nu}_\mu$ CC, $\nu_e/\bar{\nu}_e$ CC and NC. The spectrum was generated using the BANFF postfit-tuned MC templates. The spectrum shown was generated assuming the normal hierarchy, and the oscillation parameters shown in table 11. Note that the vertical axis is zoomed in by a factor of 4 compared with Fig. 6.

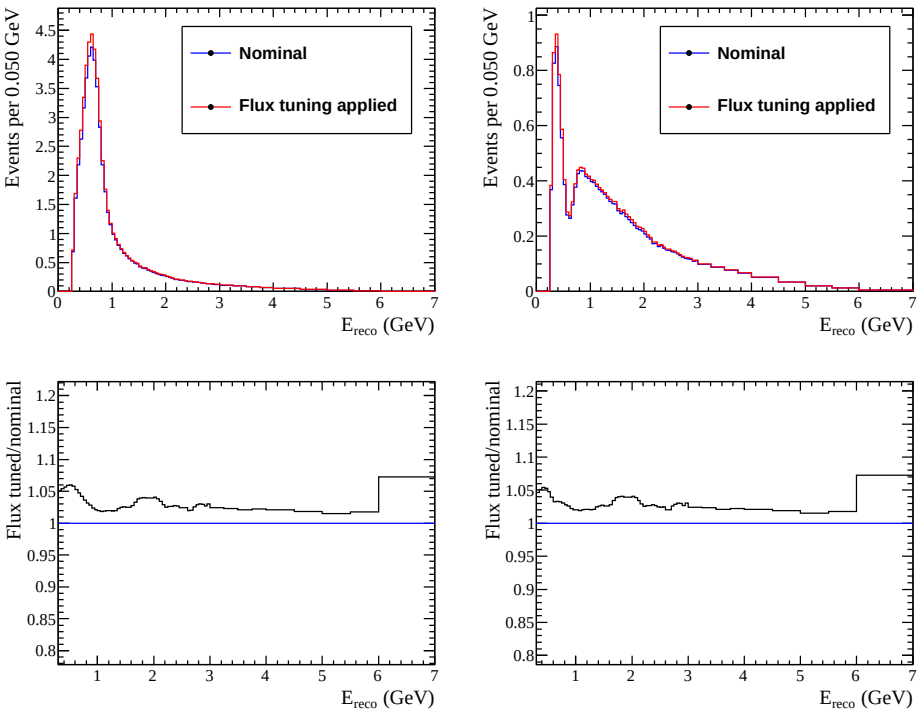


Figure 9: Top: Reconstructed-energy spectrum of 1 μ-like ring events, for an exposure of 2.315×10^{20}POT in antineutrino beam mode both with and without the effect of the NA61 flux tuning of the nominal MC templates. The spectra are shown both for no oscillations (left) and for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 11. Bottom: Ratio of flux-tuned spectrum to the nominal MC spectrum.

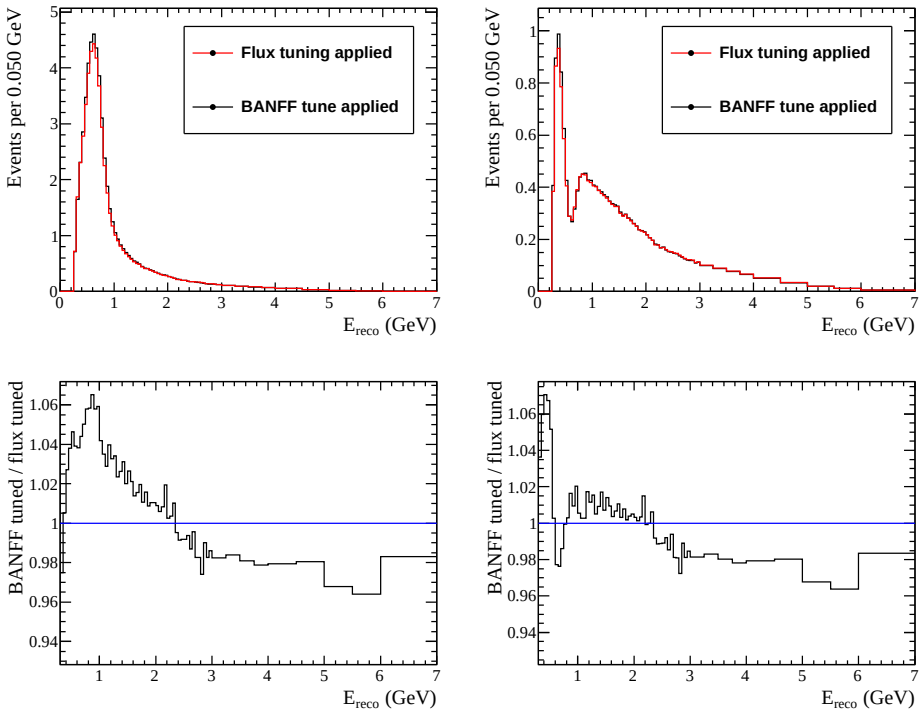


Figure 10: Top: Reconstructed-energy spectrum of 1 μ-like ring events, for an exposure of 2.315×10^{20}POT in antineutrino beam mode both with and without the effect of postfit BANFF flux and cross-section tuning on the NA61-tuned flux MC templates. The spectra are shown both for no oscillations (left) and for oscillations assuming the normal hierarchy, and the oscillation parameters shown in table 11. Bottom: Ratio of BANFF-tuned spectrum to the flux-tuned spectrum.

3. Systematic effects in the prediction of the single μ-like ring event energy spectrum

3.1. Input systematic parameters

This analysis considers 44 systematic parameters which can be grouped into 3 categories:

1. **Correlated flux and cross-section (BANFF) parameters** (37 parameters).

In this analysis, all the flux and cross-section systematic parameters are included in the ND280 data (BANFF) fit (in previous ν_μ-disappearance analyses, all the flux parameters but only a subset of the cross-section parameters were included). There are 5 ν_μ, 11 $\bar{\nu}_\mu$, 2 ν_e and 7 $\bar{\nu}_e$ flux parameters as well as 12 cross-section parameters. Best-fit values of these parameters and a covariance matrix giving their correlated uncertainties are obtained by the BANFF group from a fit of the $\bar{\nu}_\mu$ and ν_μ CC0π-like, CC1π-like and CC other-like ND280 samples as described in Ref [24]. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors is given in Tab. 4. The best-fit values are used to reweight the templates as described in Sec. 2.4, and the postfit errors in the covariance matrix are used as prior uncertainties in this analysis.

The effects of these correlated uncertainties are evaluated using the Cholesky method. The covariance matrix is "Cholesky decomposed" into the product of a lower triangular matrix (all elements above the diagonal are zero) and its transpose (in which all elements below the diagonal are zero). Random tweaks of each parameter are independently drawn from a Gaussian distribution with mean zero and standard deviation 1, and these random numbers are put into a vector. This vector is then multiplied by the lower triangular matrix to give a vector of tweaked uncertainties that have the correct correlations between them. The BANFF parameters are then varied by these tweaked uncertainties, with all parameters being varied simultaneously. This process may be repeated as many times as necessary.

Some of the cross-section parameters have a complicated, often non-linear, dependence on true and reconstructed neutrino energy; the effects of uncertainties in these parameters are evaluated by reweighting the MC templates. Weights are calculated for tweaks from -3σ to +3σ in steps of 0.5σ, and interpolations are made between those weights using cubic splines (for MaQE only, weights are calculated from -7σ to +3σ). Cubic splines are made for the following cross-section uncertainties:

- CCQE axial-mass scaling factor (MaQE)
- Fermi momentum for ^{16}O
- Binding energy for ^{16}O
- Nucleon to Δ transition axial form factor (C_5^A)
- Resonance-production axial-mass scaling factor (MaRes)
- Scale of isospin 1/2 nonresonant background
- CC other shape

The splines are made in bins of true and reconstructed energy, and the binning is given in Ref. [28].

The effects of the flux and cross-section normalisation uncertainties are also evaluated by reweighting the MC templates, but no splines are made for them since the weight is identical to the tweak for these uncertainties (each flux parameter is a normalisation for a limited range of true energy).

The BANFF parameters are applied to the MC templates as specified in Tab. 6.

Parameter	Description	Best fit	1σ pre/postfit fractional error
0	ν_μ flux normalisation, E = 0.0 - 0.7 GeV	1.142	0.094 / 0.071
1	ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.091	0.079 / 0.053
2	ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.079	0.077 / 0.052
3	ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.074	0.081 / 0.060
4	ν_μ flux normalisation, E = 2.5 - 30.0 GeV	1.050	0.080 / 0.053
5	$\bar{\nu}_\mu$ flux normalisation, E = 0.0 - 0.4 GeV	1.204	0.104 / 0.069
6	$\bar{\nu}_\mu$ flux normalisation, E = 0.4 - 0.5 GeV	1.210	0.102 / 0.058
7	$\bar{\nu}_\mu$ flux normalisation, E = 0.5 - 0.6 GeV	1.180	0.096 / 0.051
8	$\bar{\nu}_\mu$ flux normalisation, E = 0.6 - 0.7 GeV	1.128	0.085 / 0.041
9	$\bar{\nu}_\mu$ flux normalisation, E = 0.7 - 1.0 GeV	1.123	0.125 / 0.077
10	$\bar{\nu}_\mu$ flux normalisation, E = 1.0 - 1.5 GeV	1.118	0.105 / 0.068
11	$\bar{\nu}_\mu$ flux normalisation, E = 1.5 - 2.5 GeV	1.121	0.080 / 0.053
12	$\bar{\nu}_\mu$ flux normalisation, E = 2.5 - 3.5 GeV	1.121	0.074 / 0.049
13	$\bar{\nu}_\mu$ flux normalisation, E = 3.5 - 5.0 GeV	1.122	0.094 / 0.067
14	$\bar{\nu}_\mu$ flux normalisation, E = 5.0 - 7.0 GeV	1.112	0.093 / 0.064
15	$\bar{\nu}_\mu$ flux normalisation, E = 7.0 - 30.0 GeV	1.093	0.130 / 0.108
16	ν_e flux normalisation, E = 0.0 - 2.5 GeV	1.109	0.069 / 0.047
17	ν_e flux normalisation, E = 2.5 - 30.0 GeV	1.090	0.085 / 0.067
18	$\bar{\nu}_e$ flux normalisation, E = 0.0 - 0.5 GeV	1.193	0.095 / 0.054
19	$\bar{\nu}_e$ flux normalisation, E = 0.5 - 0.7 GeV	1.184	0.091 / 0.047
20	$\bar{\nu}_e$ flux normalisation, E = 0.7 - 0.8 GeV	1.167	0.091 / 0.049
21	$\bar{\nu}_e$ flux normalisation, E = 0.8 - 1.5 GeV	1.141	0.084 / 0.044
22	$\bar{\nu}_e$ flux normalisation, E = 1.5 - 2.5 GeV	1.120	0.080 / 0.054
23	$\bar{\nu}_e$ flux normalisation, E = 2.5 - 4.0 GeV	1.123	0.089 / 0.068
24	$\bar{\nu}_e$ flux normalisation, E = 4.0 - 30.0 GeV	1.180	0.156 / 0.144
25	Meson exchange current normalisation for ^{16}O	1.031	1.041 / 1.015
26	C_5^A nucleon to Δ transition axial form factor	0.854	0.119 / 0.073
27	Scale of isospin 1/2 nonresonant background	1.143	0.154 / 0.146
28	CCQE axial-mass scaling factor	0.948	0.058 / 0.028
29	Resonance-production axial-mass scaling factor	0.762	0.158 / 0.055
30	Fermi momentum for ^{16}O	0.997	0.055 / 0.054
31	CC other shape	0.023	0.400 / 0.193
32	Binding energy for ^{16}O	1.002	0.333 / 0.326
33	CC coherent for ^{16}O normalisation	1.076	1.000 / 0.972
34	NC coherent normalisation	0.980	0.300 / 0.299
35	NC other normalisation	1.413	0.300 / 0.186
36	CC ν_e normalisation	1.001	0.020 / 0.020

Table 4: Summary of flux and correlated cross-section (BANFF) systematics included in the *VALOR* Run 5+6 $\bar{\nu}_\mu$-disappearance analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [24].

Parameter	Definition	1σ error
1	ν_μ and $\bar{\nu}_\mu$ CCQE, $E_{reco} < 0.4$ GeV	0.018
2	ν_μ and $\bar{\nu}_\mu$ CCQE, 0.4 GeV $\leq E_{reco} \leq$ 1.1 GeV	0.016
3	ν_μ and $\bar{\nu}_\mu$ CCQE, $E_{reco} >$ 1.1 GeV	0.016
4	ν_μ and $\bar{\nu}_\mu$ CCnonQE	0.093
5	ν_e and $\bar{\nu}_e$ CC	1.006
6	All NC	0.598

Table 5: Binning of SuperK selection efficiency and FSI+SI systematic parameters in covariance matrices

2. **SuperK efficiencies & effects of intranuclear and secondary re-interactions** (6 parameters).

There are uncertainties in the efficiencies of the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparing 1412 days of SK-IV atmospheric data with atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using ν_μ CCQE, ν_μ CCnQE, NC, and ν_e CC-enriched control samples. The NC PID error is evaluated using a NC-enriched control sample, which does not use PID information. Each Monte Carlo control sample is fitted to the data using the efficiency as a fit parameter, and the uncertainties in the efficiencies are estimated by adding the covariance matrix containing the differences between the best-fit and nominal efficiencies to the covariance matrix containing the statistical errors in the fitted efficiencies [25].

Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [16]. The NIWG group produced 16 parameter sets that span the 1σ surface in the parameter space; this was done by varying the NEUT parameters when running the pion scattering simulations, and comparing the results with external pion-carbon scattering data. These 16 parameter sets are shown in Tab. 2 in [16]. Uncertainties due to secondary pion interactions (SI) were also evaluated. The NIWG group adapted the NEUT nuclear cascade model to simulate SI in the SuperK detector simulation (SKDETSIM) and to allow variations in the interaction probabilities. Details of this are given in [19]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. This was done by reweighting a single μ-like ring selected SK MC with the 16 FSI parameter sets using T2KReweight. The reweighting method for FSI is described in Sec. 8 in [15]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [16]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets.

We consider the SuperK detector efficiency and FSI+SI systematics together, and add their covariance matrices linearly. Each of these covariance matrices has six correlated parameters, corresponding to different modes and reconstructed energy bins, as shown in Tab. 5. The effects of these uncertainties are also evaluated using the Cholesky method.

3. **SuperK energy scale** (1 parameter).

The systematic parameter $f^{SK}_{E;r}$ is included to estimate the effects of uncertainty in the SuperK reconstructed energy scale. This uncertainty is estimated to be 2.4% [26]; its effects are calculated by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale

uncertainty is propagated to all MC templates as shown in Tab. 6.

	ν_μ CCQE	ν_μ CC 1π	ν_μ CC coherent	ν_μ CC MEC	ν_μ CC other	ν_μ/ν_τ NC $1\pi^{+/-}$	ν_μ/ν_τ NC $1\pi^0$	ν_μ/ν_τ NC coherent	ν_μ/ν_τ NC other	$\bar{\nu}_\mu$ CCQE	$\bar{\nu}_\mu$ CC 1π	$\bar{\nu}_\mu$ CC coherent	$\bar{\nu}_\mu$ CC MEC	$\bar{\nu}_\mu$ CC other	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	ν_e CCQE	ν_e CC 1π	ν_e CC coherent	ν_e CC MEC	ν_e CC other	ν_e NC $1\pi^{+/-}$	ν_e NC $1\pi^0$	ν_e NC coherent	ν_e NC other	Osc. ν_e CCQE	Osc. ν_e CC 1π	Osc. ν_e CC coherent	Osc. ν_e CC MEC	Osc. ν_e CC other	$\bar{\nu}_e$ CCQE	$\bar{\nu}_e$ CC 1π	$\bar{\nu}_e$ CC coherent	$\bar{\nu}_e$ CC MEC	$\bar{\nu}_e$ CC other	$\bar{\nu}_e$ NC $1\pi^{+/-}$	$\bar{\nu}_e$ NC $1\pi^0$	$\bar{\nu}_e$ NC coherent	$\bar{\nu}_e$ NC other	Osc. $\bar{\nu}_e$ CCQE	Osc. $\bar{\nu}_e$ CC 1π	Osc. $\bar{\nu}_e$ CC coherent	Osc. $\bar{\nu}_e$ CC MEC	Osc. $\bar{\nu}_e$ CC other
$f^{SK}_{E;r}$	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{0;t,r}RHC$	✓			✓							✓			✓																																		
$f^{SK+FSI}_{1;t,r}RHC$	✓			✓							✓			✓																																		
$f^{SK+FSI}_{2;t,r}RHC$	✓			✓							✓			✓																																		
$f^{SK+FSI}_{3;t,r}RHC$		✓	✓		✓							✓	✓		✓																																
$f^{SK+FSI}_{4;t,r}RHC$																			✓	✓	✓	✓	✓					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
$f^{SK+FSI}_{5;t,r}RHC$						✓	✓	✓	✓						✓	✓	✓	✓				✓	✓	✓	✓										✓	✓	✓	✓									
$f^{banff}_{0;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓					✓	✓	✓	✓	✓															
$f^{banff}_{1;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{2;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{3;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{4;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{5;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{6;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{7;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{8;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{9;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{10;t,r}RHC$	✓	✓	✓	✓	✓	✓	✓	✓	✓											✓	✓	✓	✓	✓																							
$f^{banff}_{11;t,r}RHC$										✓	✓	✓	✓	✓	✓	✓	✓	✓																								✓	✓	✓	✓	✓
$f^{banff}_{12;t,r}RHC$										✓	✓	✓	✓	✓	✓	✓	✓	✓																								✓	✓	✓	✓	✓
$f^{banff}_{13;t,r}RHC$										✓	✓	✓	✓	✓	✓	✓	✓	✓																								✓	✓	✓	✓	✓

Continued on next page

| | | |

3.2. Evaluation of effects of systematic parameters on the predicted single μ-like ring reconstructed energy spectrum

In this section, we present the effects of the systematics described in Sec. 3.1 on the predicted number of single μ-like ring events and the predicted reconstructed energy spectrum in SuperK. These effects are shown both without oscillations and with oscillations with the values shown in table 11. The normal mass hierarchy is assumed. All results are generated for an integrated exposure of $2.315\times10^{20}$and the BANFF-tuned MC templates are used.

The effects of systematic uncertainties on the predicted number of events are summarised in Tab. 8, both for no oscillations and for oscillations with the typical parameter values given above. In this table, systematic parameters are grouped into five categories and all parameter correlations are taken into account. Percentage uncertainties are given for both the BANFF prefit and postfit errors.

The effects of the combined systematic uncertainties on the reconstructed energy spectrum of single μ-like ring events is shown in Fig. 18 for the typical oscillation scenario used. This plot shows the total error envelope for the 1 μ-like ring event reconstructed energy spectrum; this is calculated as the $\pm1\sigma$ spread of bin contents from 100k toy MC experiments generated with randomized systematic parameters. All systematic parameters were considered and their correlations were taken into account.

The individual effect of each systematic parameter on the best-fit values of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ for the normal hierarchy was quantified by making a toy MC experiment with no statistical fluctuations and a $+1\sigma$ tweak of a single systematic parameter. A fit for only $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$, with systematics fixed are their nominal values, was made of that experiment, and the $+1\sigma$ effect of that systematic parameter on the best-fit values of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2}_{32}|$ was taken to be the difference between the best-fit values from the experiment with the $+1\sigma$ tweak and an experiment with nominal values of the systematics. Similarly, the -1σ effect of that parameter was the difference between the best-fit values from an experiment with a -1σ tweak and an experiment with nominal values of the systematics. This procedure was repeated for each systematic parameter in turn. For systematic parameters which are correlated, this procedure was repeated twice, both respecting and ignoring correlations. The study is repeated at multiple input points in the $\sin^2\overline{\theta}_{23}$-$|\overline{\Delta m^2_{32}}|$ plane.

An estimate of the order of the effect of systematics is given in Tabs. 9 and 10, ordered by the effect on $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ respectively. To make this estimate, the systematic effect is compared with the statistical error for the specific values of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$. The statistical error was evaluated using 10k toy MC datasets with no systematic fluctuations at each input point, with each dataset being fitted for only $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$. For each systematic, the largest effect seen in the study is given, whether that is from $+1\sigma$ or -1σ, including or neglecting correlations, and from any true values of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ in the study. The largest uncertainty is the SuperK detector efficiency/FSI parameter for NC events. The uncertainty in this parameter has a large effect on the oscillation parameters since the uncertainty itself is large (59.8%, please see Tab. 5) and NC events constitute a significant fraction of the number of events near the oscillation maximum (please see figure 8). The second-largest uncertainty is the MEC normalisation parameter; this has a large effect on the oscillation parameters since the uncertainty itself is large at 102% (please see table 4); the reason for the size of this uncertainty is that the rate of MEC events on oxygen is not constrained by the BANFF fit of ND280 data.

Parameter	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
All syst	19.943	26.706	33.910 %	14.442	-27.583 %
$f^{banff}_{0;t,r}RHC$	19.943	19.964	0.106 %	19.922	-0.106 %
$f^{banff}_{1;t,r}RHC$	19.943	19.970	0.133 %	19.916	-0.133 %
$f^{banff}_{2;t,r}RHC$	19.943	20.025	0.413 %	19.861	-0.413 %
$f^{banff}_{3;t,r}RHC$	19.943	20.084	0.708 %	19.802	-0.708 %
$f^{banff}_{4;t,r}RHC$	19.943	20.072	0.647 %	19.814	-0.647 %
$f^{banff}_{5;t,r}RHC$	19.943	20.026	0.417 %	19.860	-0.417 %
$f^{banff}_{6;t,r}RHC$	19.943	19.991	0.243 %	19.895	-0.243 %
$f^{banff}_{7;t,r}RHC$	19.943	19.953	0.052 %	19.933	-0.052 %
$f^{banff}_{8;t,r}RHC$	19.943	19.950	0.035 %	19.936	-0.035 %
$f^{banff}_{9;t,r}RHC$	19.943	20.074	0.658 %	19.812	-0.658 %
$f^{banff}_{10;t,r}RHC$	19.943	20.074	0.655 %	19.812	-0.655 %
$f^{banff}_{11;t,r}RHC$	19.943	20.059	0.580 %	19.827	-0.580 %
$f^{banff}_{12;t,r}RHC$	19.943	20.001	0.291 %	19.885	-0.291 %
$f^{banff}_{13;t,r}RHC$	19.943	20.008	0.327 %	19.878	-0.327 %
$f^{banff}_{14;t,r}RHC$	19.943	19.960	0.083 %	19.926	-0.082 %
$f^{banff}_{15;t,r}RHC$	19.943	19.946	0.018 %	19.940	-0.018 %
$f^{banff}_{16;t,r}RHC$	19.943	19.944	0.003 %	19.942	-0.004 %
$f^{banff}_{17;t,r}RHC$	19.943	19.944	0.005 %	19.942	-0.005 %
$f^{banff}_{18;t,r}RHC$	19.943	19.943	0.000 %	19.943	-0.000 %
$f^{banff}_{19;t,r}RHC$	19.943	19.943	0.000 %	19.943	-0.000 %
$f^{banff}_{20;t,r}RHC$	19.943	19.943	0.000 %	19.943	-0.000 %
$f^{banff}_{21;t,r}RHC$	19.943	19.943	0.001 %	19.943	-0.001 %
$f^{banff}_{22;t,r}RHC$	19.943	19.943	0.001 %	19.943	-0.001 %
$f^{banff}_{23;t,r}RHC$	19.943	19.943	0.002 %	19.943	-0.002 %
$f^{banff}_{24;t,r}RHC$	19.943	19.943	0.002 %	19.943	-0.002 %
$f^{banff}_{Norm_{MEC}}RHC$	19.943	22.176	11.196 %	17.710	-11.196 %
$f^{banff}_{C_5^A}RHC$	19.943	20.267	1.623 %	19.655	-1.446 %
$f^{banff}_{Bg_{RES}}RHC$	19.943	20.150	1.039 %	19.761	-0.914 %
$f^{banff}_{M_A^{QE}}RHC$	19.943	20.195	1.265 %	19.692	-1.258 %
$f^{banff}_{M_A^{RES}}RHC$	19.943	20.140	0.987 %	19.762	-0.907 %
$f^{banff}_{p_f}RHC$	19.943	19.761	-0.915 %	20.123	0.903 %
$f^{banff}_{Shape_{CCoth}}RHC$	19.943	20.016	0.368 %	19.870	-0.368 %
$f^{banff}_{E_B}RHC$	19.943	19.947	0.019 %	19.939	-0.018 %
$f^{banff}_{Norm_{CCcoh}}RHC$	19.943	20.394	2.259 %	19.492	-2.259 %
$f^{banff}_{Norm_{NCcoh}}RHC$	19.943	19.944	0.007 %	19.942	-0.007 %
$f^{banff}_{Norm_{NCother}}RHC$	19.943	20.031	0.444 %	19.855	-0.443 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}RHC$	19.943	19.944	0.003 %	19.942	-0.003 %
$f^{SK}_{E;r}RHC$	19.943	19.943	0.000 %	19.943	-0.000 %
$f^{SK+FSI}_{0;t,r}RHC$	19.943	19.970	0.137 %	19.916	-0.137 %
$f^{SK+FSI}_{1;t,r}RHC$	19.943	20.016	0.364 %	19.870	-0.364 %
$f^{SK+FSI}_{2;t,r}RHC$	19.943	20.069	0.631 %	19.817	-0.631 %
$f^{SK+FSI}_{3;t,r}RHC$	19.943	20.394	2.260 %	19.492	-2.260 %
$f^{SK+FSI}_{4;t,r}RHC$	19.943	19.974	0.156 %	19.912	-0.156 %
$f^{SK+FSI}_{5;t,r}RHC$	19.943	20.640	3.496 %	19.246	-3.496 %

Table 7: $\pm1\sigma$ effect of the systematic parameters on the total number of events, using 2.315×10^{20}POT, and the oscillation parameters in 11.

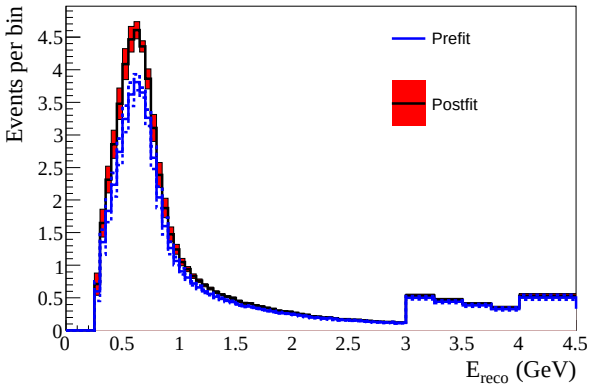


Figure 11: Error envelopes for the energy spectrum as a result of SuperK detector and FSI/SI parameters. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillations were applied.

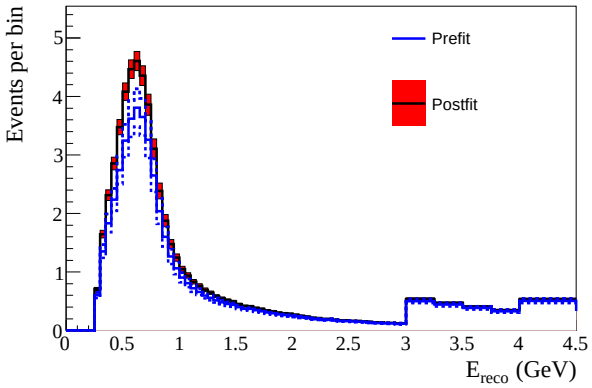


Figure 12: Error envelopes for the energy spectrum as a result of BANFF flux systematics. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillations were applied.

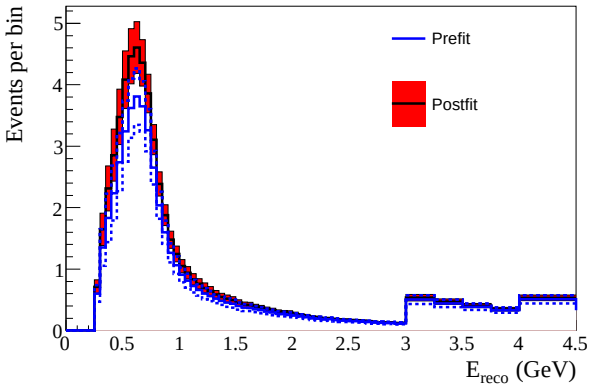


Figure 13: Error envelopes for the energy spectrum as a result of only cross-section BANFF parameters. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillations were applied.

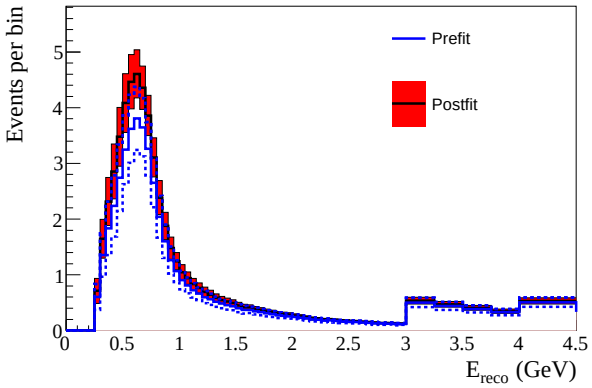


Figure 14: Error envelopes for the energy spectrum as a result of all systematic parameters. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. No oscillations were applied.

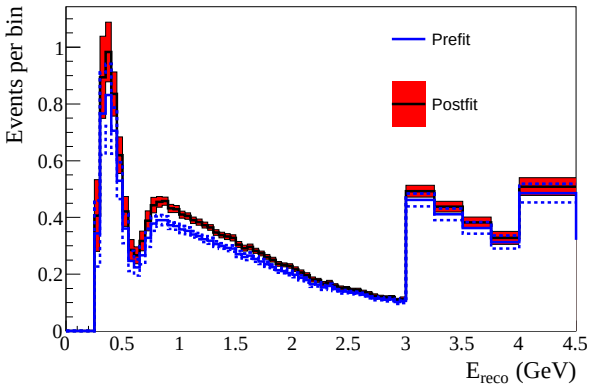


Figure 15: Error envelopes for the energy spectrum as a result of SuperK detector and FSI/SI parameters. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 11.

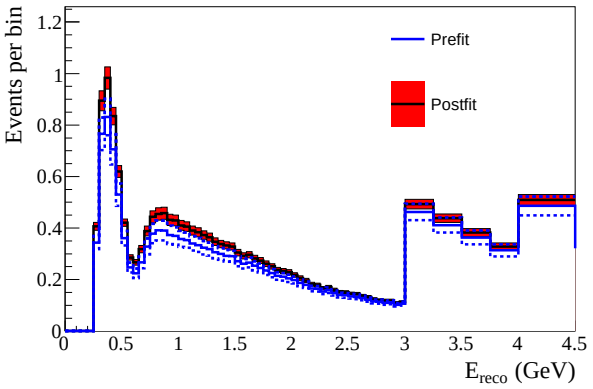


Figure 16: Error envelopes for the energy spectrum as a result of BANFF flux systematics. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 11.

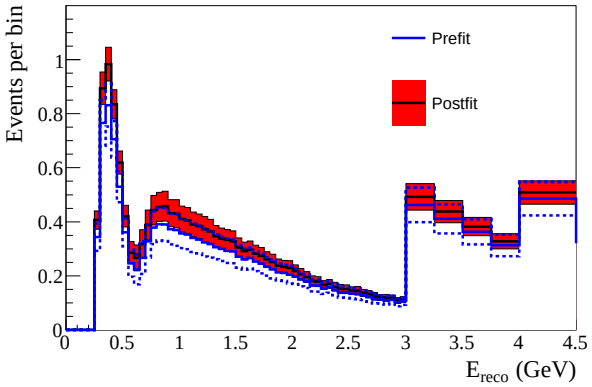


Figure 17: Error envelopes for the energy spectrum as a result of only cross-section BANFF parameters. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 11.

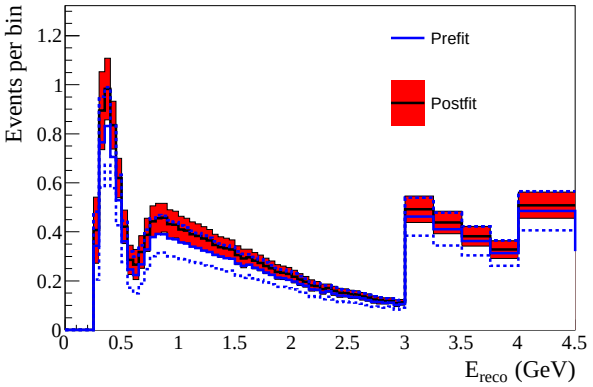


Figure 18: Error envelopes for the energy spectrum as a result of all systematic parameters. 100k toy MC experiments were generated with randomized systematic parameters and the 1σ spread of bin contents was calculated. BANFF tuned MC templates were used. All systematic parameters were considered and their correlations were taken into account. Oscillations were applied with the parameter values shown in table 11.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (no oscillations)	$\delta N_{SK}/N_{SK}$ (oscillations with typical parameter values)
SuperK detector + FSI + SI	2.34%	4.11%
BANFF (prefit)	13.19%	13.00%
BANFF (postfit)	9.53%	9.26%
BANFF (Non-oxygen, postfit)	2.95%	2.96%
BANFF (Oxygen-specific postfit)	9.25%	8.89%
Total (BANFF prefit)	13.40%	13.63%
Total (BANFF postfit)	9.81%	10.13%

Table 8: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events in the reverse-horn current sample. The BANFF parameters considered to be oxygen-specific are Oxygen MEC normalisation, Fermi Energy, Binding Energy, CC Coherent Normalisation and the nue/numu normalisation.

Parameter	$\frac{\delta_{syst} sin^2\overline{\theta_{23}}}{\delta_{stat} sin^2\overline{\theta_{23}}}$	$\frac{\delta_{syst}\lvert\overline{\Delta m^2_{32}}\rvert(eV^2/c^4)}{\delta_{stat}\lvert\overline{\Delta m^2_{32}}\rvert(eV^2/c^4)}$
SKDet + FSI/SI 5; E_r = 0.00 - 30.00 GeV; all NC; RHC	1.040	0.225
Meson exchange current normalisation for ^{16}O	0.918	0.127
CC coherent for ^{16}O normalisation	0.491	0.028
C_5^A nucleon to Δ transition axial form factor	0.422	0.024
SK energy scale; RHC	0.383	0.101
NC other normalisation	0.347	0.029
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	0.326	0.048
SKDet + FSI/SI 3; E_r = 0.00 - 30.00 GeV; $\overset{(-)}{\nu_\mu}$ CCnQE; RHC	0.315	0.022
Scale of isospin 1/2 nonresonant background	0.315	0.012
Resonance-production axial-mass scaling factor	0.247	0.011
Fermi momentum for ^{16}O	0.230	0.024
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.4 GeV	0.228	0.021
SKDet + FSI/SI 1; E_r = 0.40 - 1.10 GeV; $\overset{(-)}{\nu_\mu}$ CCQE; RHC	0.225	0.023
SKDet + FSI/SI 4; E_r = 0.00 - 30.00 GeV; $\overset{(-)}{\nu_e}$ CC; RHC	0.211	0.009
RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	0.202	0.016
CCQE axial-mass scaling factor	0.200	0.021
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.4 - 0.5 GeV	0.190	0.030
RHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	0.150	0.009
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.6 - 0.7 GeV	0.142	0.009
SKDet + FSI/SI 0; E_r = 0.00 - 0.40 GeV; $\overset{(-)}{\nu_\mu}$ CCQE; RHC	0.116	0.006
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.5 - 0.6 GeV	0.115	0.025
RHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	0.113	0.006
RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	0.108	0.005
RHC ν_μ flux normalisation, E = 0.0 - 0.7 GeV	0.100	0.009
CC other shape	0.096	0.003
RHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	0.087	0.006
RHC ν_μ flux normalisation, E = 2.5 - 30.0 GeV	0.085	0.004
SKDet + FSI/SI 2; E_r = 1.10 - 30.00 GeV; $\overset{(-)}{\nu_\mu}$ CCQE; RHC	0.070	0.005
Binding energy for ^{16}O	0.063	0.007
RHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 3.5 GeV	0.051	0.001
RHC $\bar{\nu_\mu}$ flux normalisation, E = 3.5 - 5.0 GeV	0.021	0.002
NC coherent normalisation	0.019	0.000
RHC $\bar{\nu_\mu}$ flux normalisation, E = 7.0 - 30.0 GeV	0.009	0.000
CC ν_e normalisation	0.008	0.000
RHC $\bar{\nu_\mu}$ flux normalisation, E = 5.0 - 7.0 GeV	0.007	0.000
RHC ν_e flux normalisation, E = 2.5 - 30.0 GeV	0.006	0.000
RHC ν_e flux normalisation, E = 0.0 - 2.5 GeV	0.004	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 4.0 - 30.0 GeV	0.003	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 4.0 GeV	0.002	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 1.5 - 2.5 GeV	0.001	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 0.8 - 1.5 GeV	0.000	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 0.5 - 0.7 GeV	0.000	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 0.5 GeV	0.000	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 0.7 - 0.8 GeV	0.000	0.000

Table 9: Maximum deviations, as a fraction of statistical error at the oscillation parameter grid point, from the input values of fits for only $\sin^2\overline{\theta_{23}}$ and $|\overline{\Delta m^2_{32}}|$ to datasets with ±1 sigma individual systematic variations at a range of 23-sector oscillation points. $|\overline{\Delta m^2_{32}}|$ values of 2.0×10^{-3} eV^2/c^4, 2.5×10^{-3} eV^2/c^4, and 3.0×10^{-3} eV^2/c^4 are used with each of $\sin^2\overline{\theta_{23}}$ = 0.370, 0.500, 0.524, 0.650. All other oscillation parameters are fixed at the values in table 11. These maximum deviations usually occur at maximum disappearance, where the statistical error is smaller, as explained in Sec. 5. The values are sorted by the size of the deviation in $\sin^2\overline{\theta_{23}}$.

Parameter	$\frac{\delta_{syst} sin^2\overline{\theta_{23}}}{\delta_{stat} sin^2\overline{\theta_{23}}}$	$\frac{\delta_{syst}\overline{\lvert\Delta m^2_{32}\rvert}(eV^2/c^4)}{\delta_{stat}\overline{\lvert\Delta m^2_{32}\rvert}(eV^2/c^4)}$
SKDet + FSI/SI 5; E_r = 0.00 - 30.00 GeV; all NC; RHC	1.040	0.225
Meson exchange current normalisation for ^{16}O	0.918	0.127
SK energy scale; RHC	0.383	0.101
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	0.326	0.048
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.4 - 0.5 GeV	0.190	0.030
NC other normalisation	0.347	0.029
CC coherent for ^{16}O normalisation	0.491	0.028
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.5 - 0.6 GeV	0.115	0.025
C_5^A nucleon to Δ transition axial form factor	0.422	0.024
Fermi momentum for ^{16}O	0.230	0.024
SKDet + FSI/SI 1; E_r = 0.40 - 1.10 GeV; $\overset{(-)}{\nu_\mu}$ CCQE; RHC	0.225	0.023
SKDet + FSI/SI 3; E_r = 0.00 - 30.00 GeV; $\overset{(-)}{\nu_\mu}$ CCnQE; RHC	0.315	0.022
CCQE axial-mass scaling factor	0.200	0.021
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.4 GeV	0.228	0.021
RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	0.202	0.016
Scale of isospin 1/2 nonresonant background	0.315	0.012
Resonance-production axial-mass scaling factor	0.247	0.011
RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.6 - 0.7 GeV	0.142	0.009
SKDet + FSI/SI 4; E_r = 0.00 - 30.00 GeV; $\overset{(-)}{\nu_e}$ CC; RHC	0.211	0.009
RHC ν_μ flux normalisation, E = 0.0 - 0.7 GeV	0.100	0.009
RHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	0.150	0.009
Binding energy for ^{16}O	0.063	0.007
RHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	0.087	0.006
SKDet + FSI/SI 0; E_r = 0.00 - 0.40 GeV; $\overset{(-)}{\nu_\mu}$ CCQE; RHC	0.116	0.006
RHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	0.113	0.006
RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	0.108	0.005
SKDet + FSI/SI 2; E_r = 1.10 - 30.00 GeV; $\overset{(-)}{\nu_\mu}$ CCQE; RHC	0.070	0.005
RHC ν_μ flux normalisation, E = 2.5 - 30.0 GeV	0.085	0.004
CC other shape	0.096	0.003
RHC $\bar{\nu_\mu}$ flux normalisation, E = 3.5 - 5.0 GeV	0.021	0.002
RHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 3.5 GeV	0.051	0.001
RHC $\bar{\nu_\mu}$ flux normalisation, E = 5.0 - 7.0 GeV	0.007	0.000
NC coherent normalisation	0.019	0.000
RHC $\bar{\nu_\mu}$ flux normalisation, E = 7.0 - 30.0 GeV	0.009	0.000
CC ν_e normalisation	0.008	0.000
RHC ν_e flux normalisation, E = 2.5 - 30.0 GeV	0.006	0.000
RHC ν_e flux normalisation, E = 0.0 - 2.5 GeV	0.004	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 4.0 GeV	0.002	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 4.0 - 30.0 GeV	0.003	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 1.5 - 2.5 GeV	0.001	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 0.5 GeV	0.000	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 0.5 - 0.7 GeV	0.000	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 0.7 - 0.8 GeV	0.000	0.000
RHC $\bar{\nu_e}$ flux normalisation, E = 0.8 - 1.5 GeV	0.000	0.000

Table 10: Maximum deviations, as a fraction of statistical error at the oscillation parameter grid point, from the input values of fits for only $\sin^2\overline{\theta_{23}}$ and $\overline{|\Delta m^2_{32}|}$ to datasets with ±1 sigma individual systematic variations at a range of 23-sector oscillation points. $\overline{|\Delta m^2_{32}|}$ values of 2.0×10^{-3} eV^2/c^4, 2.5×10^{-3} eV^2/c^4, and 3.0×10^{-3} eV^2/c^4 are used with each of $\sin^2\overline{\theta_{23}}$ = 0.370, 0.500, 0.524, 0.650. All other oscillation parameters are fixed at the values in table 11. These maximum deviations usually occur at maximum disappearance, where the statistical error is smaller, as explained in Sec. 5. The values are sorted by the size of the deviation in $\overline{|\Delta m^2_{32}|}$.

Parameter(s)	Fixed value in fits (NH)	Fixed value in fits (IH)
$\sin^2\theta_{23}$	0.527	0.533
$\sin^2\overline{\theta}_{13}$ and $\sin^2\theta_{13}$	0.0248	0.0252
$\sin^2\overline{\theta}_{12}$ and $\sin^2\theta_{12}$	0.304	0.304
$\lvert\Delta m^2_{32}\rvert$ (NH only)	2.51×10^{-3} eV^2/c^4	-
$\lvert\Delta m^2_{31}\rvert$ (IH only)	-	2.48×10^{-3} eV^2/c^4
$\overline{\Delta m^2_{21}}$ and Δm^2_{21}	7.53×10^{-5} eV^2/c^4	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.55	-1.56

Table 11: Values of oscillation parameters that are fixed in fits (NH = normal mass hierarchy, IH = inverted mass hierarchy)[11]. These are the values with the reactor constraint, and are used in the real data fits.

Parameter(s)	Fixed value in fits (NH)	Fixed value in fits (IH)
$\sin^2\theta_{23}$	0.524	0.534
$\sin^2\overline{\theta}_{13}$ and $\sin^2\theta_{13}$	0.0422	0.0491
$\sin^2\overline{\theta}_{12}$ and $\sin^2\theta_{12}$	0.304	0.304
$\lvert\Delta m^2_{32}\rvert$ (NH only)	2.51×10^{-3} eV^2/c^4	-
$\lvert\Delta m^2_{31}\rvert$ (IH only)	-	2.49×10^{-3} eV^2/c^4
$\overline{\Delta m^2_{21}}$ and Δm^2_{21}	7.53×10^{-5} eV^2/c^4	7.53×10^{-5} eV^2/c^4
δ_{CP}	1.91	1.01

Table 12: Values of oscillation parameters that are fixed in fits (NH = normal mass hierarchy, IH = inverted mass hierarchy)[11]. These are the values without the reactor constraint, and used are for certain validation plots. This is for historical reasons in the analysis - the fake-data fits in appendix B were fitted before the oscillation group agreed to use the values in table 11.

4. Neutrino oscillation fitting

4.1. Fit method

One of the motivations for this analysis is to determine whether the oscillation parameters for $\bar{\nu}_\mu$ disappearance are different from those for ν_μ disappearance. For this reason, two separate sets of oscillation parameters are used. The first set consists of $\sin^2\overline{\theta}_{23}$, $\sin^2\overline{\theta}_{12}$, $\sin^2\overline{\theta}_{13}$, $|\overline{\Delta m^2_{32}}|$ (normal hierarchy) or $|\overline{\Delta m^2_{31}}|$ (inverted hierarchy), $\overline{\Delta m^2_{21}}$ and δ_{CP}, and these parameters are applied to the $\bar{\nu}_\mu$, $\bar{\nu}_e$ and oscillated $\bar{\nu}_e$ interaction modes. The other set of oscillation parameters is $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), Δm^2_{21} and δ_{CP}, and they are applied to the ν_μ, ν_e and oscillated ν_e interaction modes. All these parameters are fixed in the fits of the Super K data except $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ (normal hierarchy) or $|\overline{\Delta m^2_{31}}|$ (inverted hierarchy); these fixed values are the T2K+reactor best-fit values and are shown in Tab. 11.

The *VALOR* disappearance analysis uses a binned likelihood-ratio method. Measurements of the oscillation parameters $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}|$ are obtained by comparing the observed and predicted SuperK reconstructed energy spectra for 1-ring μ-like events.

Let N be the number of reconstructed energy bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the SuperK reconstructed energy spectrum prediction (e.g parameters that control the beam, neutrino interaction simulations and the detector response). Best-fit values are obtained by minimizing:

$$-2\ln\lambda(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp} - n_i^{obs})\right) + (\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0}) \quad (3)$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of the systematics parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, n_i^{obs} is the observed

number of events in the ith bin, and $n_i^{exp} = n_i^{exp}(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a})$ is the corresponding expected number of events.

The advantage of the likelihood ratio method over the extended maximum-likelihood method is that, in the large-sample limit, the quantity $-2\ln\lambda(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a})$ in Eq.3 has a χ^2 distribution and it can therefore be used as a goodness-of-fit test. In this note, $-2\ln\lambda(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a})$ and $\chi^2(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a})$ are used interchangeably.

The minimization is performed with Minuit [40], using the Migrad method. If Migrad does not converge, the starting values of the oscillation parameters are re-initialised: the new initial values are randomly drawn from uniform distributions in the ranges 1.5×10^{-3} eV^2/c^4 $\leq |\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}| \leq 2.0\times10^{-2}$ eV^2/c^4 and either $0.35 \leq \sin^2\overline{\theta}_{23} \leq 0.50$ or $0.53 \leq \sin^2\overline{\theta}_{23} \leq 0.68$ (the new initial value of $\sin^2\overline{\theta}_{23}$ is in the same octant as the previous initial value). The fit is retried with the new initial values, and this is repeated up to 10 times. If the fit still fails to converge after 10 attempts, the Simplex algorithm is used. For the data and Asimov dataset analysis described in this document, this fallback was never triggered. The search for the minimum value of $-2\ln\lambda(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a})$ (see Eq. 3) is performed in the range $0 \leq \sin^2\overline{\theta}_{23} \leq 1$ and 1×10^{-3} eV^2/c^4 $\leq |\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}| \leq 2\times10^{-2}$ eV^2/c^4. After a successful fit, The Hesse algorithm is called to improve Minuit's estimates of the errors on the fitted parameters.

Each fit is repeated twice: in the first fit, the initial values are $|\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}| = 2.4\times10^{-3}$ eV^2/c^4 and $\sin^2\overline{\theta}_{23} = 0.6$, while the second fit uses as initial values the best fit of $|\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}|$ and the mirror point in the other octant of the best fit of $\sin^2\overline{\theta}_{23}$ from the first fit.

All the systematic parameters are allowed to float in the fit, and are restricted to the range [-3σ_s, +3σ_s] where σ_s is the one standard deviation error assigned to each systematic parameter.

Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin in any interaction mode are not allowed. If a systematic throw gives a negative number of events, that toy experiment is rejected.

4.2. Goodness-of-fit tests

In this analysis, the goodness-of-fit tests are done using the same coarse binning as in our ν_μ-disappearance analyses [4], [5] and [6]. The choice of this binning followed a decision by the oscillation analysis group that p-values should be calculated using the rate and the reconstructed energy distribution, and that the distribution will use a very coarse binning scheme to ensure that there are sufficient numbers of events in each bin: this scheme has 5 bins from 0-0.4, 0.4-0.7, 0.7-1.0, 1.0-2.0 and 2.0-30.0 GeV. The Asimov spectrum using this binning can be seen in figure 19. These goodness-of-fit tests are done in two separate ways: in the first method, both systematic variations and statistical fluctuations are applied to the toy datasets, and all 44 systematic parameters are included in the fits of those datasets, while in the second method only statistical fluctuations are applied to the toy datasets and the systematic parameters are not included in the fits.

4.3. Construction of confidence regions

In this analysis, uncertainties in the jointly measured parameters are quoted using the constant-$\Delta\chi^2$ [36] (pp. 390-401).

One finds the best-fit point and calculates the confidence contours as lines of *constant* $\Delta\chi^2$ from this point[2].

The procedure for constructing the $\Delta\chi^2(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|)$ surface itself from the T2K Run 5+6 data is as follows: assuming that N nuisance parameters are allowed to float in the oscillation fit, the values of $\chi^2(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a})$ are calculated via a fit at *each* point of the ($\sin^2\overline{\theta}_{23}$, $|\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}|$) 2-dimensional grid, where only the nuisance parameters are allowed to float and $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ or $|\overline{\Delta m^2_{31}}|$ are fixed to the true values for the given grid point. The following grid spacing is used : $\delta\sin^2\overline{\theta}_{23}$ = 5×10^{-3} and $\delta|\overline{\Delta m^2_{32}}|$ or $\delta|\overline{\Delta m^2_{31}}|$ = 5×10^{-5} eV^2/c^4.

[2] For 1 fit parameter, the critical values of $\Delta\chi^2$ used are 1.00 (68% CL), 2.71 (90% CL), 3.84 (95% CL), 6.63 (99% CL). For 2 parameters fitted simultaneously, the critical values of $\Delta\chi^2$ used are 2.30 (68% CL), 4.61 (90% CL), 5.99 (95% CL), 9.21 (99% CL).

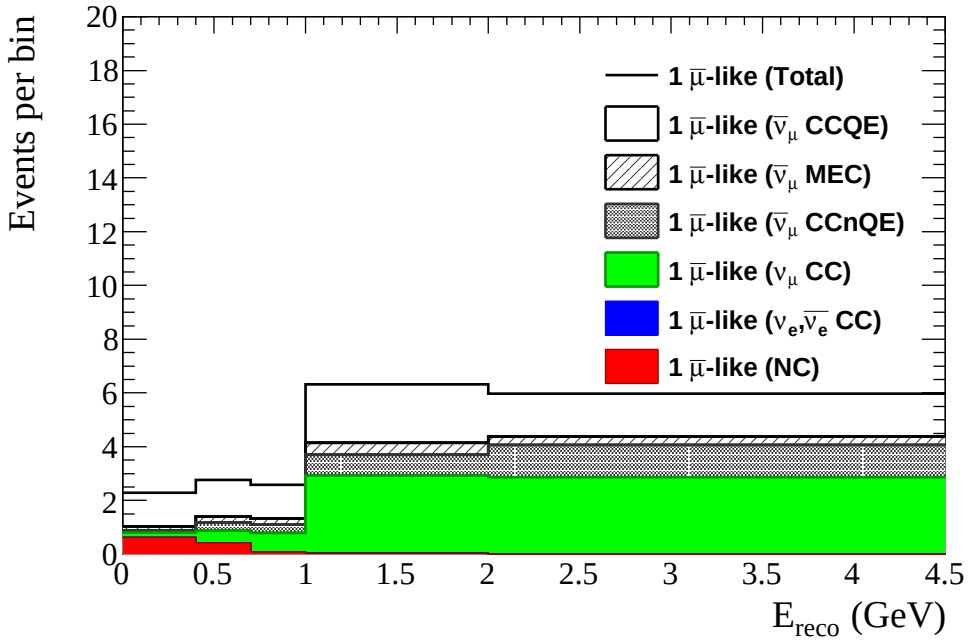


Figure 19: Predicted reconstructed-energy spectrum of 1 μ-like ring events, for the postfit Asimov dataset, using the Goodness-of-Fit binning, for an exposure of 2.315×10^{20}POT in antineutrino beam mode. Event counts are shown as total per bin, without accounting for bin width. Oscillations are applied at the values from table 11. Note that the highest bin here, from 2.0 GeV onwards contains all events at with energies greater than 2.0 GeV.

The allowed region in the 2D oscillation parameter space is then given by

$$\Delta\chi^2(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|) = \chi^2(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a}) - \chi^2_{min} < \Delta\chi^2{}_{critical} \quad (4)$$

where χ^2_{min} is the minimum value of $\chi^2(\sin^2\overline{\theta}_{23}, |\overline{\Delta m^2_{32}}| \text{ or } |\overline{\Delta m^2_{31}}|; \mathbf{a})$ from the fit of the T2K Run 5+6 data over the oscillation parameter space considered.

5. Validation of the *VALOR* Run 5+6 $\bar{\nu}_\mu$-disappearance analysis

The performance of the fitter was evaluated using ensembles of 3k or 5k toy MC experiments, and these were generated with statistical fluctuations and variations of all 44 systematic parameters. Toy ensembles were generated for both the normal and the inverted mass hierarchies, and were fitted using MINUIT and the MINUIT HESSE errors. Distributions of pulls of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ are shown in figures 20 for both 2.500×10^{20} POT and full (7.800×10^{21}) POT, with true input values of $\sin^2\overline{\theta}_{23} = 0.527$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4. The distributions for $|\overline{\Delta m^2_{32}}|$ are as expected, with a mean close to 0 and a RMS that is very near 1.0 for full POT. The tails in the distributions for $\sin^2\overline{\theta}_{23}$ are caused by toys with upward statistical fluctuations in the bins near the oscillation maximum, which are fitted with a value of $\sin^2\overline{\theta}_{23}$ that is less than maximal; these best-fit values can be in either octant since $P(\nu_\mu \to \nu_\mu)$ is almost identical between a value of $\sin^2\overline{\theta}_{23}$ and its mirror value in the other octant. The spike near zero is due to toys which have either small or downward statistical fluctuations in the bins near the oscillation maximum; the spike is smaller at full POT since there are fewer toys with downward statistical fluctuations.

The pulls of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ are also shown in figures 21 and 22 from 3k toy MC experiments with input values $\sin^2\overline{\theta}_{23} = 0.35$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4. In these figures, the pulls for $|\overline{\Delta m^2_{32}}|$ are as expected, with a mean near 0 and a width that approaches 1.0 at full POT. The pulls for $\sin^2\overline{\theta}_{23}$ appear to have a positive bias at 3.0×10^{20} POT, and this is caused by toys with downward statistical fluctuations in bins near the oscillation maximum; these toys are fitted with maximal $\sin^2\overline{\theta}_{23}$, and this is seen more clearly in the distribution of residuals (figure 23). The bias disappears at full POT since there are less toys with downward statistical fluctuations at higher statistics. Similar distributions are seen in the pulls of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ from 3k toys with input values $\sin^2\overline{\theta}_{23} = 0.65$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4 in figures 24 and 25; here there appears to be negative bias in the pulls of $\sin^2\overline{\theta}_{23}$ at 3.0×10^{20} POT, which is again due to downward statistical fluctuations in bins near the oscillation maximum and disappears at full POT.

The pull distributions of these two oscillation parameters are also shown from 3k toys experiments with input values $\sin^2\overline{\theta}_{23} = 0.527$ and $|\overline{\Delta m^2_{32}}| = 2.2\times10^{-3}$ eV^2/c^4 in figures 27 and 28 and from 3k toys with input values $\sin^2\overline{\theta}_{23} = 0.527$ and $|\overline{\Delta m^2_{32}}| = 2.8\times10^{-3}$ eV^2/c^4 in figures 29 and 30. These pull distributions are as expected except that there are small biases of +0.25 and -0.13 in the pull means when the input value of $|\overline{\Delta m^2_{32}}|$ is 2.2×10^{-3} eV^2/c^4 and 2.8×10^{-3} eV^2/c^4 respectively for 2.315×10^{20} POT; these biases are reduced to +0.23 and -0.08 respectively for 7.800×10^{21} POT.

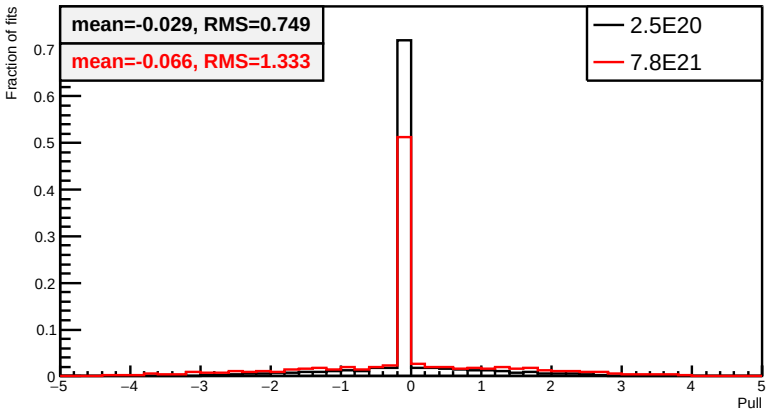
A check was also made of the pull distributions of the systematic parameters, and, in these studies, two different types of systematic parameter throws were used: in one set of fits, systematic parameter throws were restricted to physically-allowed values, and were rethrown if a thrown value was outside these limits. Thus the throw distribution formed a truncated Gaussian. In the second set of fits, throws of systematic parameters were allowed to be outside their physical limits, leading to a full Gaussian distribution of thrown values.

A summary of the mean and RMS of pulls from the unrestricted fits can be seen in figures 31 and 32. No problems were identified with the fitter, which performed as expected, and there are only small deviations from the expected mean of 1 and RMS of 0.

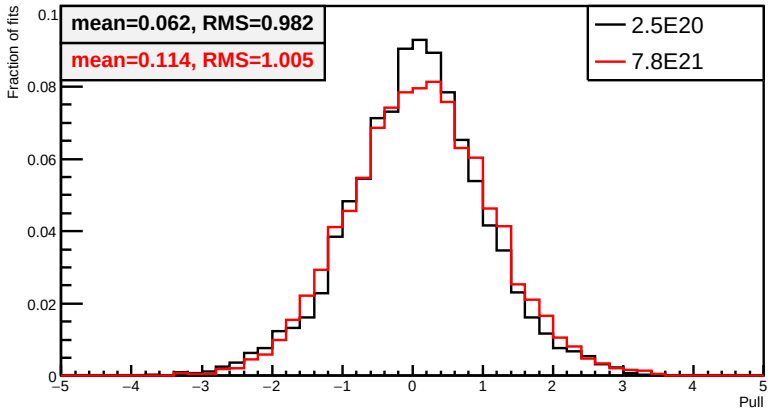
Pulls for the restricted fits can be seen in figure 33 and 34. These pulls are Gaussian with mean 1 and RMS 0 for most systematic parameters, but show small differences from this for a few parameters. The differences from a Gaussian with mean 1 and RMS 0 for these parameters are understood:

- **SuperK energy scale.** This systematic is different from the others in that it migrates events from bin to bin instead of increasing or decreasing the numbers of events in the bins of reconstructed energy. Its pull is not expected to be a Gaussian with zero mean since the shape of the predicted energy spectrum is not symmetric, and, consequently, a positive tweak of this parameter has a different effect on that spectrum from a negative tweak. This can lead to a discontinuity in the derivative of the χ^2 distribution at the zero point, which also contributes to a larger number of fits with this best-fit point. An example of this discontinuity can be seen in figure 36.

- **Fermi Momentum.** This systematic has a physical limit between 200 MeV and 275 MeV. The unconstrained parameter value is 225 MeV and the 1σ error is 31 MeV. Consquently, the parameter has physical limits at around -1 and $+2\sigma$. The fitter cannot reach the true values when the parameter



(a) $\sin^2\overline{\theta}_{23}$



(b) $|\overline{\Delta m^2_{32}}|$

Figure 20: Distributions of pulls of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ from 5k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.527$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 2.500×10^{20} POT (Runs 5 and 6, black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

is thrown beyond these limits. The asymmetry of the allowed region around the central value results in a small downward bias in the pulls.

- **MEC Normalisation** The MEC normalisation on ^{16}O has a nominal value of 1.0 and a prefit error of 73.3%. This means that some unrestricted throws of this parameter are < 0, which is unphysical. Hence the restricted throws have a Gaussian distribution that is truncated at its lower end. The throw values that are truncated are the lowest values, and these would be expected to have positive pulls due to the penalty term pulling them towards the nominal value. This leads to a pull distribution that is narrower on the plus side than on the minus side.

- **CC Coherent Normalisation** The CC coherent normalisation has a nominal value of 1.0 and a prefit error of 100%. This means that some unrestricted throws of this parameter are < 0, which is unphysical. Hence the restricted throws have a Gaussian distribution that is truncated at its lower end. The throw

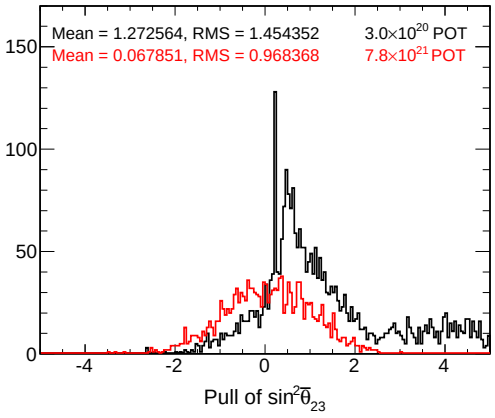


Figure 21: Distributions of pulls of $\sin^2\overline{\theta}_{23}$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.35$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

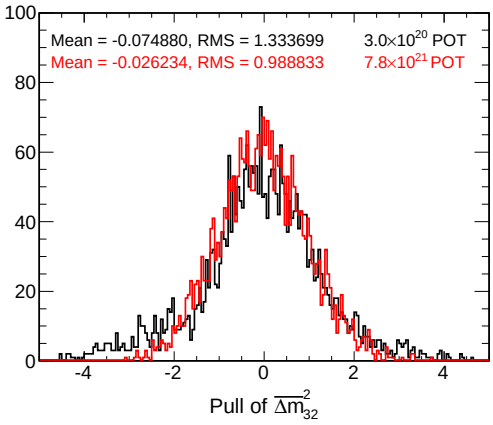


Figure 22: Distributions of pulls of $|\overline{\Delta m^2_{32}}|$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.35$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

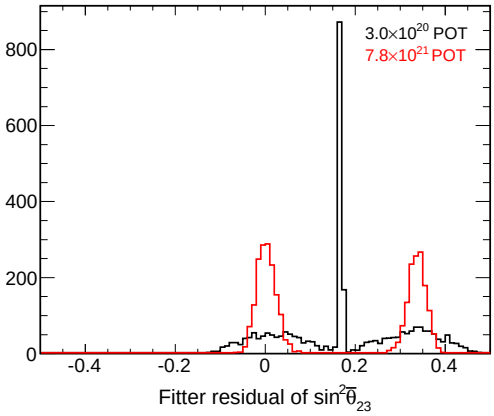


Figure 23: Distributions of fitter residuals of $\sin^2\overline{\theta}_{23}$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.35$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

values that are truncated are the lowest values, and these would be expected to have positive pulls due to the penalty term pulling them towards the nominal value. This leads to a pull distribution that is narrower on the plus side than on the minus side.

- **SuperK Detector + FSI + SI Parameter 6 (all NC events)** The bias comes from the low statistics at 2.500×10^{20} POT. When Poisson fluctuations are made of the bin contents of the E_{reco} distributions, a downward fluctuation is more likely than an upward fluctuation. For example, a Poisson distribution with mean 3 has P(1) = 0.149 and P(2) = 0.224, whereas P(4) = 0.168 and P(5) = 0.101. The NC events are mainly around the oscillation maximum where there are few events, and downward Poisson fluctuations are more likely than upward ones. Fitting these downward fluctuations involves moving this parameter down from zero, and this explains the negative bias in its pull.

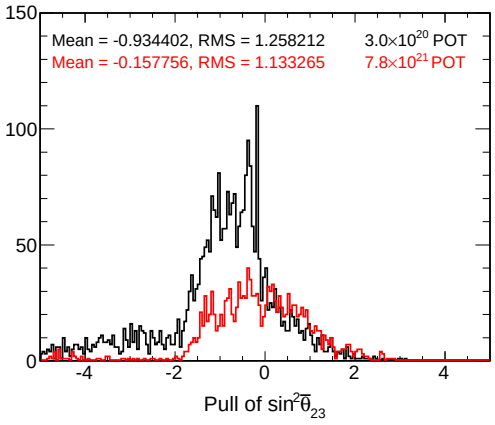


Figure 24: Distributions of pulls of $\sin^2\overline{\theta}_{23}$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.65$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

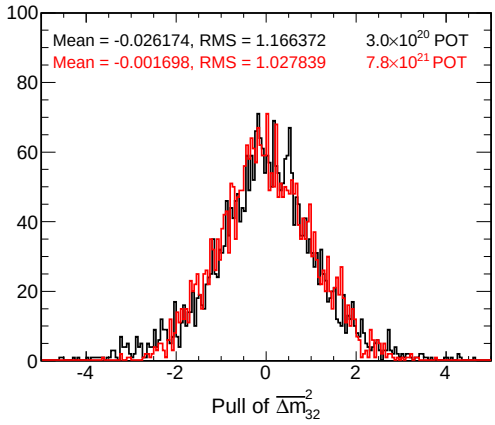


Figure 25: Distributions of pulls of $|\overline{\Delta m^2_{32}}|$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.65$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

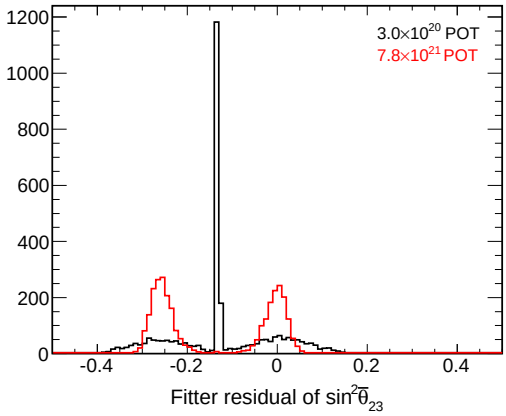


Figure 26: Distributions of fitter residuals of $\sin^2\overline{\theta}_{23}$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.65$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}$ eV^2/c^4, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

The pull distributions for four of the systematics with biases are shown in figures 35.

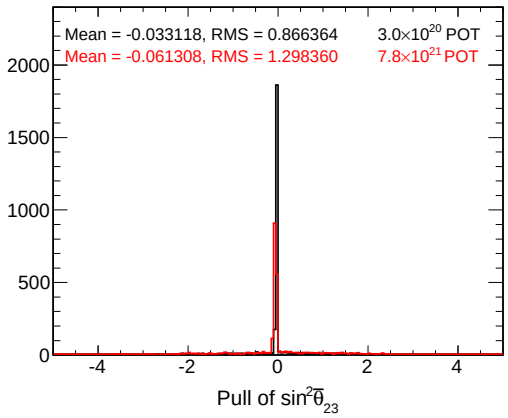


Figure 27: Distributions of pulls of $\sin^2\overline{\theta}_{23}$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.524$ and $|\overline{\Delta m^2_{32}}| = 2.2\times10^{-3}\mathrm{eV}^2/\mathrm{c}^4$, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

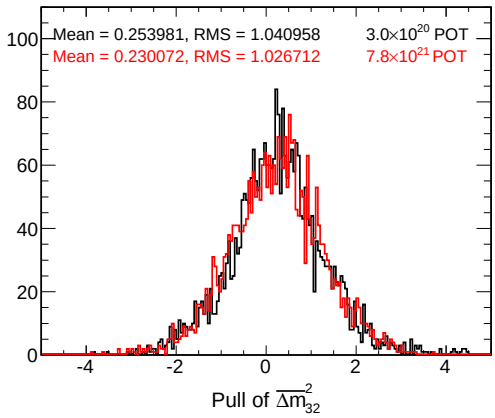


Figure 28: Distributions of pulls of $|\overline{\Delta m^2_{32}}|$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.524$ and $|\overline{\Delta m^2_{32}}| = 2.2\times10^{-3}\mathrm{eV}^2/\mathrm{c}^4$, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

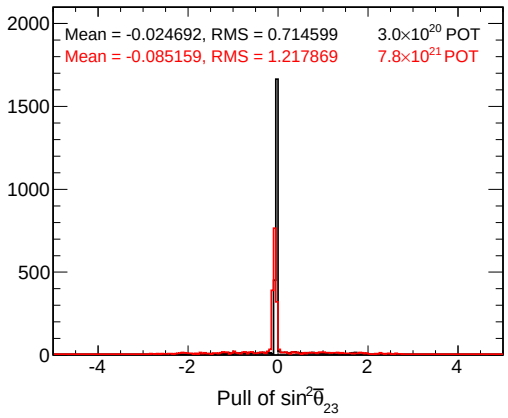


Figure 29: Distributions of pulls of $\sin^2\overline{\theta}_{23}$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.524$ and $|\overline{\Delta m^2_{32}}| = 2.8\times10^{-3}\mathrm{eV}^2/\mathrm{c}^4$, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

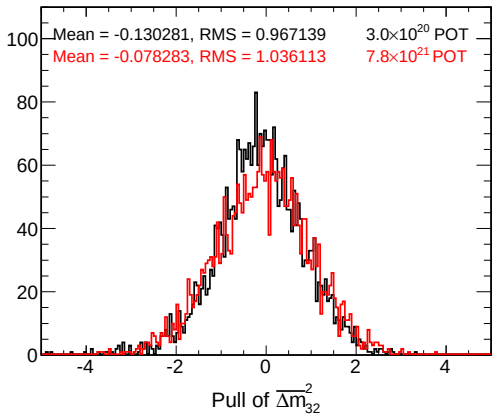


Figure 30: Distributions of pulls of $|\overline{\Delta m^2_{32}}|$ from 3k toy MC experiments, generated at $\sin^2\overline{\theta}_{23} = 0.524$ and $|\overline{\Delta m^2_{32}}| = 2.8\times10^{-3}\mathrm{eV}^2/\mathrm{c}^4$, including statistical fluctuations and with randomized systematic parameters. The other oscillation parameters are fixed to the values in Tab. 12, and the normal mass hierarchy is assumed. The distribution is shown for both 3.0×10^{20} POT (black histogram) and 7.800×10^{21} POT (the T2K ultimate goal, red histogram).

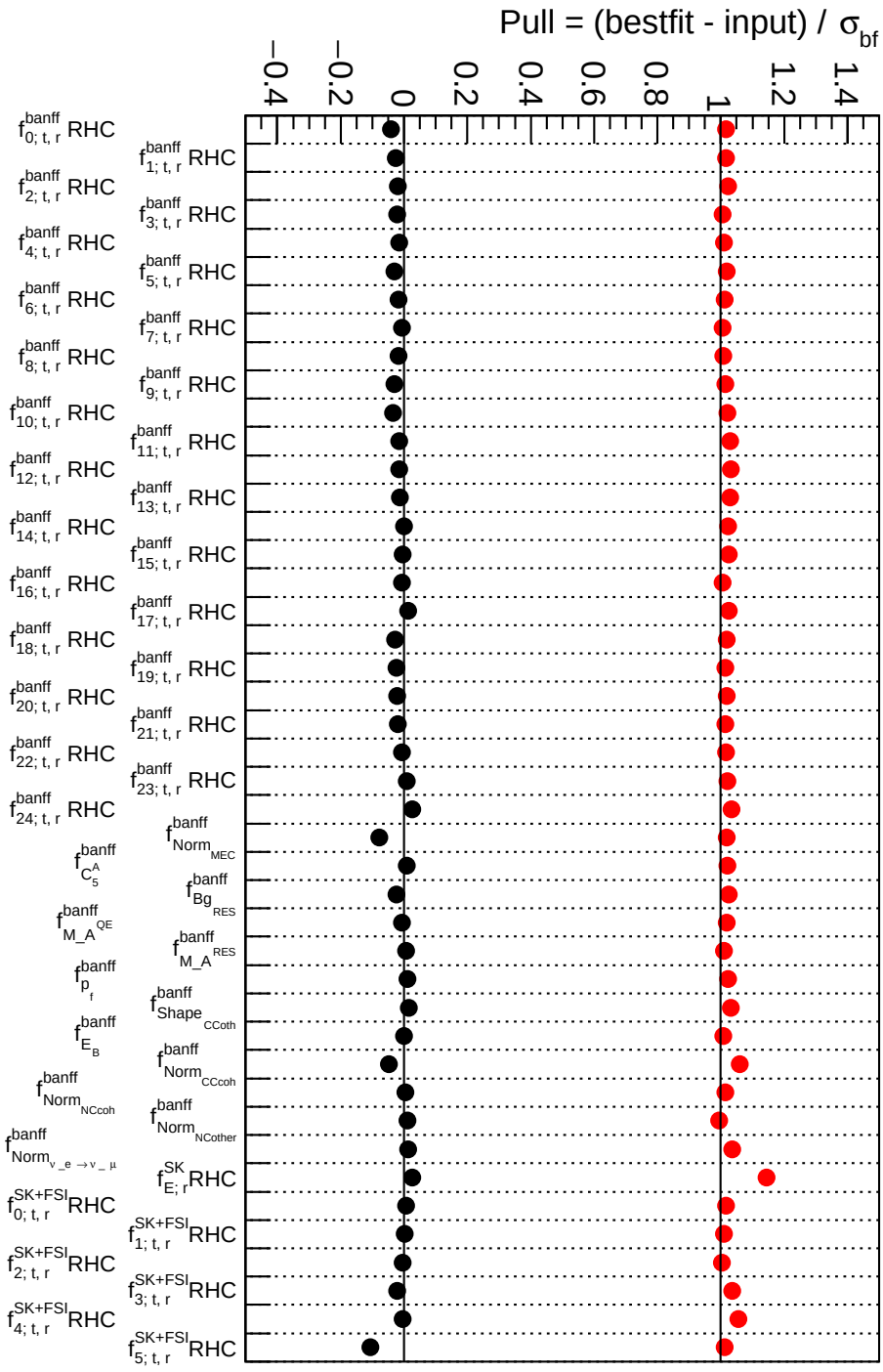


Figure 31: Summary of systematic pull distributions for all systematic parameters for 5k toy MC experiments at 2.500×10^{20} POT including statistical fluctuations and with randomized systematic parameters. Throws of systematic parameters were allowed to be thrown outside the physical region. All other oscillation parameters were fixed at their values from table 12. The normal mass hierarchy is assumed.

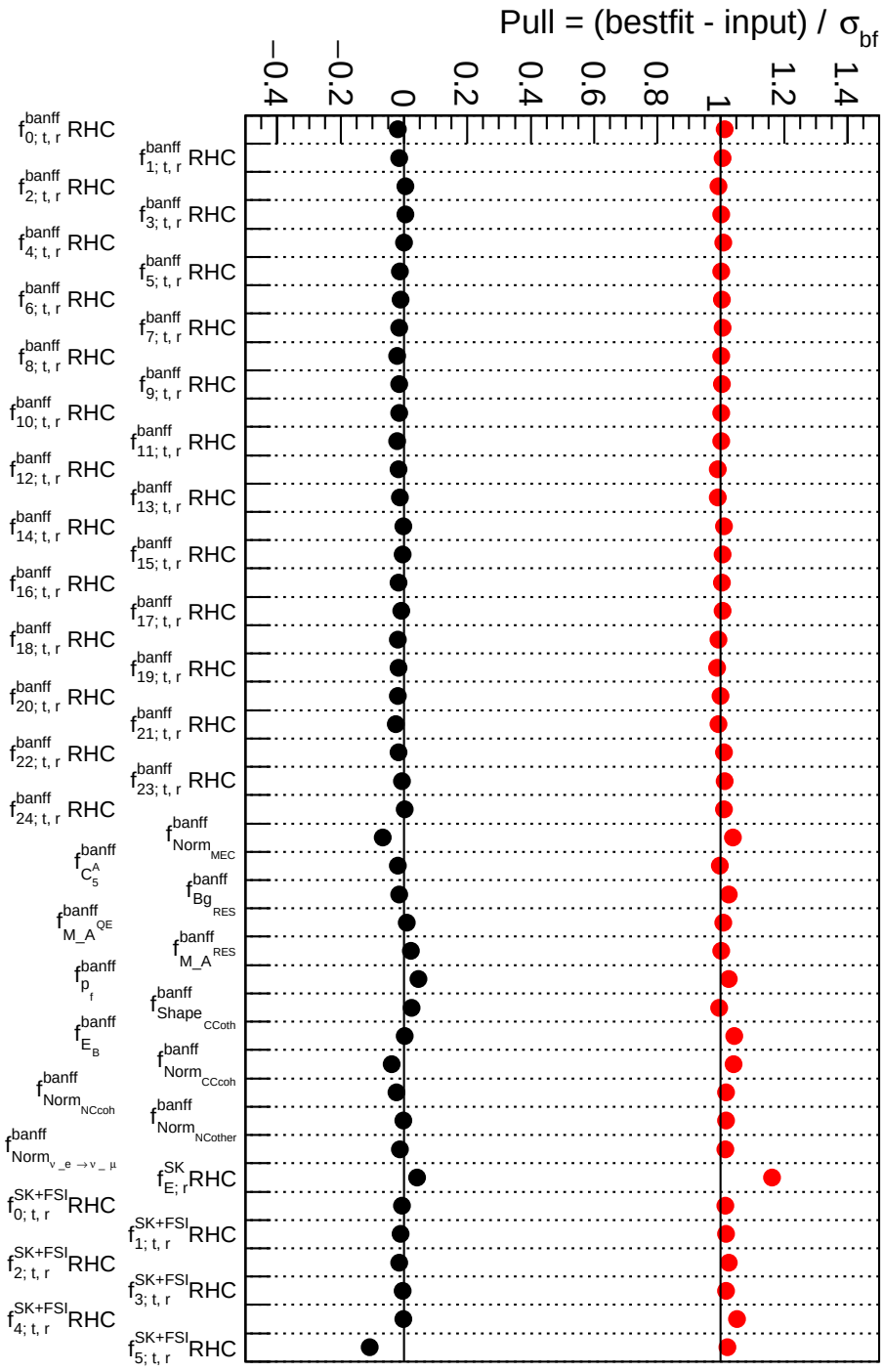


Figure 32: Summary of systematic pull distributions for all systematic parameters for 5k toy MC experiments at 2.500×10^{20} POT including statistical fluctuations and with randomized systematic parameters. Throws of systematic parameters were allowed to be thrown outside the physical region. All other oscillation parameters were fixed at their values from table 12. The inverted mass hierarchy is assumed.

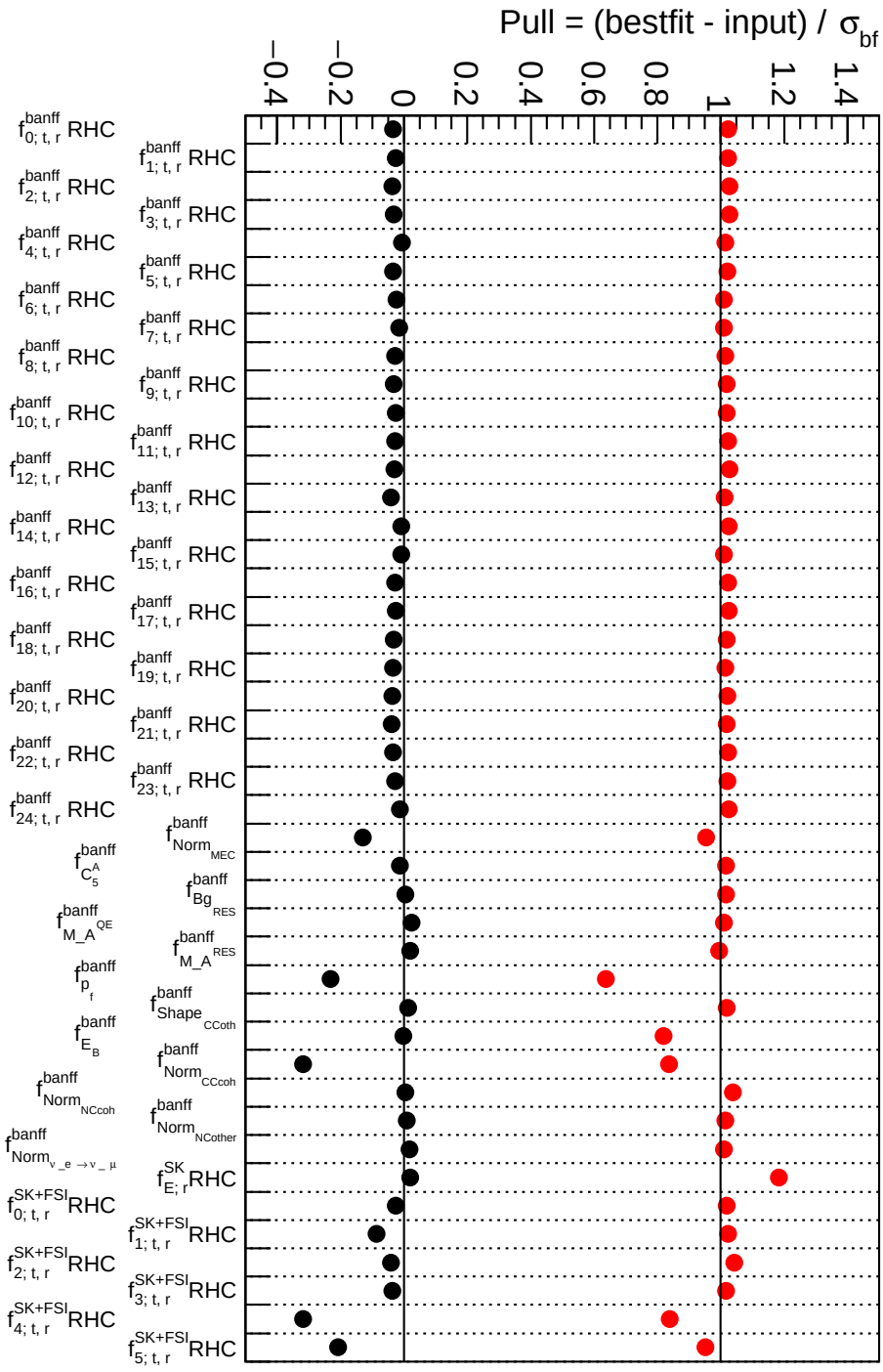


Figure 33: Summary of systematic pull distributions for all systematic parameters for 5k toy MC experiments at 2.500×10^{20} POT including statistical fluctuations and with randomized systematic parameters. Systematic parameters were rethrown if an unphysical thrown value was obtained. All other oscillation parameters were fixed at their values from table 12. The normal mass hierarchy is assumed.

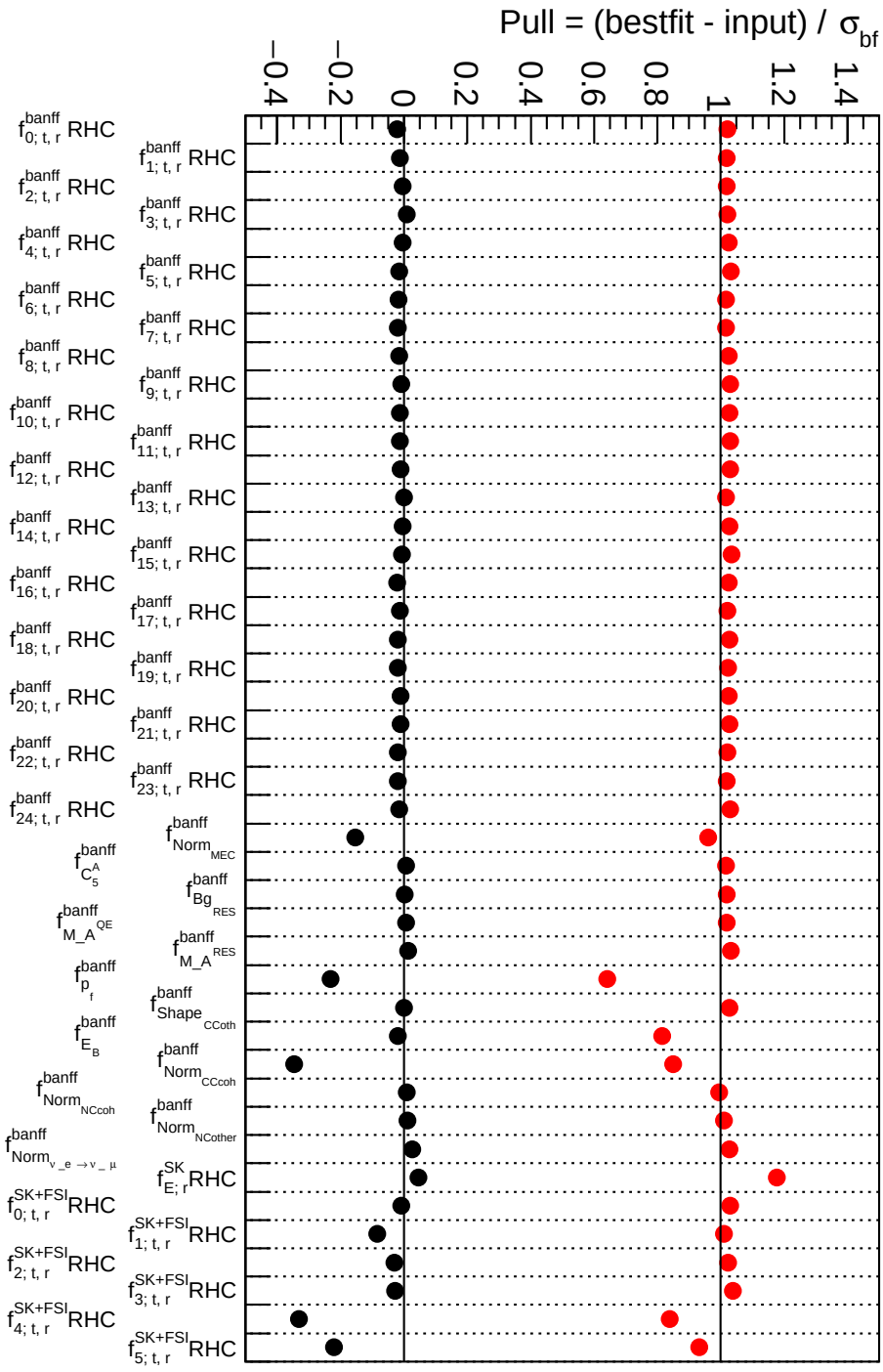
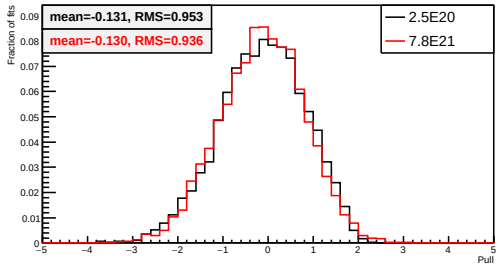
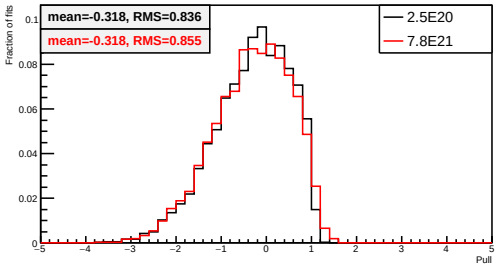


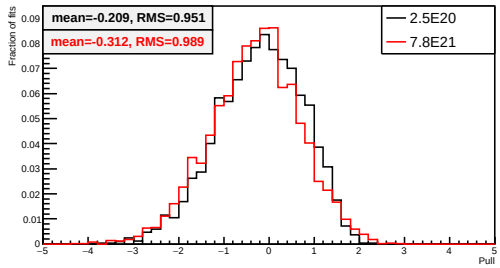
Figure 34: Summary of systematic pull distributions for all systematic parameters for 5k toy MC experiments at 2.500×10^{20} POT including statistical fluctuations and with randomized systematic parameters. Systematic parameters were rethrown if an unphysical thrown value was obtained. All other oscillation parameters were fixed at their values from table 12. The inverted mass hierarchy is assumed.



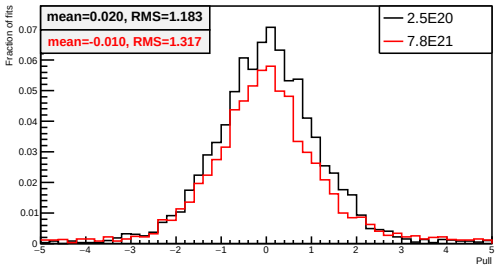
(a) MEC Normalization on ^{16}O



(b) CC Coherent Normalization



(c) SuperK NC Normalization



(d) SuperK Energy Scale

Figure 35: Distributions of pulls for four systematic parameters that have small biases. These distributions are shown for both 2.500×10^{20} POT (Runs 5 and 6, black histograms) and 7.800×10^{21} POT (the T2K ultimate goal, red histograms).

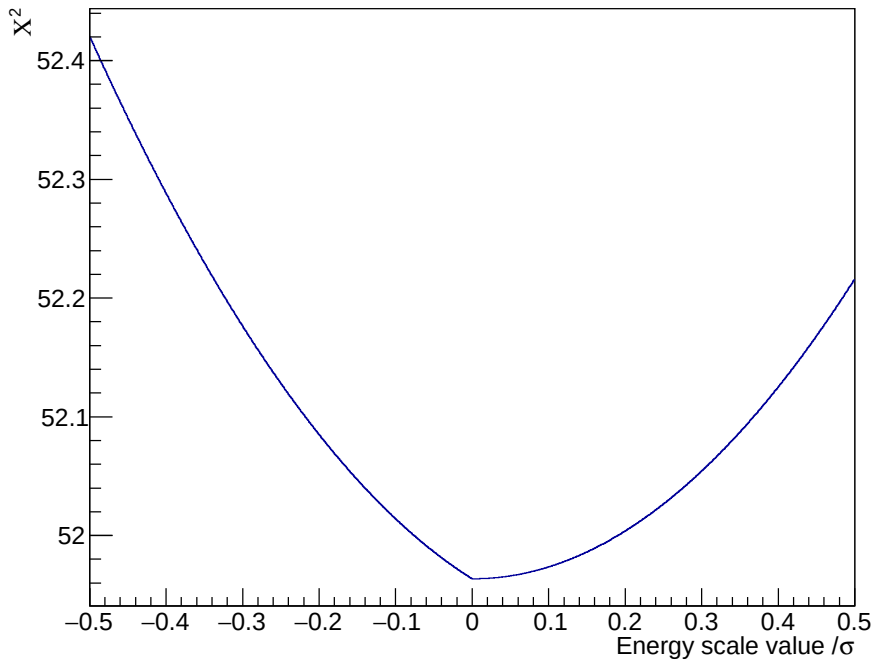


Figure 36: The chi-square distribution when varying SuperK energy scale for a single randomized toy experiment. Note that the distribution is different on the two sides of the zero point, leading to a kink in the plot at zero. In the vast majority of datasets, this kink is imperceptible; the toy plotted here was selected specifically from our validation toys as a clear demonstration of the effect.

6. Results of the $\bar{\nu}_\mu$-disappearance analysis with the Run 5+6 dataset

6.1. Best-fit oscillation parameters

Two 3-flavour $\bar{\nu}_\mu$-disappearance fits were performed on the combined 2.315×10^{20} POT Run 5+6 dataset, using the method described in Sec. 4.1. In the first fit, the normal mass hierarchy was assumed, and $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ were allowed to float, while, in the second fit, the inverted mass hierarchy was assumed and $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{31}}|$ were allowed to float. In both fits, all 44 systematic parameters considered in this analysis were allowed to float.

The reconstructed energy distribution of the Run 5+6 single μ-like ring 2.315×10^{20} POT dataset is shown in figure 37 along with the best-fit energy spectrum assuming the normal mass hierarchy. The same Run 5+6 data spectrum is also shown with the best-fit energy spectrum assuming the inverted mass hierarchy in figure 38. A summary is given of the best-fit oscillation parameters, χ^2 and p-values and the observed and expected numbers of events in SuperK in table 13.

Two sets of comparative best-fit spectra are also included, which use a coarser binning in reconstructed energy. The bin edges used are {0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0, 30.0 GeV}. Figures 39 and 40 show a comparison between the best-fit spectrum from this analysis and the spectrum given by the T2K run 1-4 joint fit result. Figures 41 and 42 show the VALOR spectra compared to the other oscillation groups.

Analysis	Run period	$\sin^2\overline{\theta}_{23}$	$\lvert\overline{\Delta m^2_{32}}\rvert/\lvert\overline{\Delta m^2_{31}}\rvert$	N_{obs}	N_{exp}	χ^2_{bf}/ndf	p-value
2015 3-flavour Normal	Run 5+6	0.511	2.318×10^{-3}	17	19.663	44.893/71	0.40
2015 3-flavour Inverted	Run 5+6	0.514	2.291×10^{-3}	17	20.439	44.899/71	0.42

Table 13: Summary of best-fit parameters from the fits of the combined Run 5+6 dataset. The quoted best-fit χ^2 values (χ^2_{bf}) were computed from Eq. 3 in Sec. 4.1. using the reconstructed energy binning scheme given in Sec. 2.2.5. The table also shows the p-values obtained from the goodness-of-fit test described in Sec. 4.2 for all fits.

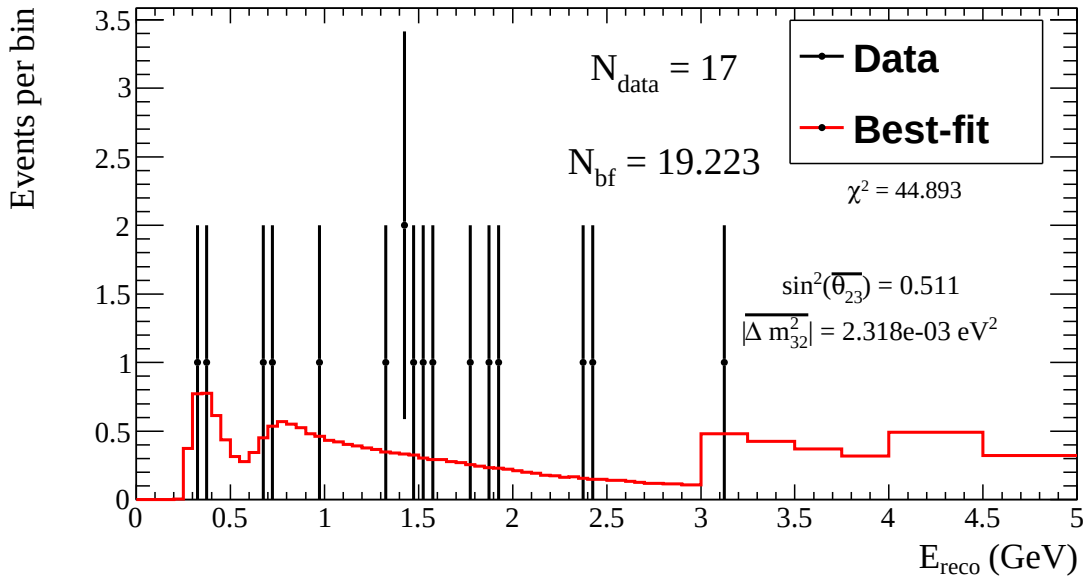


Figure 37: The Run 5+6 single μ-like ring 2.315×10^{20} POT dataset and the best-fit spectrum. The normal mass hierarchy is assumed. Note that total number of events per bin is shown, without taking into account bin width.

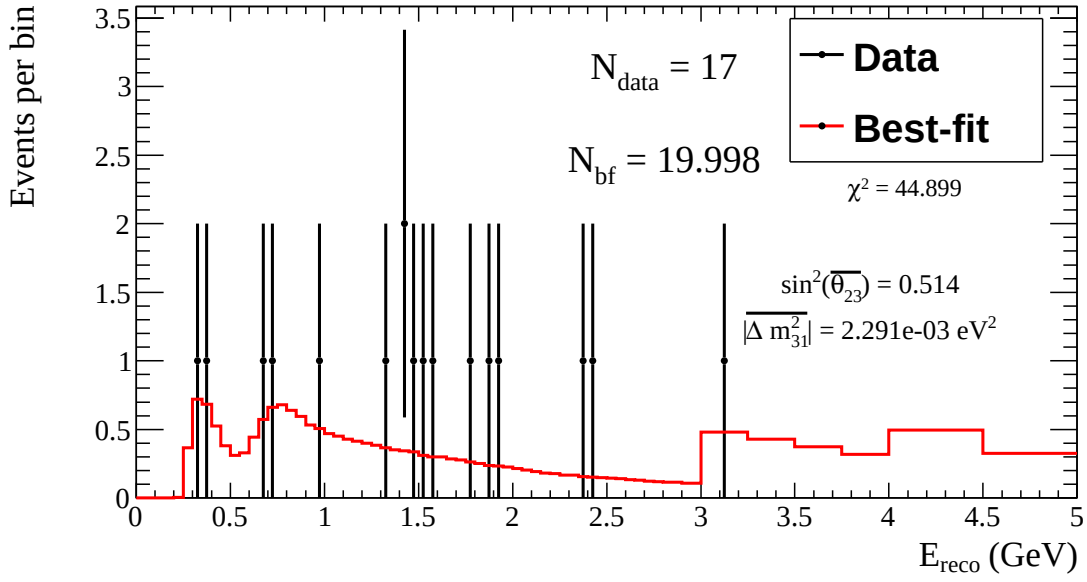


Figure 38: The Run 5+6 single μ-like ring 2.315×10^{20} POT dataset and the best-fit spectrum. The inverted mass hierarchy is assumed. Note that total number of events per bin is shown, without taking into account bin width.

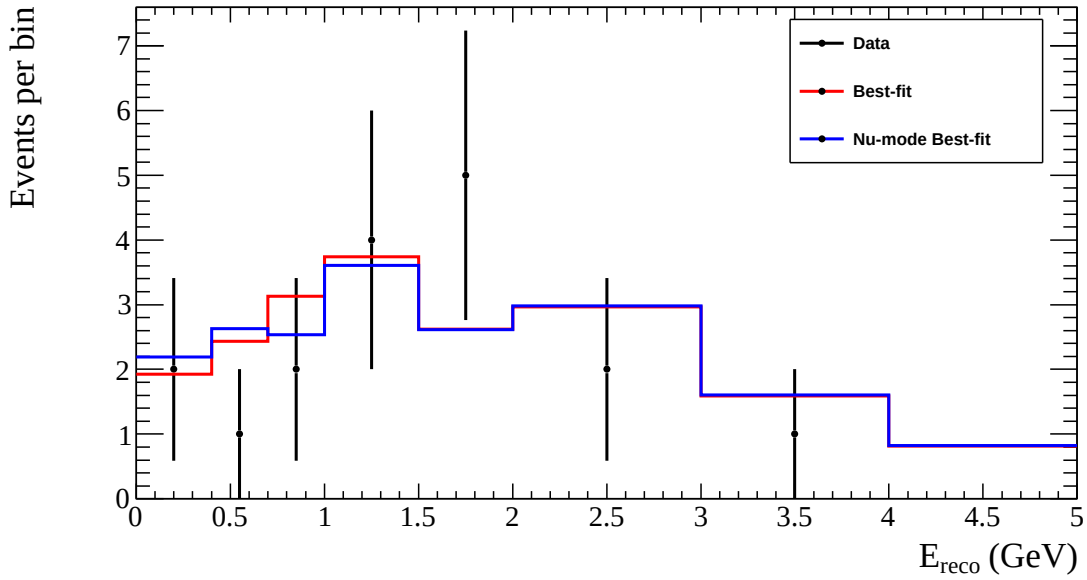


Figure 39: The Run 5+6 single μ-like ring 2.315×10^{20} POT dataset and the best-fit spectrum. The binning is coarser than that used in the actual analysis (0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0, 30.0 GeV) and event counts are shown per bin. The VALOR normal mass hierarchy result is shown, and is compared with the best-fit spectrum from the $\bar{\nu}_\mu$ dataset fixing *all* the ν and $\bar{\nu}$ oscillation paramters to the T2K best-fit values from the Run 1-4 joint analysis as shown in table 11, while still profiling the systematics.

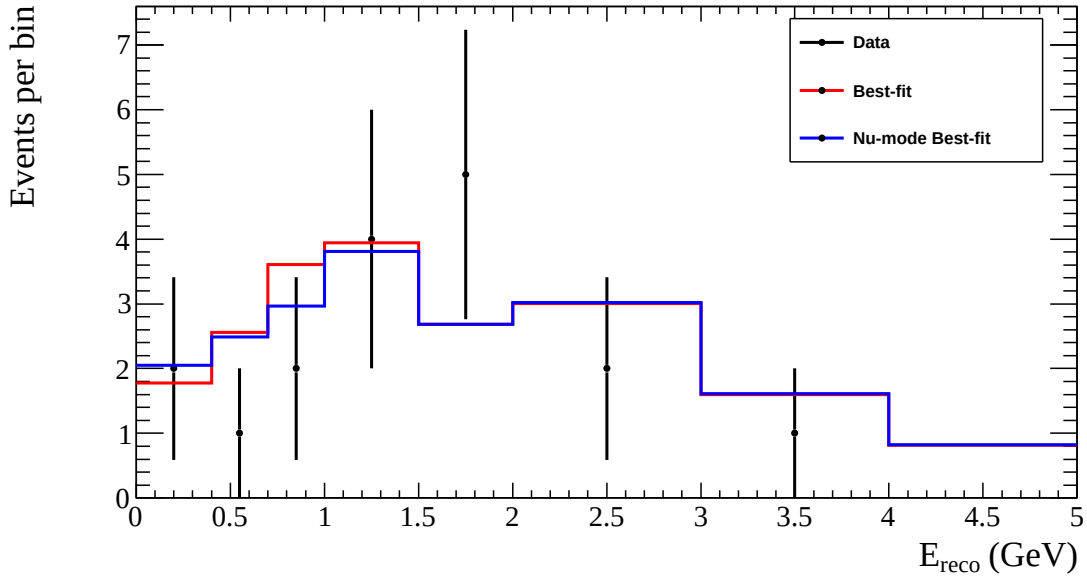


Figure 40: The Run 5+6 single μ-like ring 2.315×10^{20} POT dataset and the best-fit spectrum. The binning is coarser than that used in the actual analysis (0.0, 0.4, 0.7, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0, 30.0 GeV) and event counts are shown per bin. The VALOR inverted mass hierarchy result is shown, and is compared with the best-fit spectrum from the $\bar{\nu}_\mu$ dataset fixing *all* the ν and $\bar{\nu}$ oscillation paramters to the T2K best-fit values from the Run 1-4 joint analysis as shown in table 11, while still profiling the systematics.

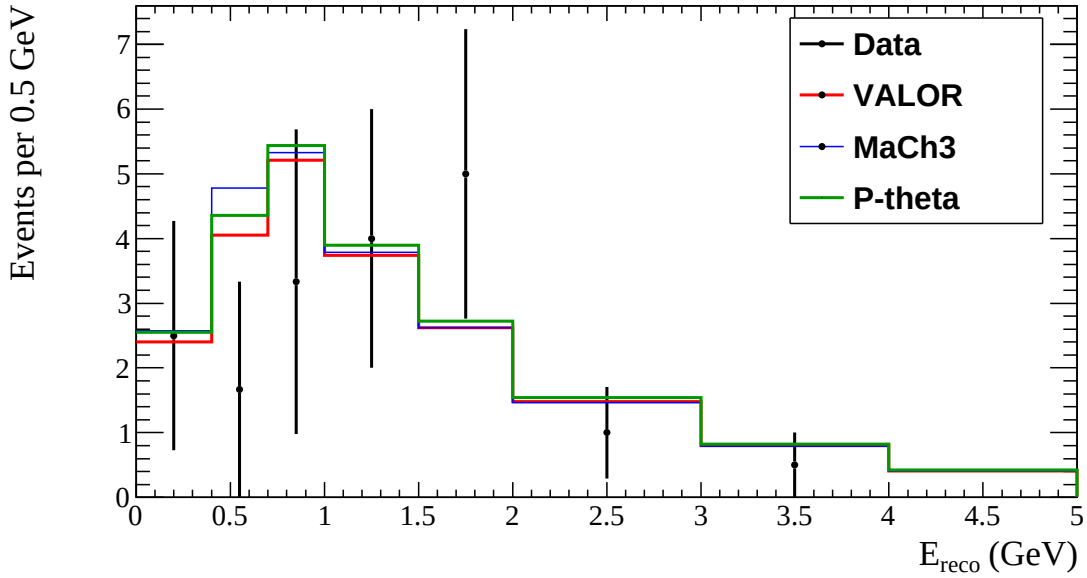


Figure 41: The Run 5+6 single μ-like ring 2.315×10^{20} POT dataset and the best-fit spectrum. This plot is more coarsely binned than the actual analysis and event counts are shown per unit energy. The VALOR normal mass hierarchy result is shown, compared to the other oscillation groups.

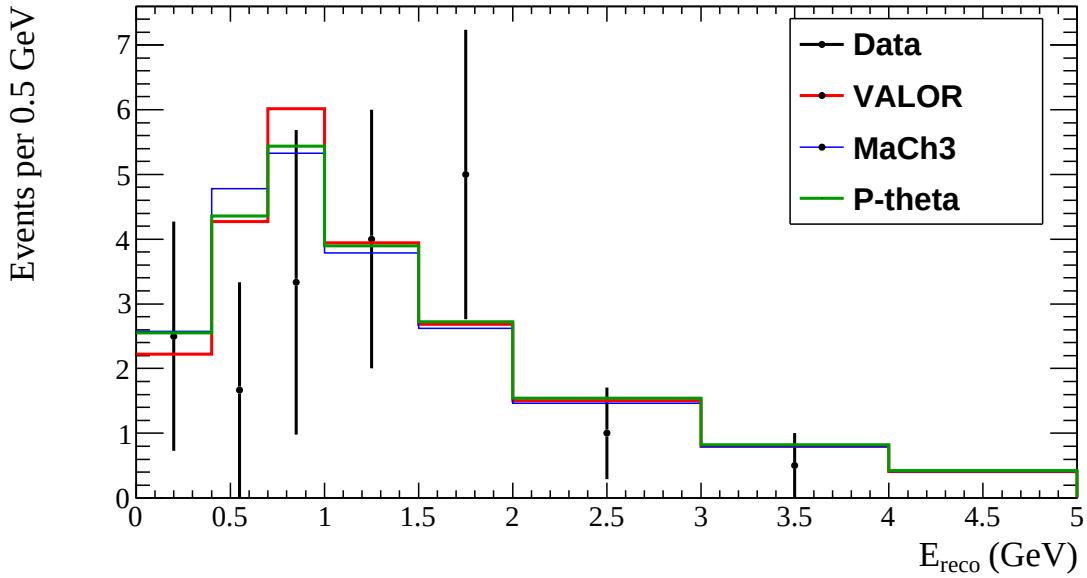


Figure 42: The Run 5+6 single μ-like ring 2.315×10^{20} POT dataset and the best-fit spectrum. This plot is more coarsely binned than the actual analysis and event counts are shown per unit energy. The VALOR inverted mass hierarchy result is shown, compared to the other oscillation groups.

564 *6.2. Goodness-of-fit tests*

565 Goodness-of-fit tests were performed for the fits of the real data as described in Sec. 4.2, The p-values
566 from these tests show that both the fits that include the 44 systematic parameters and those that do not
567 include them are plausible for both mass hierarchies.

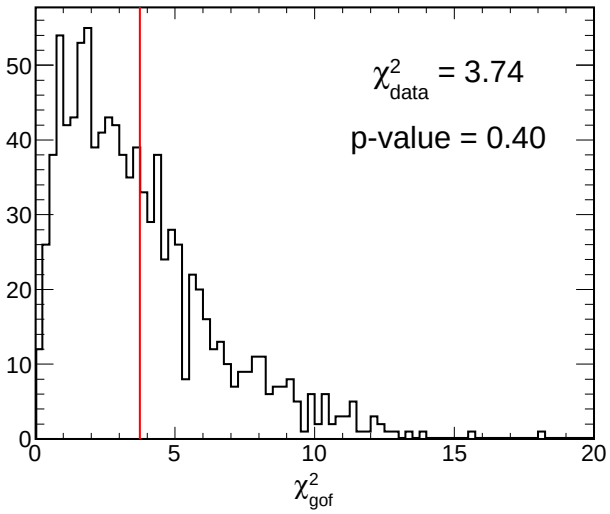


Figure 43: Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments whose true input values are the best-fit oscillation parameters from the fit of the real data ($\sin^2\overline{\theta}_{23} = 0.511$, $|\overline{\Delta m^2_{32}}| = 2.318\times10^{-3}\mathrm{eV}^2/\mathrm{c}^4$). Systematic variations and statistical fluctuations are applied to the toy datasets, and all 44 systematic parameters are included in the fits. The toy experiments are generated for 2.315×10^{20} POT and the normal mass hierarchy is assumed. The χ^2 value from the fit of the real data (with all 44 systematic parameters included in the fit) is shown as a red line.

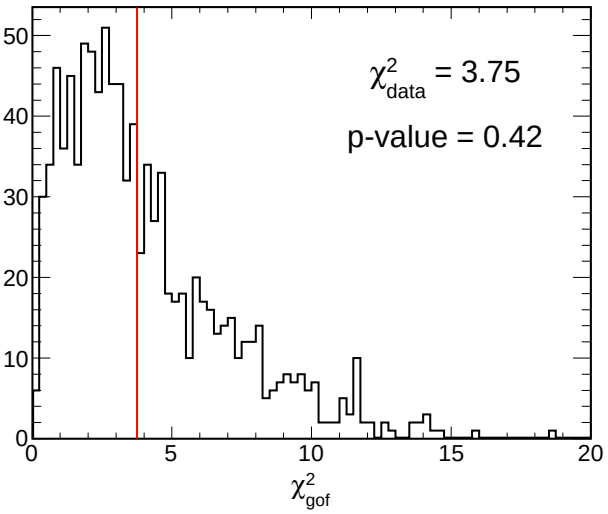


Figure 44: Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments whose true input values are the best-fit oscillation parameters from the fit of the real data ($\sin^2\overline{\theta}_{23} = 0.514$, $|\overline{\Delta m^2_{31}}| = 2.291\times10^{-3}\mathrm{eV}^2/\mathrm{c}^4$). Systematic variations and statistical fluctuations are applied to the toy datasets, and all 44 systematic parameters are included in the fits. The toy experiments are generated for 2.315×10^{20} POT and the inverted mass hierarchy is assumed. The χ^2 value from the fit of the real data (with all 44 systematic parameters included in the fit) is shown as a red line.

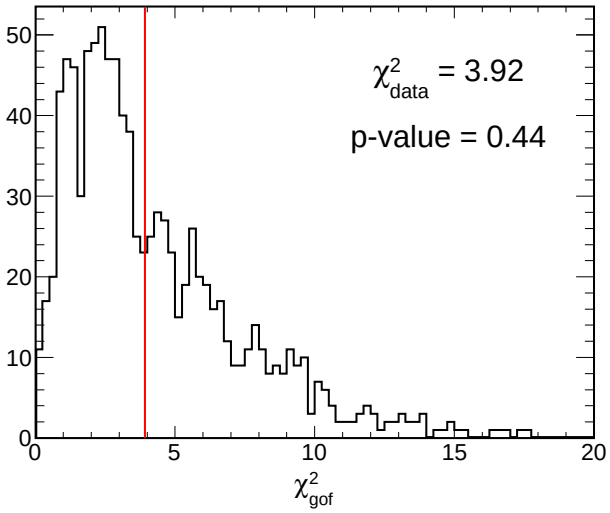


Figure 45: Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments whose true input values are the best-fit oscillation parameters from the fit of the real data ($\sin^2\overline{\theta}_{23} = 0.511$, $|\overline{\Delta m^2_{32}}| = 2.318\times10^{-3}\mathrm{eV}^2/\mathrm{c}^4$). Statistical fluctuations only are applied to the toy datasets, and the systematic parameters are not included in the fits. The toy experiments are generated for 2.315×10^{20} POT and the normal mass hierarchy is assumed. The χ^2 value from the fit of the real data (with the systematic parameters not included in the fit) is shown as a red line.

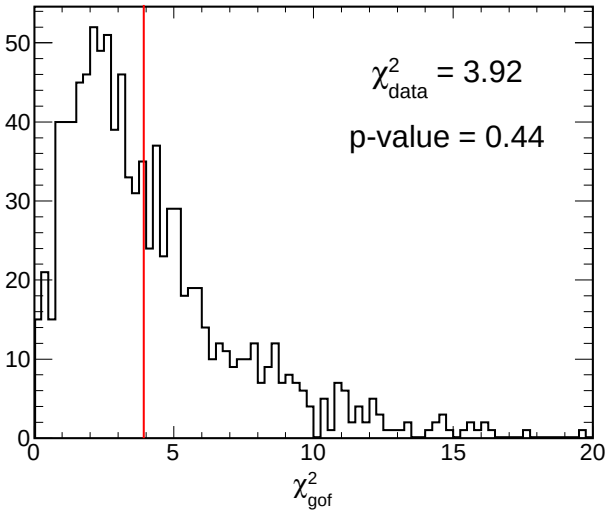


Figure 46: Distribution of χ^2_{gof}, the goodness-of-fit (gof) χ^2, from 1k toy MC experiments whose true input values are the best-fit oscillation parameters from the fit of the real data ($\sin^2\overline{\theta}_{23} = 0.514$, $|\overline{\Delta m^2_{31}}| = 2.291\times10^{-3}\mathrm{eV}^2/\mathrm{c}^4$). Statistical fluctuations only are applied to the toy datasets, and the systematic parameters are not included in the fits. The toy experiments are generated for 2.315×10^{20} POT and the inverted mass hierarchy is assumed. The χ^2 value from the fit of the real data (with the systematic parameters not included in the fit) is shown as a red line.

6.3. $\Delta\chi^2$ surfaces and confidence regions

$\Delta\chi^2$ surfaces from the fits of the combined Run 5+6 dataset are shown in figure 47 for the normal mass hierarchy and in figure 48 for the inverted mass hierarchy. The 68% and 90% allowed regions are shown using the constant-$\Delta\chi^2$ method in figures 49 - 52: the first of these figures shows a comparison of the stat+syst confidence contour with the stat-only contour, and figures 51 and 52 show comparisons between the allowed regions from the fits of the combined Run 5+6 dataset with those from the fits of the Asimov dataset.

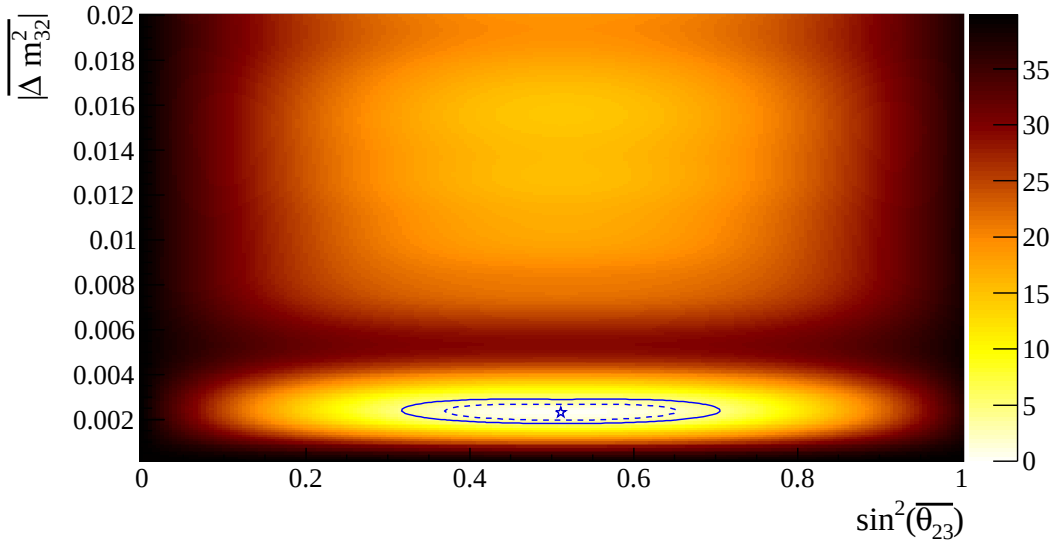


Figure 47: $\Delta\chi^2$ surface from the fit of the 2.315×10^{20} POT run 5+6 dataset. The normal mass hierarchy is assumed. 68% and 90% confidence contours are shown.

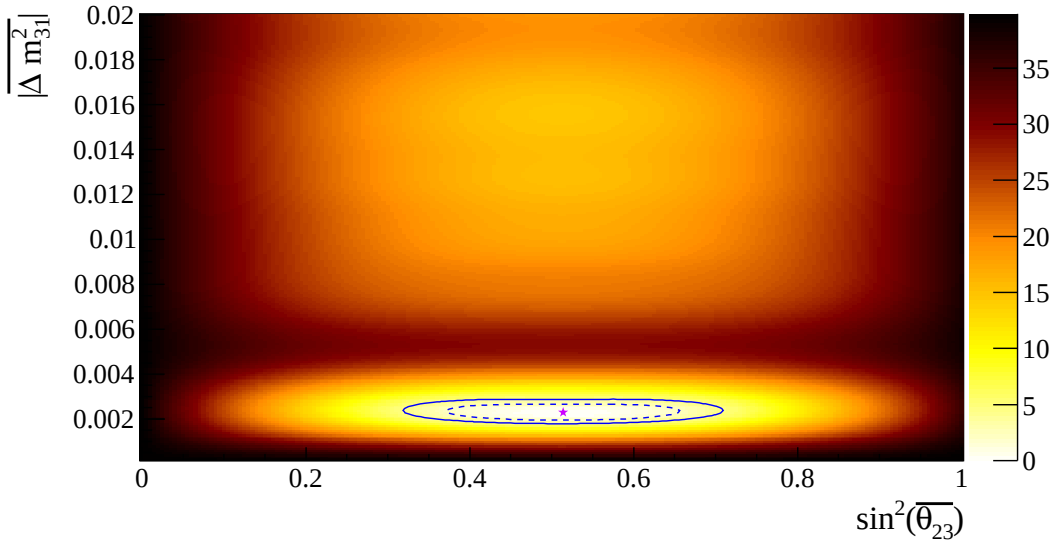


Figure 48: $\Delta\chi^2$ surface from the fit of the 2.315×10^{20} POT Run 5+6 dataset. The inverted mass hierarchy is assumed. 68% and 90% confidence contours are shown.

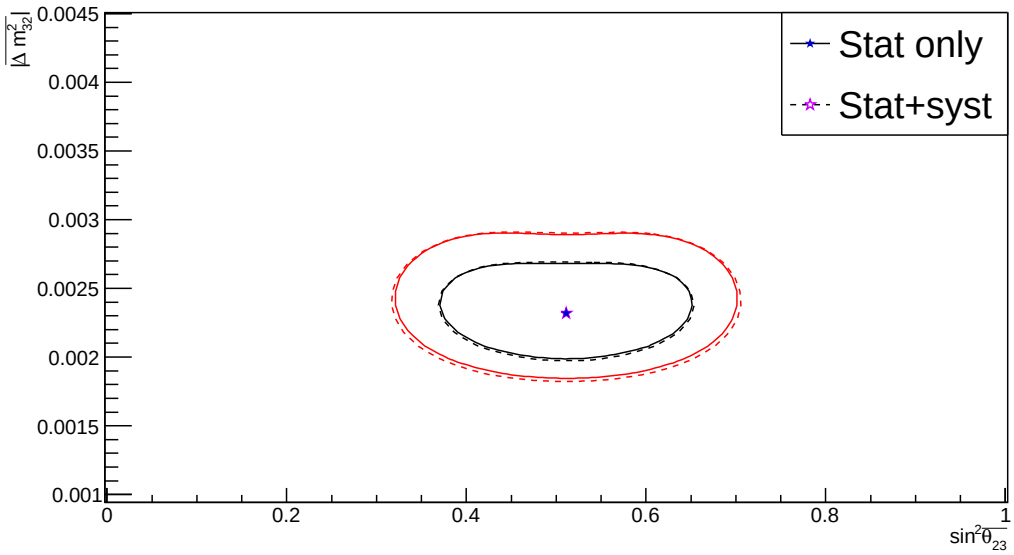


Figure 49: 68% and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to the Run 5+6 dataset. A comparison is shown between the full stat+syst contour and the contour with the systematic parameters fixed at their best-fit values. It can be seen that the fit is statistically limited. The normal hierarchy is assumed.

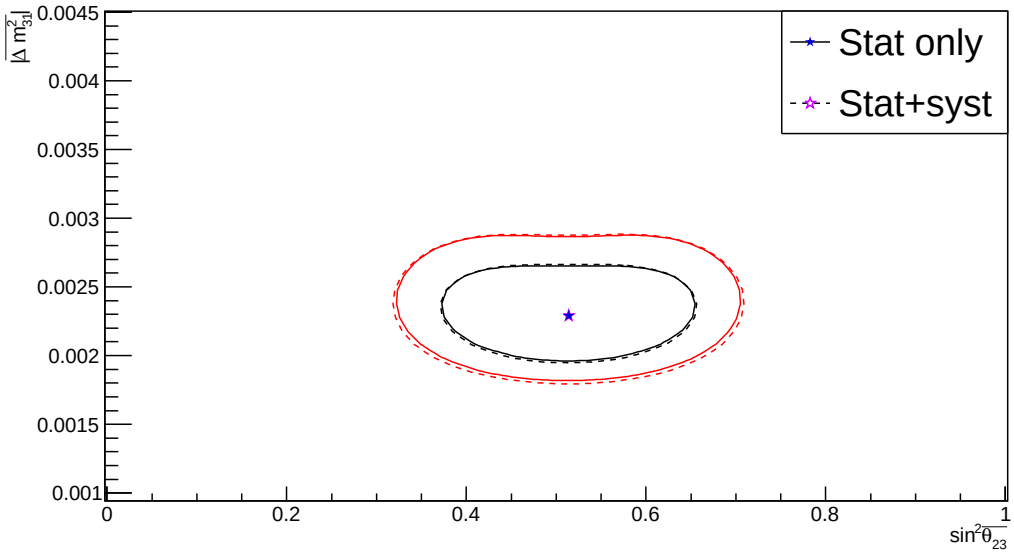


Figure 50: 68% and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to the Run 5+6 dataset. A comparison is shown between the full stat+syst contour and the contour with the systematic parameters fixed at their best-fit values. It can be seen that the fit is statistically limited. The inverted hierarchy is assumed.

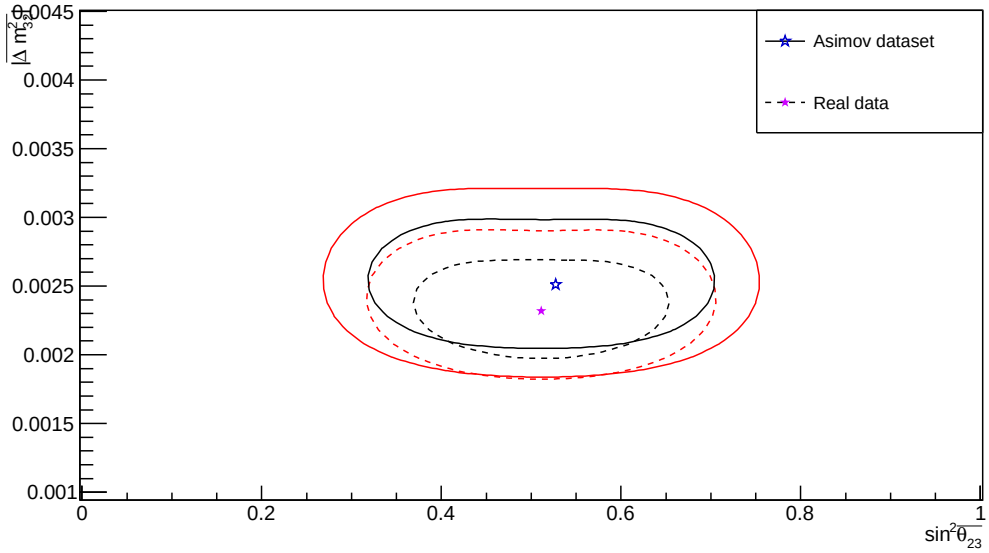


Figure 51: 68% and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to the Run 5+6 dataset, with the effects of statistical fluctuations and postfit systematic variations being included. These contours compare a fit to the Asimov dataset with the fit to the real data. The normal hierarchy is assumed.

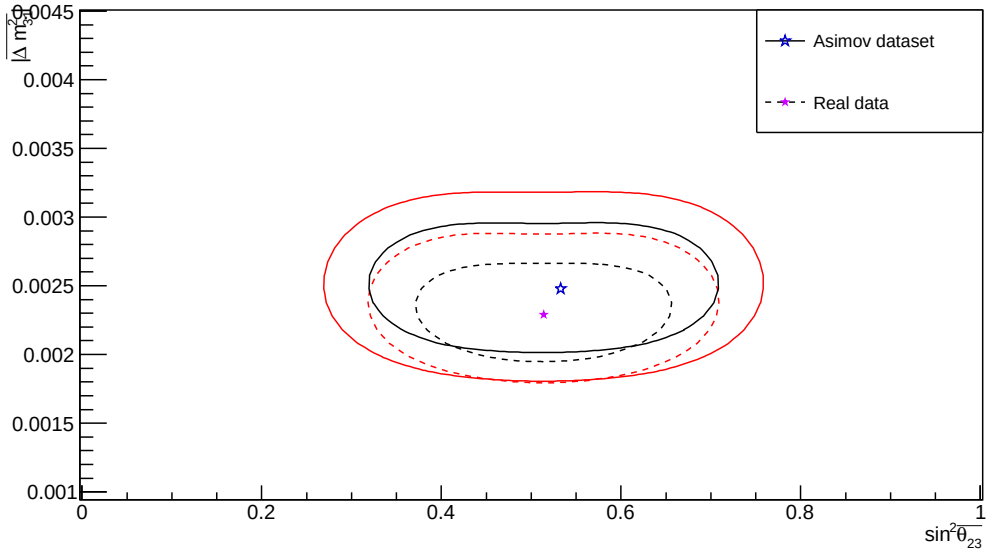


Figure 52: 68% and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to Run 5+6 dataset, with the effects of statistical fluctuations and postfit systematic variations being included. These contours compare a fit to the Asimov dataset with the fit to the real data. The inverted hierarchy is assumed.

$\Delta\chi^2$ surfaces and confidence intervals are shown for each of $\sin^2\overline{\theta}_{23}$ and $\overline{\Delta m^2_{32}}$ separately in figures 53 and 54 respectively. These figures were made using separate fits of the Run 5+6 dataset in which the oscillation parameter not shown was profiled. Similar fits were performed where this parameter was profiled without a penalty term, and these gave extremely similar results.

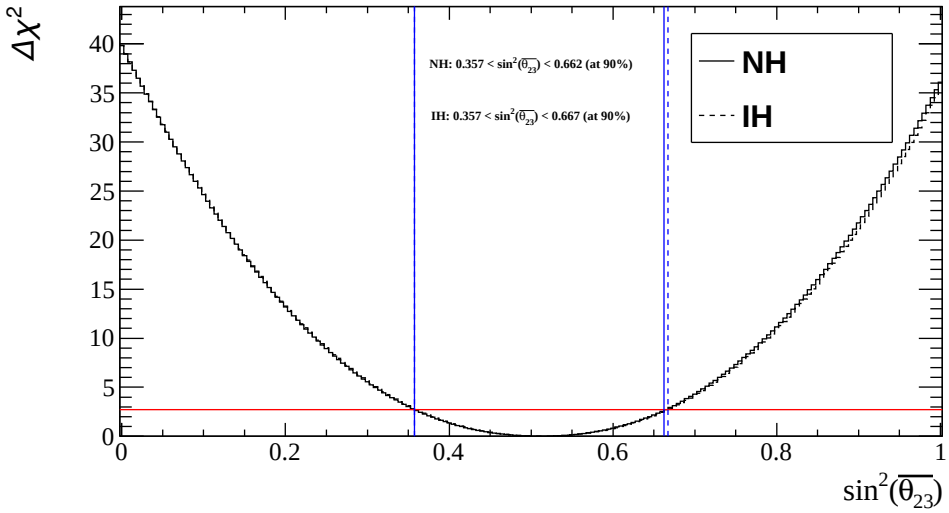


Figure 53: Plots of 1D $\Delta\chi^2$ for $\sin^2\overline{\theta}_{23}$ with $\overline{\Delta m^2_{32}}$ (normal hierarchy) or $\overline{\Delta m^2_{31}}$ (inverted hierarchy) being profiled. These are from the fit of the 2.315×10^{20}real data with the effects of statistical fluctuations and postfit systematic variations being included. Using the constant-$\Delta\chi^2$ method, the 1D 90% CL allowed intervals are $0.357 < \sin^2\overline{\theta}_{23} < 0.662$ (normal hierarchy) and $0.357 < \sin^2\overline{\theta}_{23} < 0.667$ (inverted hierarchy).

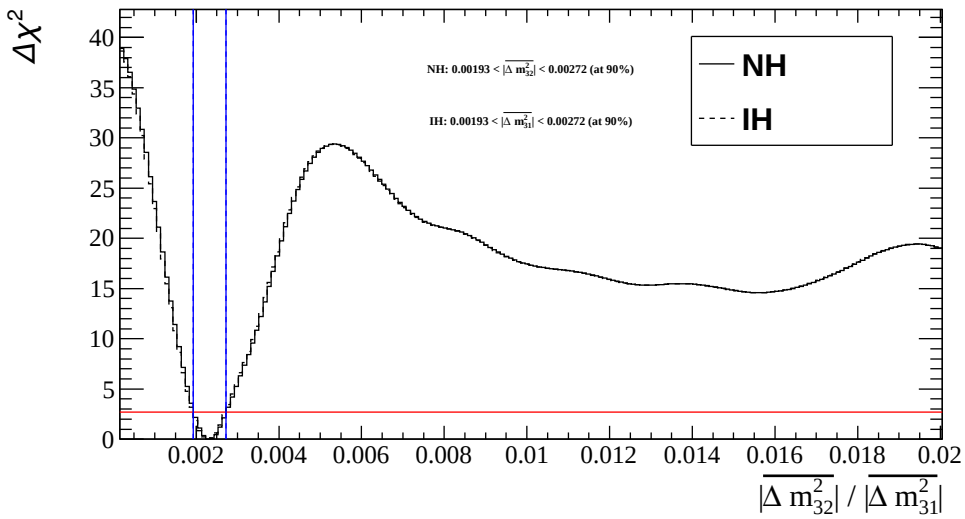


Figure 54: Plots of 1D $\Delta\chi^2$ for $\overline{\Delta m^2_{32}}$ (normal hierarchy) and $\overline{\Delta m^2_{31}}$ (inverted hierarchy), with $\sin^2\overline{\theta}_{23}$ being profiled. These are from the fit of the 2.315×10^{20}real data with the effects of statistical fluctuations and postfit systematic variations being included. Using the constant-$\Delta\chi^2$ method, the 1D 90% CL allowed intervals are $1.93\times10^{-3}\text{eV}^2/\text{c}^4 < \overline{\Delta m^2_{32}} < 2.72\times10^{-3}\text{eV}^2/\text{c}^4$ (normal hierarchy) and $1.93\times10^{-3}\text{eV}^2/\text{c}^4 < \overline{\Delta m^2_{31}} < 2.72\times10^{-3}\text{eV}^2/\text{c}^4$ (inverted hierarchy).

6.4. Comparisons of confidence regions

Figure 55 shows a comparison between the VALOR result, and the other T2K oscillation analysis groups. These analyses appear highly consistent.

In figure 56, we show a comparison between the 90% CL allowed regions from the fits of the combined Run 5+6 dataset using the constant-$\Delta\chi^2$ method and a recent result from MINOS.

Finally, we have produced a comparison with the official Run 1-4 T2K neutrino mode fit result for the normal mass hierarchy, which is shown in figure 57. The distribution of χ^2 from 3k toy MC experiments whose true input values of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ are equal to the best-fit values of the equivalent parameters for the normal mass hierarchy from the official T2K Run 1-4 result in neutrino beam mode is shown in figure 58. The value of χ^2 from a fit of the Run 5-6 antineutrino beam mode data made while fixing $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ to those same Run 1-4 best-fit values is 45.6; this gives a p-value for the antineutrino beam mode oscillation parameters being as different or more different from the Run 1-4 neutrino mode best-fit than the Run 5-6 best-fit parameters of 0.74. This is a quantitative measure of the difference between the neutrino mode and antineutrino mode best-fit oscillation parameters. It means that the results of the Run 1-4 neutrino beam mode and Run 5-6 antineutrino beam mode analyses considered together are consistent with CPT being conserved.

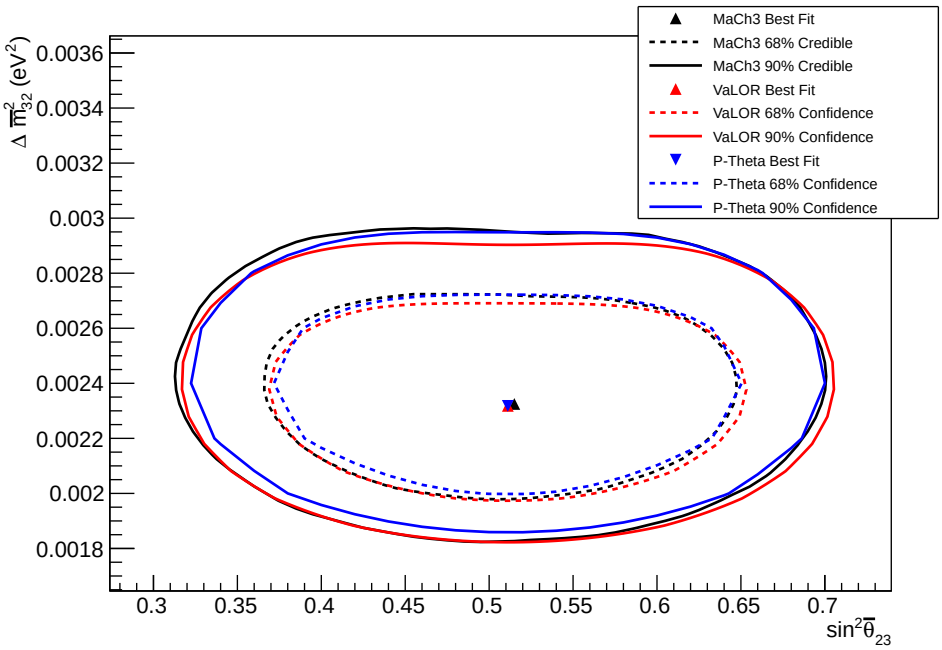


Figure 55: Confidence regions for the VALOR normal hierarchy fit to the Run 5+6 dataset compared to the other oscillation groups. This figure was produced by the MaCh3 group.

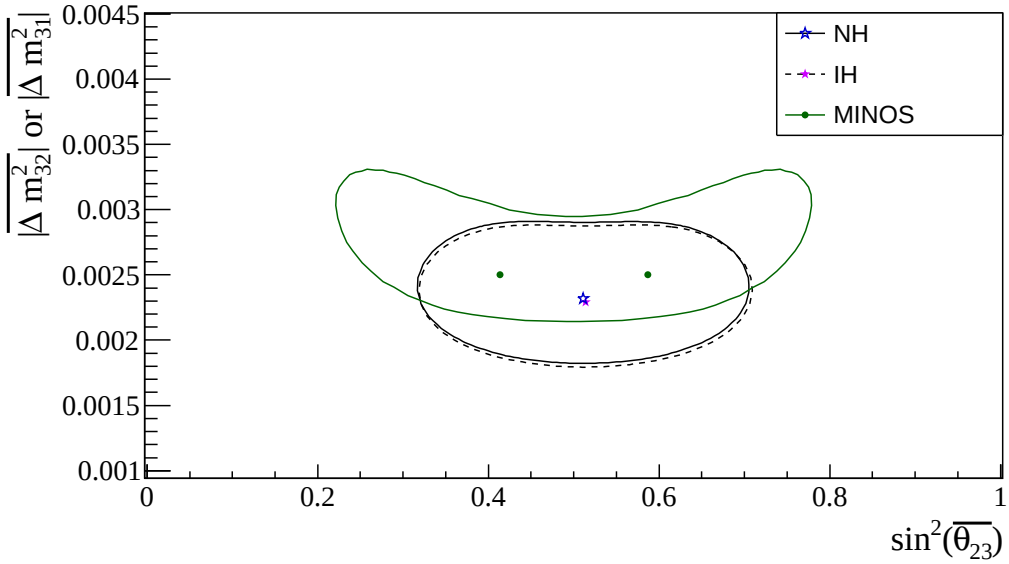


Figure 56: 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to the Run 5+6 dataset, with the effects of statistical fluctuations and postfit systematic variations being included. The best-fit point for each hypothesis is shown, with the larger point representing the better of the two fits. A MINOS contour has been added for comparison[54]. This MINOS contour was fitted in $\sin^2 2\theta_{23}$ but has been converted here to $\sin^2 \theta_{23}$.

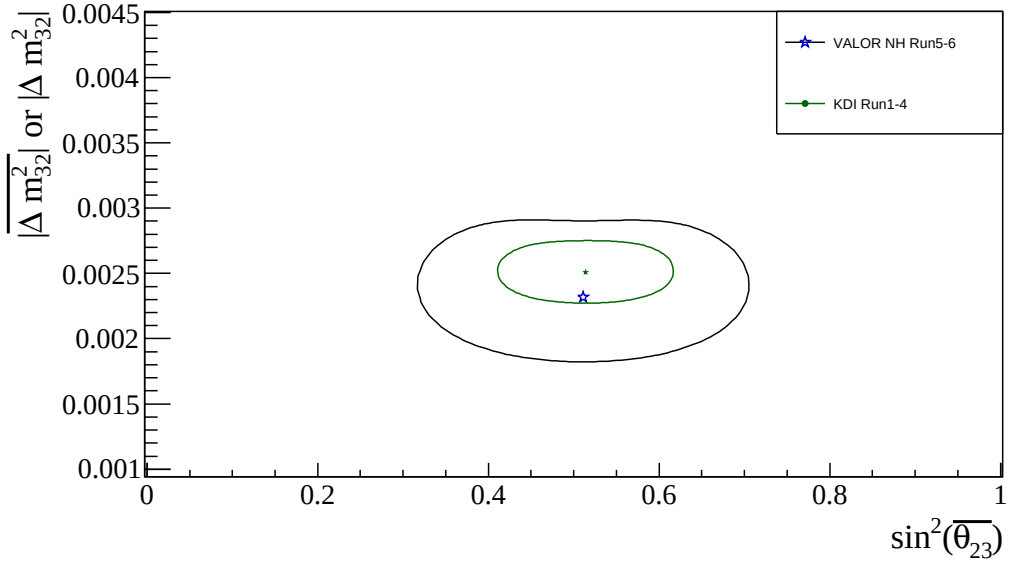


Figure 57: 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to the Run 5+6 dataset, with the effects of statistical fluctuations and postfit systematic variations being included. The normal hierarchy is assumed, and the official T2K numu disappearance contour from the run1-4 neutrino mode fits was drawn.

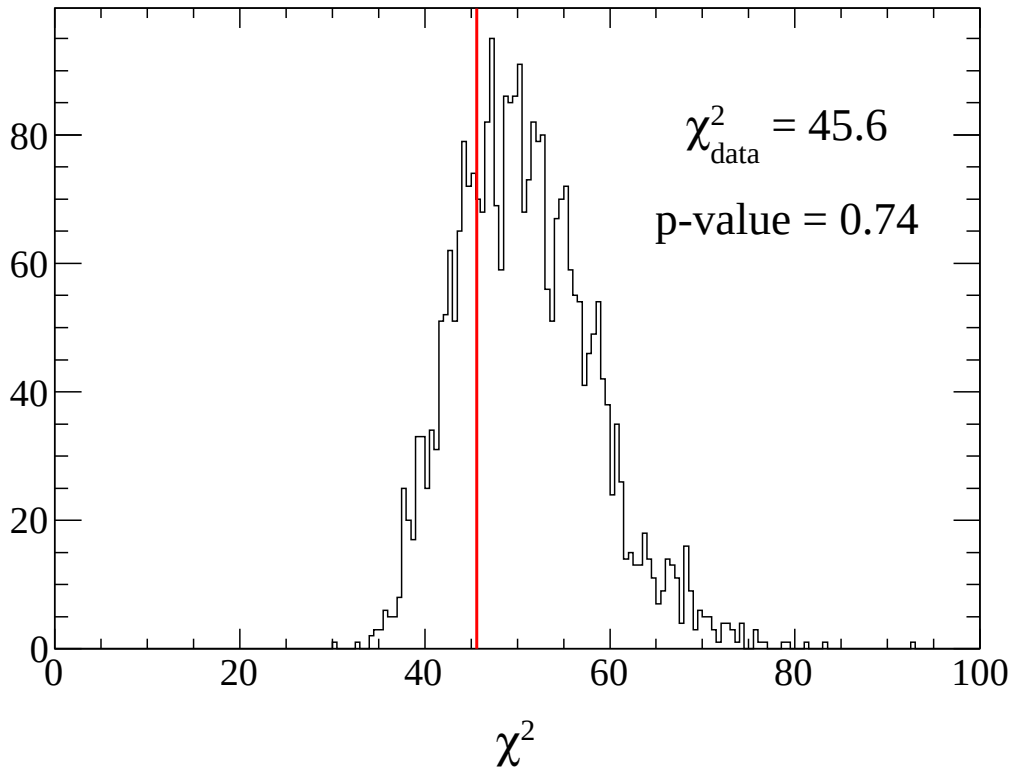


Figure 58: Distribution of best-fit χ^2 from 3k toy MC experiments whose true input values of $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ are equal to the best-fit equivalent parameters for the normal mass hierarchy from the official T2K Run 1-4 result in neutrino beam mode ($\sin^2\theta_{23} = 0.514$, $|\Delta m^2_{32}| = 2.51\times10^{-3}\text{eV}^2/\text{c}^4$). The input values of the other oscillation parameters are given in the normal hierarchy column of Tab. 11. The toy experiments are generated for 2.315×10^{20} POT and the normal mass hierarchy is assumed. Systematic variations and statistical fluctuations are applied to the toy datasets, and all 44 systematic parameters are included in the fits. The value of χ^2 from a fit of the 2.315×10^{20} POT Run 5-6 antineutrino mode data fixing $\sin^2\overline{\theta}_{23} = 0.514$ and $|\overline{\Delta m^2_{32}}| = 2.51\times10^{-3}\text{eV}^2/\text{c}^4$ (the best-fit equivalent parameters from the Run 1-4 neutrino mode analysis) is shown as a red line.

7. Summary

In this technical note, we have presented the results of a 3-flavour $\bar{\nu}_\mu$-disappearance analysis, performed by the *VALOR* group, on the combined Run 5+6 2.315×10^{20} dataset, using only data taken in antineutrino beam mode. This analysis has adapted the work presented in Refs. [1], [2], [3], [4], [5] and [6] to this dataset. The Run 5+6 dataset corresponds to an integrated J-PARC neutrino beam exposure of 2.315×10^{20} POT.

This analysis predicts 59.8 ± 9.2 (syst) single μ-like ring events in Super K in the absence of any oscillation, but only 17 were observed. The observed deficit has a strong energy dependence: the ratio of the number of observed events to the number expected under the no-oscillation hypothesis is $\sim$18% $<$ 0.5 GeV, $\sim$9% between 0.5 and 1 GeV and $\sim$69% $>$ 1 GeV.

One of the motivations for this analysis is to investigate whether the oscillation parameters for $\bar{\nu}_\mu$ disappearance are different from those for ν_μ disappearance. In order to allow for this possibility, two separate sets of oscillation parameters are used. One set is applied to the $\bar{\nu}_\mu$, $\bar{\nu}_e$ and oscillated $\bar{\nu}_e$ interaction modes, while the other set is applied to the ν_μ, ν_e and oscillated ν_e modes.

A $\bar{\nu}_\mu$-disappearance analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single μ-like ring events was fitted, and separate fits were made for the normal and the inverted mass hierarchies. In these fits, $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ were allowed to float, while all other oscillation parameters in both sets were fixed to T2K best-fit or PDG values. All 44 systematic parameters considered in this analysis were also allowed to float in the fits.

For the normal mass hierarchy, the 3-flavour ν_μ-disappearance fit of the combined Run 5+6 dataset gives $\sin^2\overline{\theta}_{23} = 0.511$ and $|\overline{\Delta m^2_{32}}| = 2.318\times10^{-3}$ eV^2/c^4 as the best-fit values (p-value = 0.40). The fit for the inverted mass hierarchy yields $\sin^2\overline{\theta}_{23} = 0.514$ and $|\overline{\Delta m^2_{31}}| = 2.291\times10^{-3}$ eV^2/c^4 as the best-fit values (p-value = 0.42). The p-value for the normal hierarchy best-fit values given a null hypothesis of the normal hierarchy best-fit values of the equivalent parameters from the official Run 1-4 neutrino mode analysis is 0.74; thus the results of this analysis considered together with the Run 1-4 neutrino mode results are consistent with CPT being conserved.

Confidence regions in the parameter space $\sin^2\overline{\theta}_{23}$ - $|\overline{\Delta m^2_{32}}|$ are shown using the constant-$\Delta\chi^2$ method .

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Appendices

A. Results from fits of Asimov dataset

$\Delta\chi^2$ surfaces are shown from fits of the Asimov dataset in figures 59 and 60 for the normal and inverted hierarchies respectively. A constant-$\Delta\chi^2$ confidence contour is also shown from these fits in figure 61. These Asimov datasets use oscillation parameters from the T2K run 1-4 joint fit with reactor constraint, which can be seen in table 11.

Additionally, figure 62 shows a comparison between the T2K Asimov datasets for those oscillation parameters, and for the MINOS disappearance parameter best-fit points[54], converted to the single-angle convention and selecting the upper octant ($\sin^2\theta_{23} = 0.587$, $\Delta m^2_{23} = 2.5 \times 10^{-3}\text{eV}^2$). The non-23 parameters are left fixed at their T2K run1-4 values.

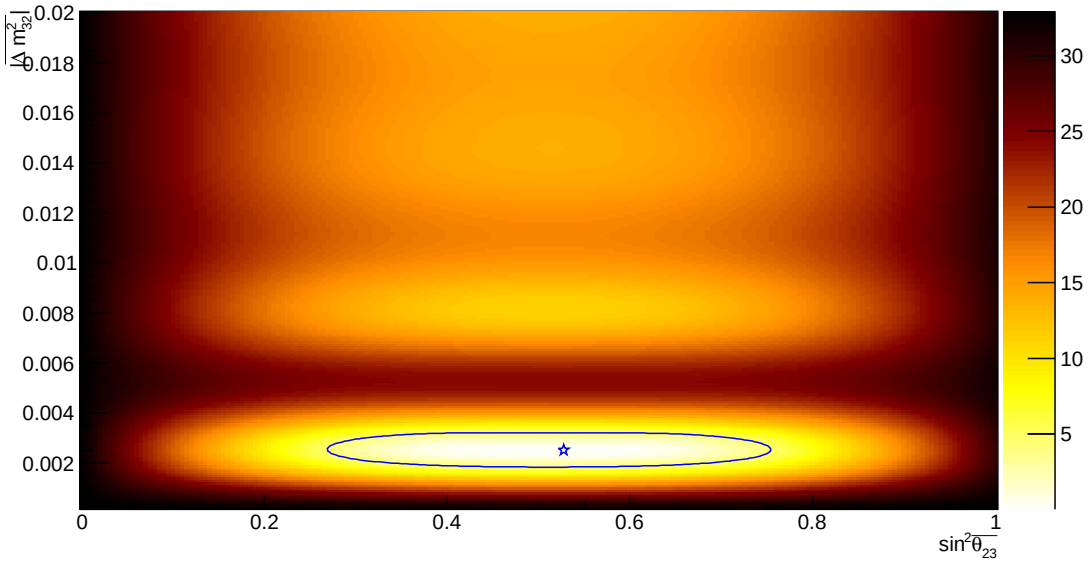


Figure 59: $\Delta\chi^2$ surface from the fit of the 2.315×10^{20}Asimov dataset. The normal mass hierarchy is assumed.

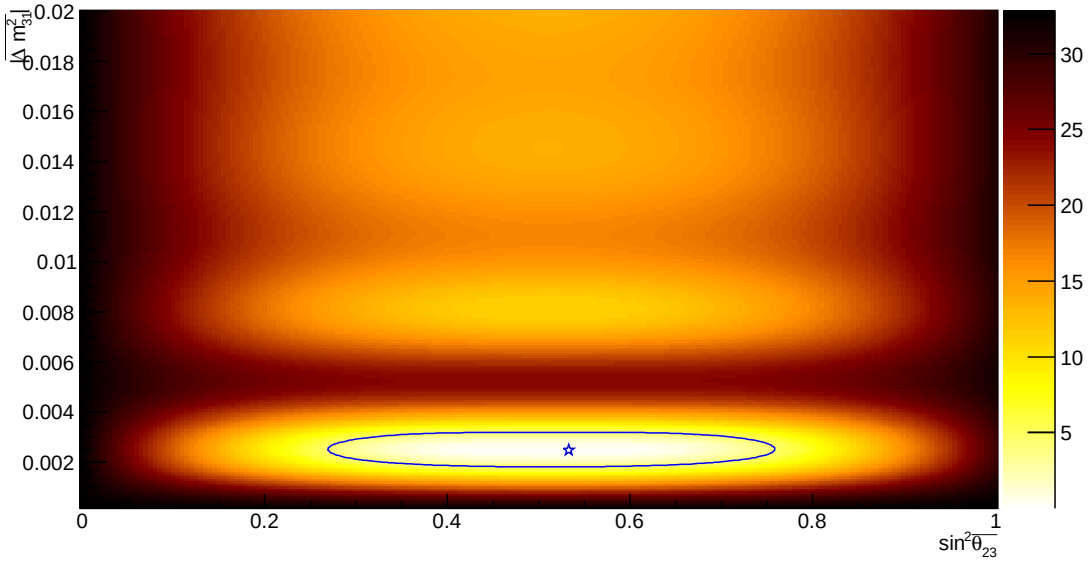


Figure 60: $\Delta\chi^2$ surface from the fit of the 2.315×10^{20}Asimov dataset. The inverted mass hierarchy is assumed.

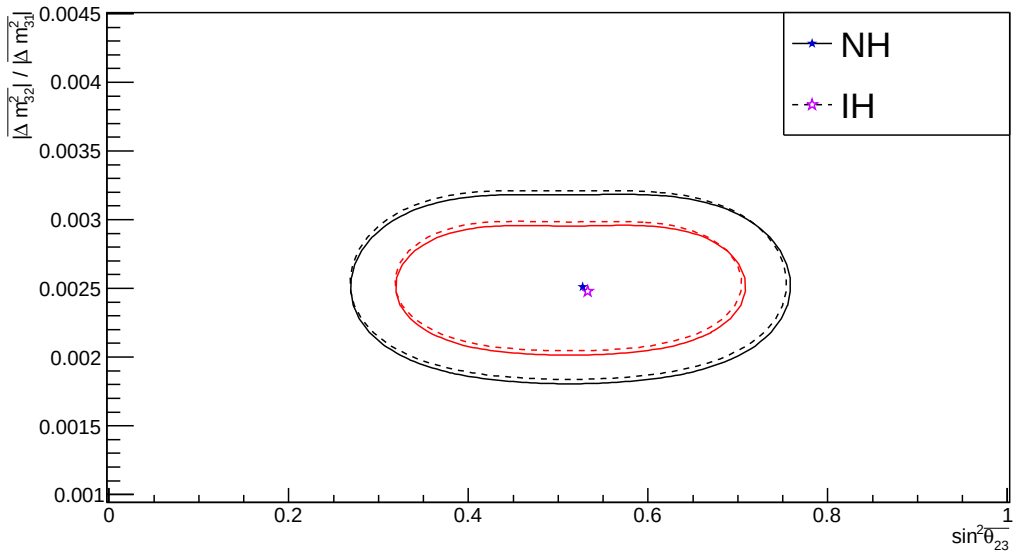


Figure 61: 68% CL and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit of the 2.315×10^{20} Asimov dataset, with the effects of statistical fluctuations and postfit systematic variations being included. The best-fit point for each hypothesis is shown, with the larger point representing the better of the two fits.

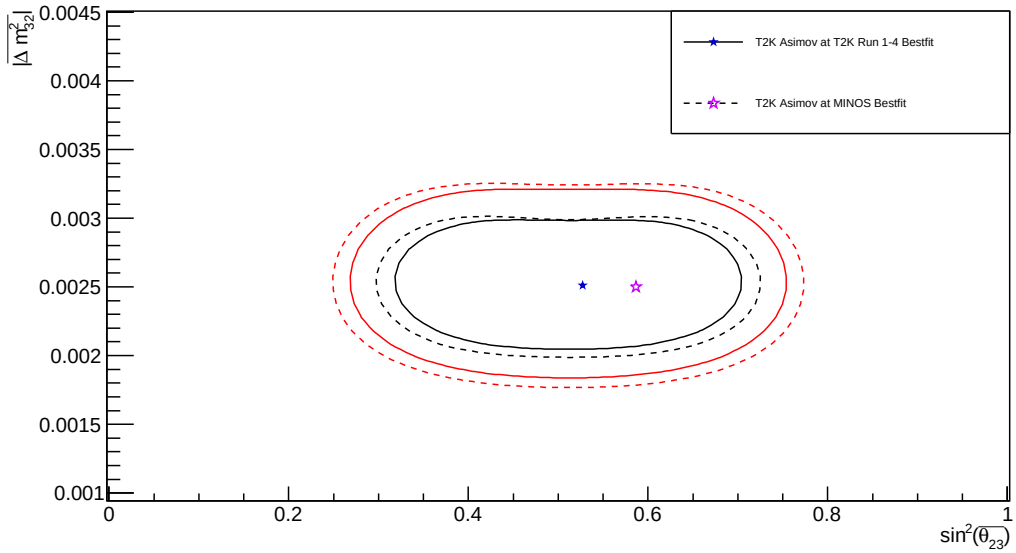


Figure 62: 68% CL and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit of the 2.315×10^{20} Asimov dataset, with the effects of statistical fluctuations and postfit systematic variations being included. Contours are drawn with both the T2K run 1-4 oscillation parameters, and the MINOS antineutrino mode best-fit parameters ($\sin^2\theta_{23} = 0.587$, $\Delta m^2_{23} = 2.5 \times 10^{-3}\text{eV}^2$). The normal hierarchy is assumed.

Fake	True input values				VALOR best-fit values		
dataset	$\sin^2\theta_{23}$	$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	$\sin^2\theta_{13}$	δ_{CP}	$\sin^2\overline{\theta}_{23}$	$\lvert\overline{\Delta m^2_{32}}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	Hierarchy
A	0.42	2.5×10^{-3}	0.0238	4.712	0.675	2.774×10^{-3}	Inverted
B	0.56	3.0×10^{-3}	0.012	0.0	0.640	2.873×10^{-3}	Normal
C	0.52	2.1×10^{-3}	0.0414	3.0	0.360	2.081×10^{-3}	Normal
D	0.63	-4.0×10^{-3}	0.0314	6.01	0.400	4.160×10^{-3}	Inverted
E	0.512	2.5×10^{-3}	0.0238	4.712	0.825	2.576×10^{-3}	Normal
F	0.47	1.8×10^{-3}	0.001	0.0	0.525	2.081×10^{-3}	Inverted

Table 14: True input and VALOR best-fit values of oscillation parameters for SuperK fake datasets A-F. None of this information was known by analyzers pre-fit. All fake data used $\Delta m^2_{21} = 7.6\times10^{-5}\mathrm{eV}^2$ and $\sin^2\theta_{12}$.

B. Fake data fit Results

Six fake SuperK datasets at 2.500×10^{20}POT were kindly made by Roger Wendell. The true input parameters were not revealed to the analysers until after they had been fitted by all the the groups performing a $\bar{\nu}_\mu$-disappearance analysis. The true input parameters and VALOR best-fit values for $\sin^2\overline{\theta}_{23}$ and $|\overline{\Delta m^2_{32}}|$ for each of these datasets are shown in table 14. The $\Delta\chi^2$ surfaces and confidence regions in the $\sin^2\overline{\theta}_{23}$-$|\overline{\Delta m^2_{32}}|$ parameter space from the fits of these datasets are shown in figures 63-80.

For these contours, the oscillation parameters were fixed at their values from table 12. These plots used the results of the BANFF fit to nominal MC, and version 2 of the SK Detector Errors (v3 was used for the data fit).

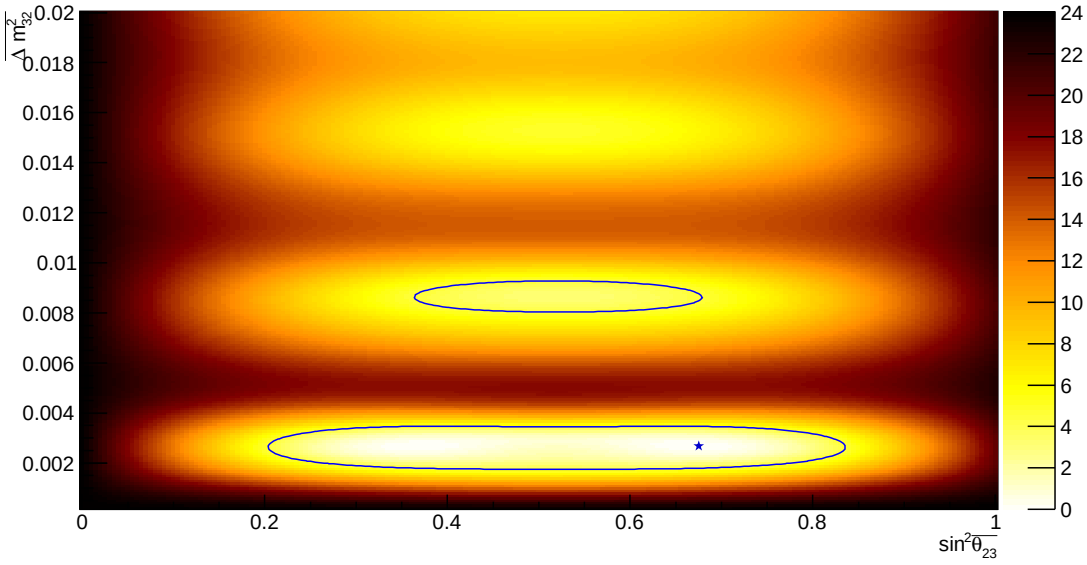


Figure 63: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset A. The normal mass hierarchy is assumed.

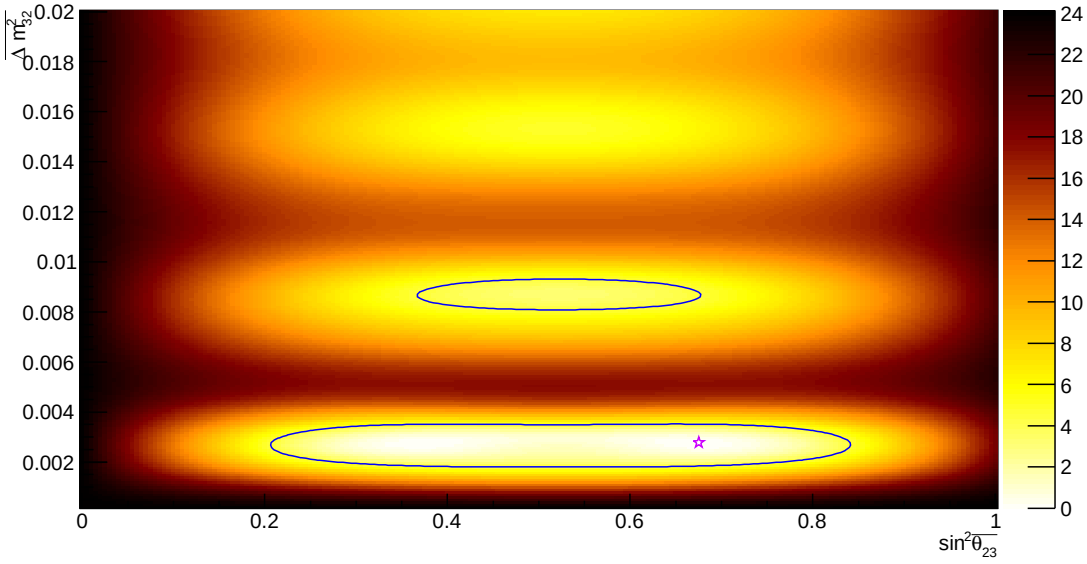


Figure 64: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset A. The inverted mass hierarchy is assumed.

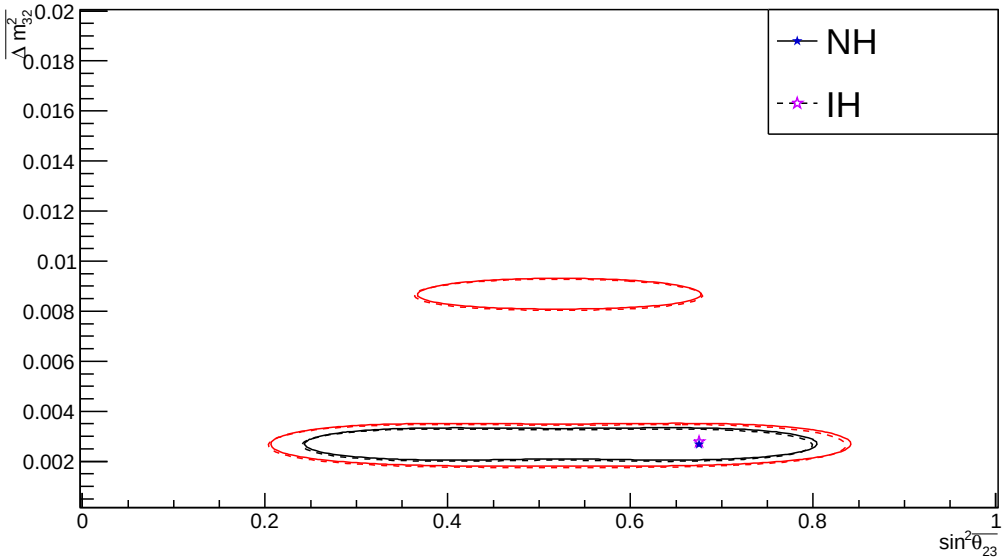


Figure 65: 68% CL and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to FakePot POT fake dataset A, with the effects of statistical fluctuations and postfit systematic variations being included. The best-fit point for each hypothesis is shown, with the larger point representing the better of the two fits.

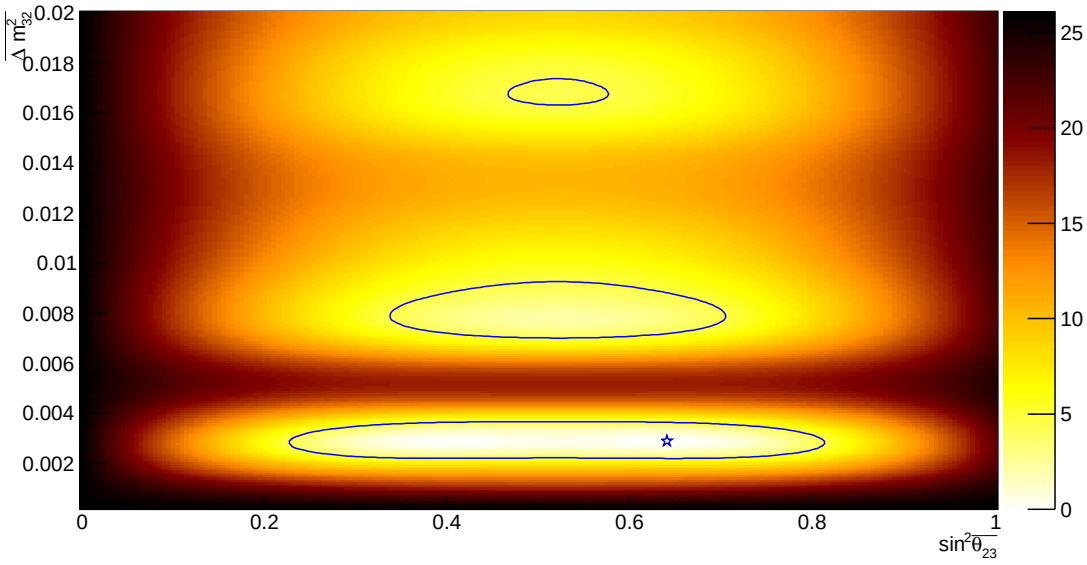


Figure 66: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset B. The normal mass hierarchy is assumed.

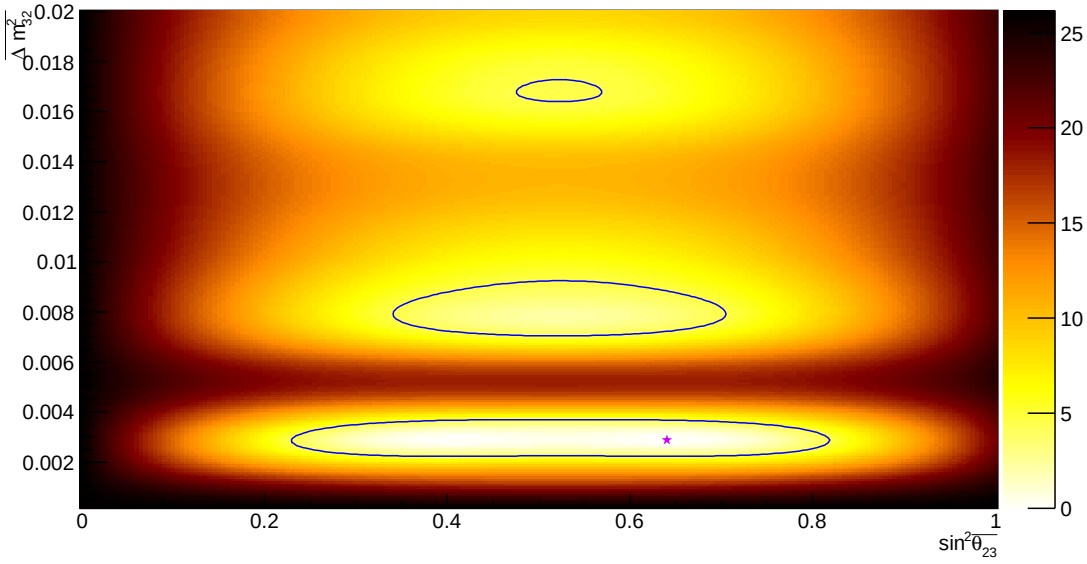


Figure 67: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset B. The inverted mass hierarchy is assumed.

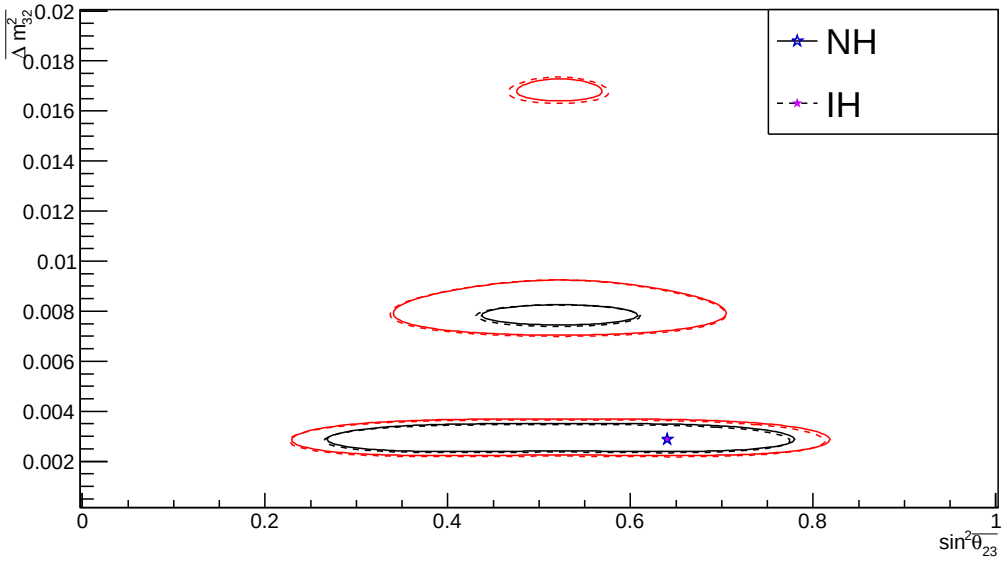


Figure 68: 68% CL and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to FakePot POT fake dataset B, with the effects of statistical fluctuations and postfit systematic variations being included. The best-fit point for each hypothesis is shown, with the larger point representing the better of the two fits.

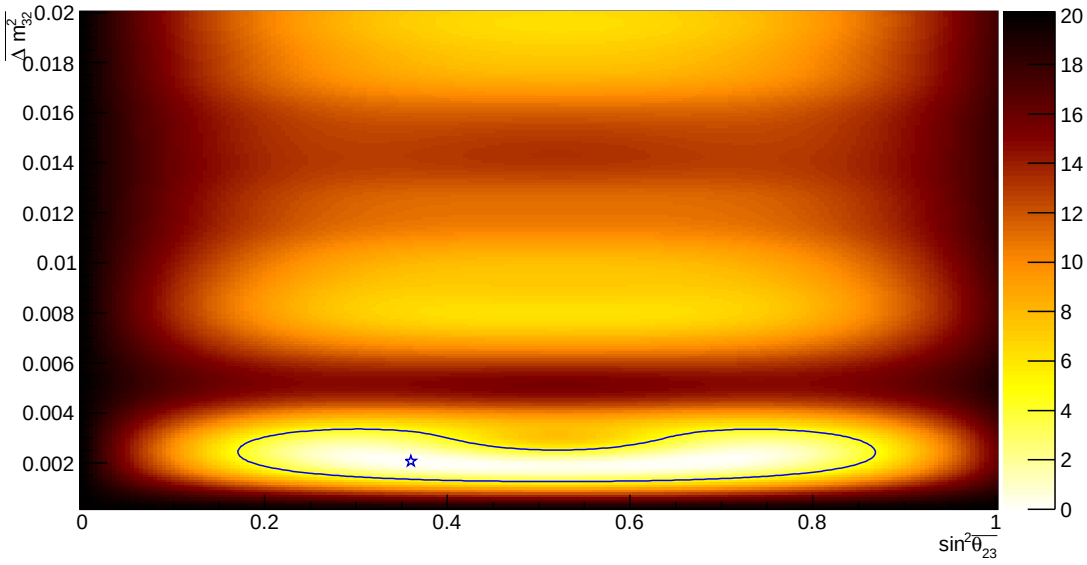


Figure 69: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset C. The normal mass hierarchy is assumed.

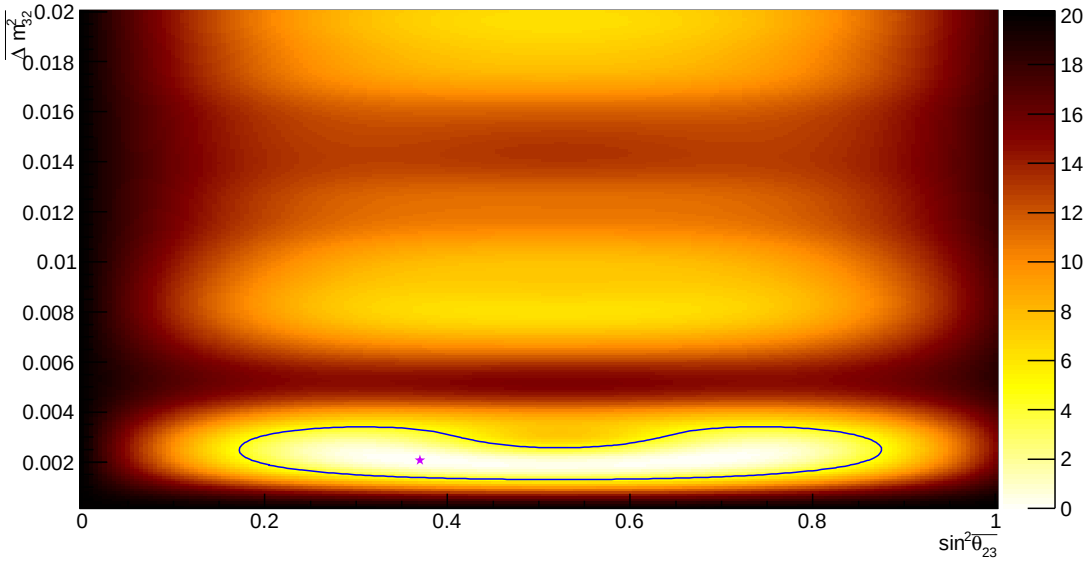


Figure 70: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset C. The inverted mass hierarchy is assumed.

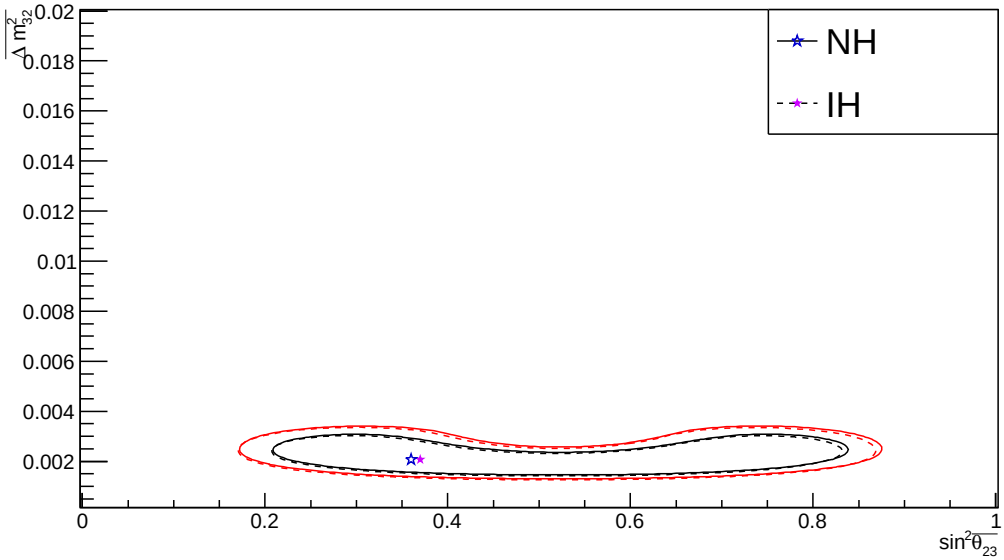


Figure 71: 68% CL and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to FakePot POT fake dataset C, with the effects of statistical fluctuations and postfit systematic variations being included. The best-fit point for each hypothesis is shown, with the larger point representing the better of the two fits.

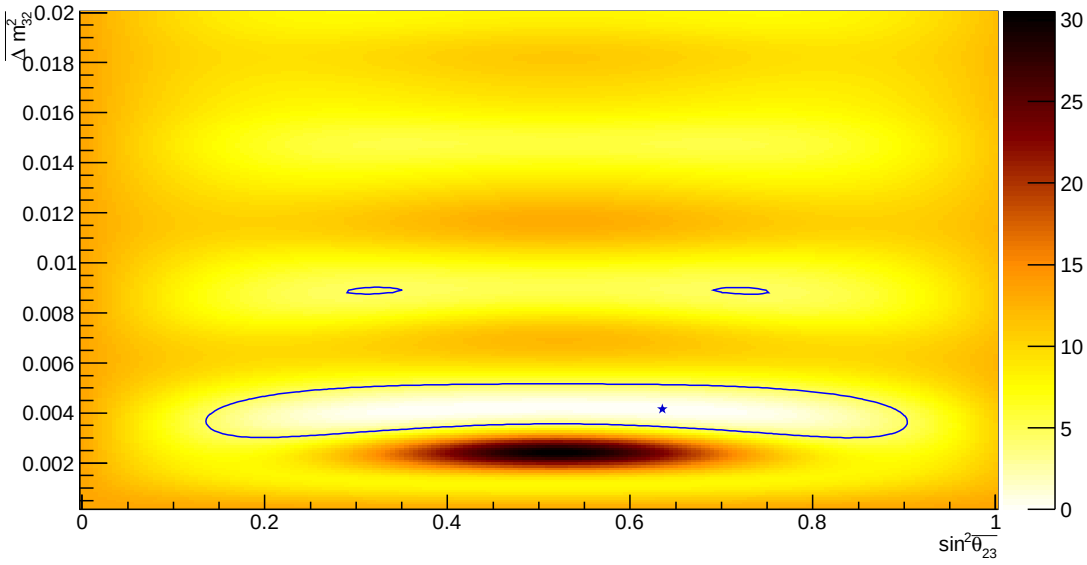


Figure 72: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset D. The normal mass hierarchy is assumed.

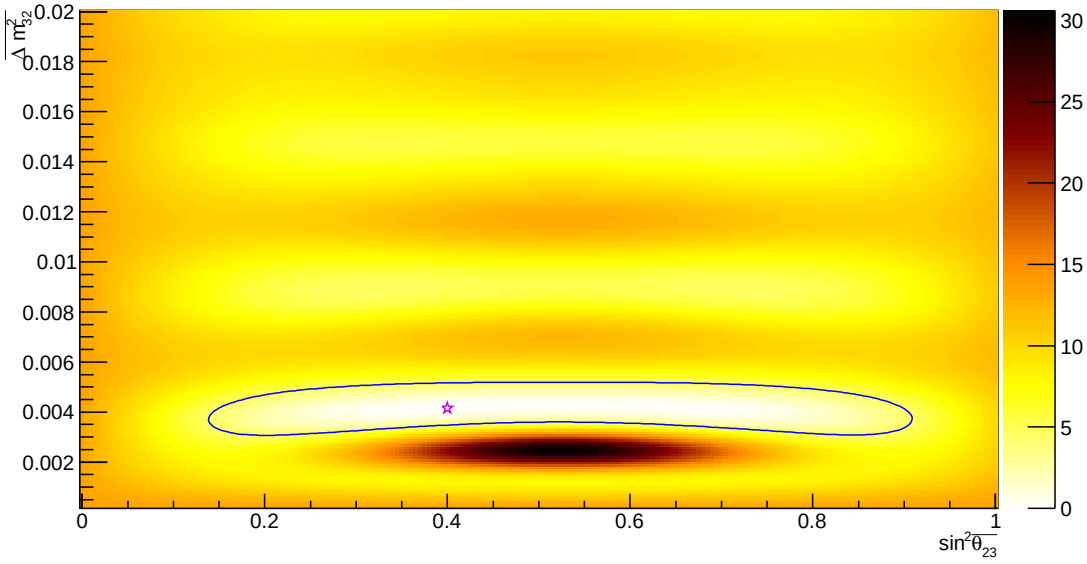


Figure 73: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset D. The inverted mass hierarchy is assumed.

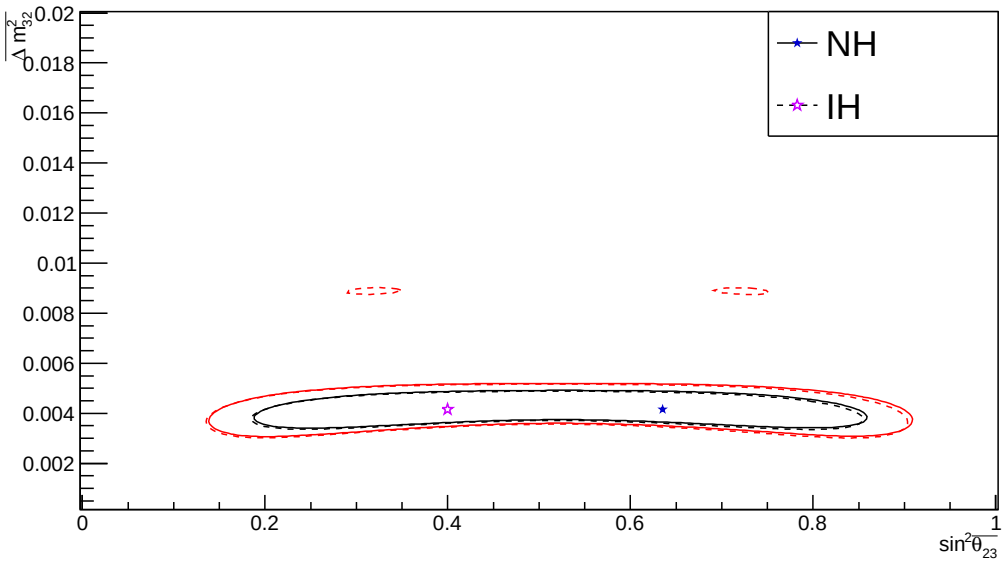


Figure 74: 68% CL and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to FakePot POT fake dataset D, with the effects of statistical fluctuations and postfit systematic variations being included. The best-fit point for each hypothesis is shown, with the larger point representing the better of the two fits.

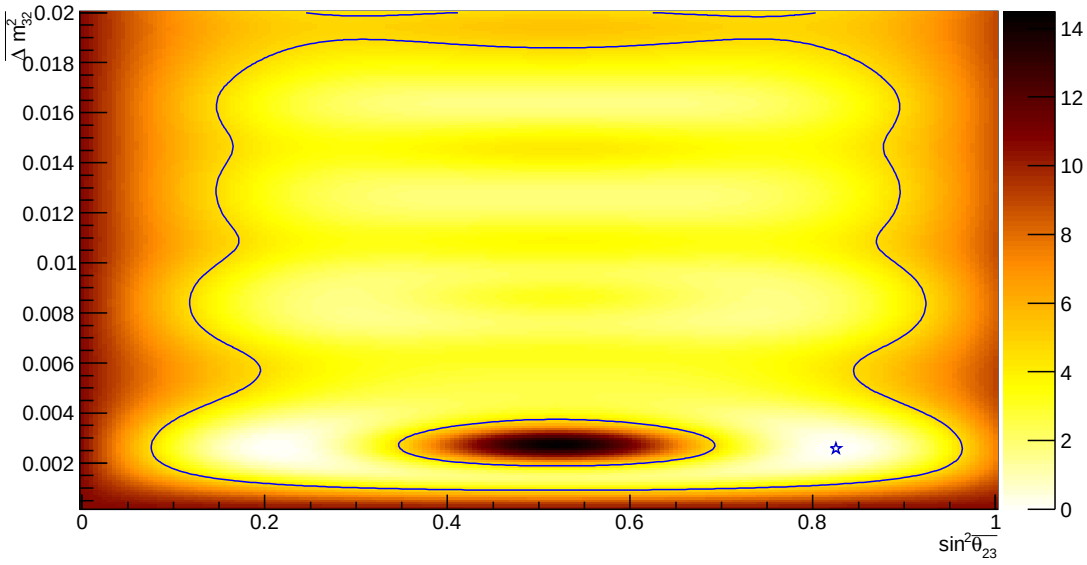


Figure 75: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset E. The normal mass hierarchy is assumed.

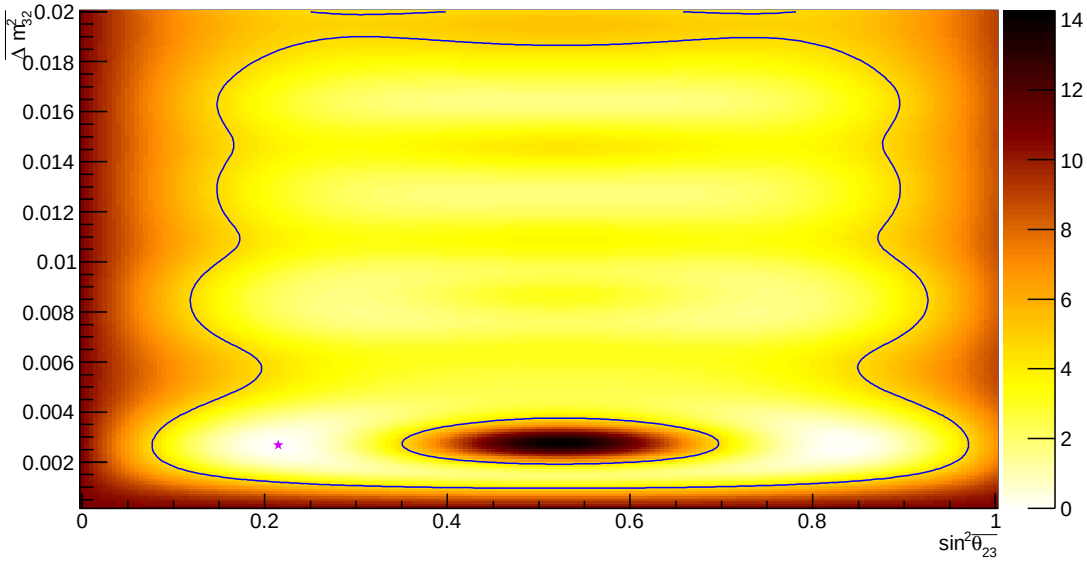


Figure 76: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset E. The inverted mass hierarchy is assumed.

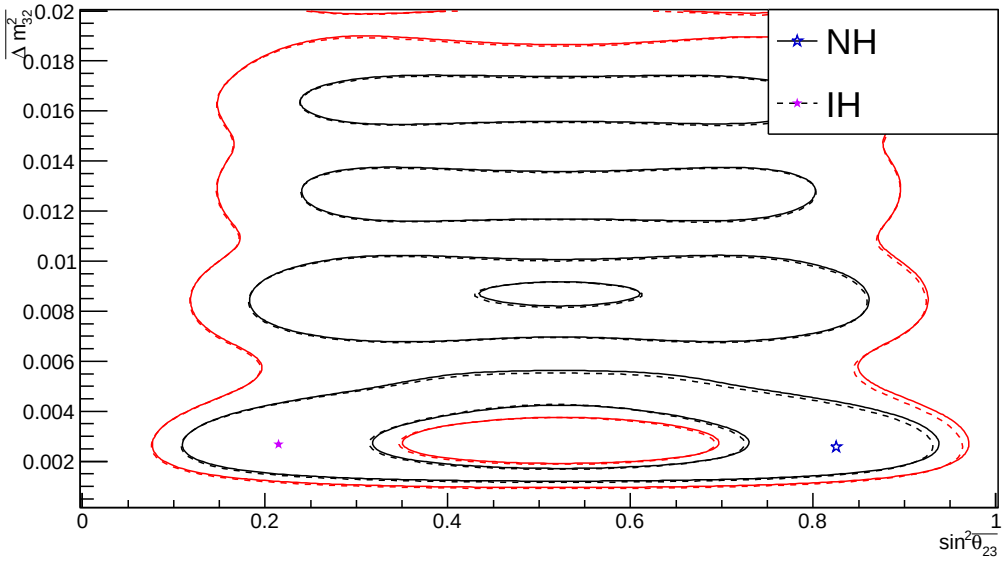


Figure 77: 68% CL and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to FakePot POT fake dataset E, with the effects of statistical fluctuations and postfit systematic variations being included. The best-fit point for each hypothesis is shown, with the larger point representing the better of the two fits.

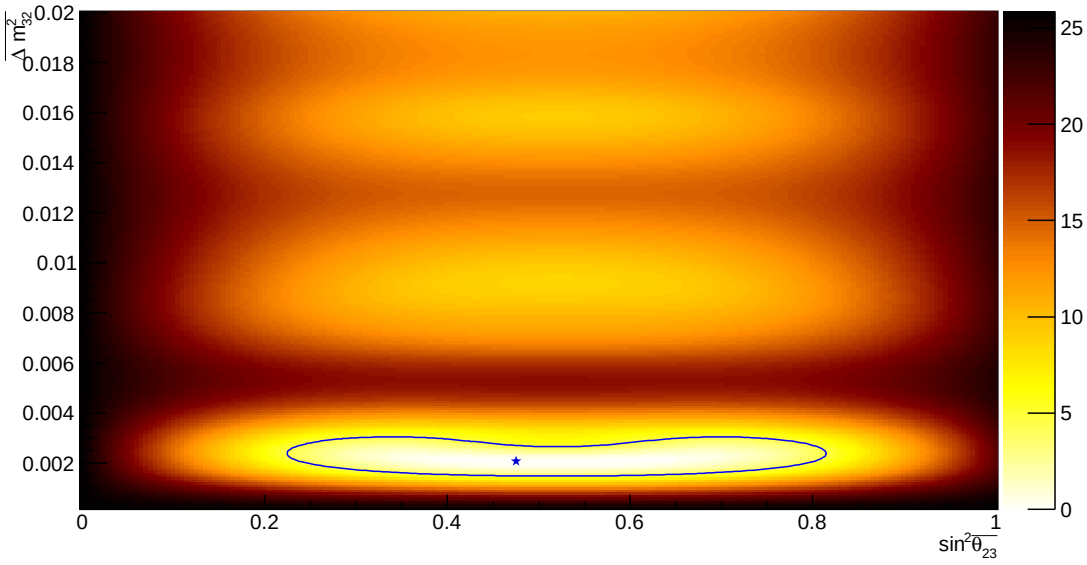


Figure 78: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset F. The normal mass hierarchy is assumed.

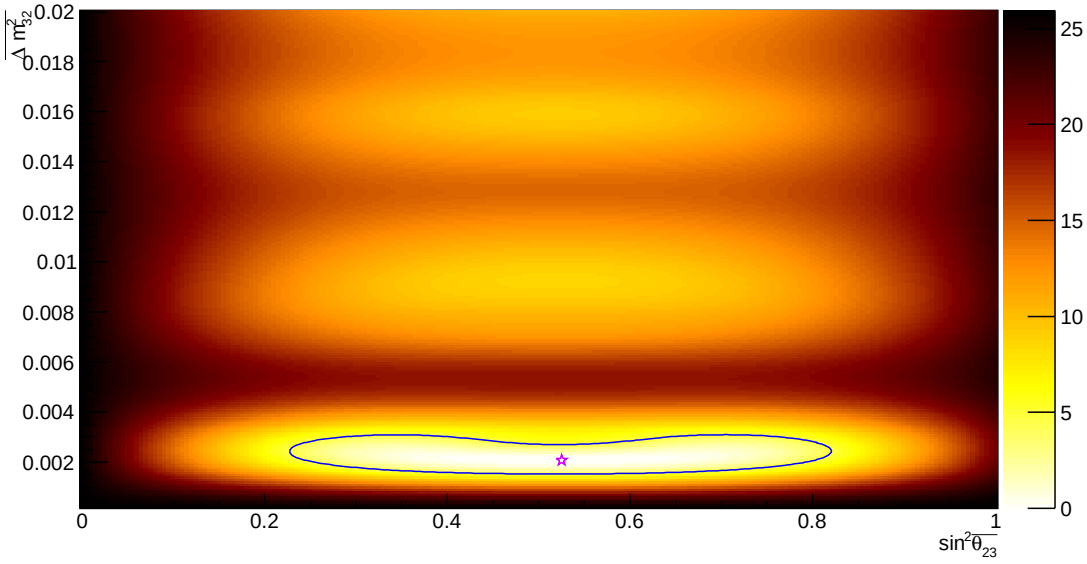


Figure 79: $\Delta\chi^2$ surface from the fit of the 2.500×10^{20}POT fake dataset F. The inverted mass hierarchy is assumed.

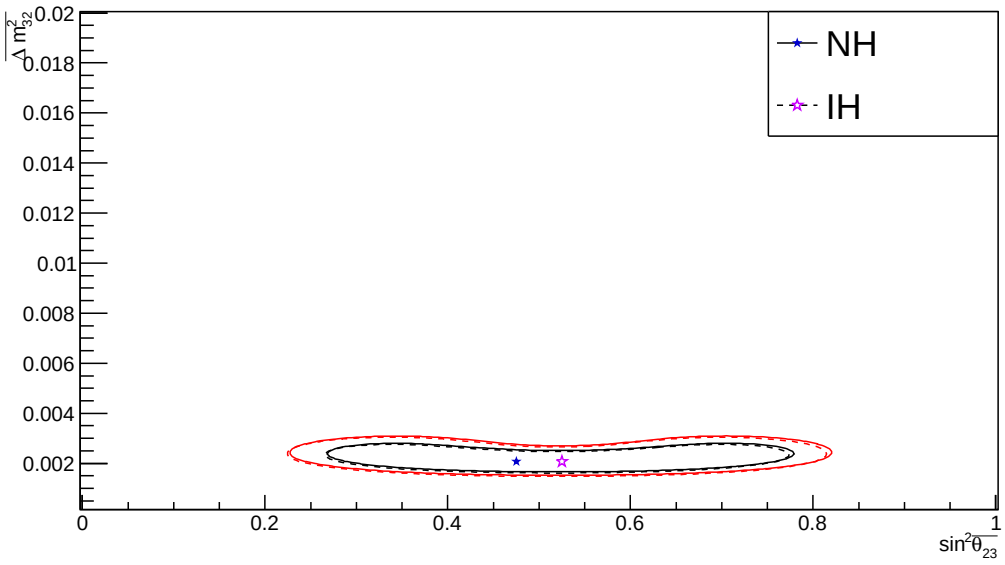


Figure 80: 68% CL and 90% CL allowed regions, constructed with the constant-$\Delta\chi^2$ method, from the fit to FakePot POT fake dataset F, with the effects of statistical fluctuations and postfit systematic variations being included. The best-fit point for each hypothesis is shown, with the larger point representing the better of the two fits.